

A STRIP INEQUALITY FOR PROJECTIONS, TWO-VARIABLE BMV POSITIVITY, AND OPTIMAL CGLMP MEASUREMENTS IN EVERY DIMENSION

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ABSTRACT. We prove that for every number of outcomes $d \geq 2$ the measurements of Durt, Kaszlikowski and Żukowski (DKZ) maximise the CGLMP Bell expression I_d over all projective measurements on maximally entangled states of any local dimension D , the maximal value being $I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d}$, and that they are the only maximisers: equality forces $d \mid D$ and, up to a local unitary $u \otimes \bar{u}$, the strategy is the DKZ strategy tensored with the identity. This settles, for projective measurements, the maximally entangled clause of Part B of Open Quantum Problem 27 of IQOQI Vienna.

The proof has an analytic part and a computer-assisted part. The analytic part is an inequality for a projection B and a Hermitian matrix g of arbitrary size: the sum of an explicit harmonic function h_λ of the strip $0 < \operatorname{Re} z < 1$ over the eigenvalues of $B + ig$ dominates $\operatorname{Tr}[(P(g - \lambda)P)_+]$, $P = \mathbf{1} - B$, with an exact formula for the defect. It follows from a two-variable positivity theorem of Bessis–Moussa–Villani (Stahl) type for projections with an explicit density $\frac{1}{2\pi} \sum_i |\operatorname{Im} \xi_i(s, \tau)|$, where the ξ_i are the roots of $\det(g - s - \xi(P - \tau))$; the proof uses the Fourier-slice mechanism of Heinävaara’s tracial joint spectral measures. Through a noncommutative continuum version of the Stein–Weiss rearrangement theorem and a family of cell embeddings, the strip inequality reduces the CGLMP problem to a statement about Clausen kernels: an explicit vector lies in a convex cone generated by explicit cell vectors. We verify this cone condition for every d by interval arithmetic, with linear-programming certificates on rational cells for $d \leq 200$, certified Gaussian-modulated Dirichlet constructions for $201 \leq d \leq 2000$, and a uniform interval Taylor-model argument for $d \geq 2001$.

We also give a second, distribution-free proof of the two-variable positivity theorem, through a Radon identity for matrix pencils whose proof rests on a complex inviscid Burgers equation for the pencil roots. The argument has been formalised in Lean 4 with Mathlib: for $2 \leq d \leq 20$ the formal proof of optimality and rigidity is complete, including the cone condition; for $d \geq 21$ it is complete except for the cone condition, which is established by the interval-arithmetic certificates; and the strip inequality and the positivity theorem are formalised in full.

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1. INTRODUCTION

1.1. The problem. The CGLMP inequalities of Collins, Gisin, Linden, Massar and Popescu [6] are Bell inequalities for two parties with two measurement settings each and d outcomes per measurement. Write A_1, A_2 and B_1, B_2 for the outcomes (elements of $\mathbb{Z}_d$) of Alice's and Bob's measurements and $P(A_x = B_y + k)$ for the probability that $A_x - B_y \equiv k \pmod{d}$ when the settings x and y are used. The CGLMP expression is

$$(1.1) \quad I_d = \sum_{k=0}^{\lfloor d/2 \rfloor - 1} \left(1 - \frac{2k}{d-1}\right) \left([P(A_1 = B_1 + k) + P(B_1 = A_2 + k + 1) \right. \\ \left. + P(A_2 = B_2 + k) + P(B_2 = A_1 + k)] - [P(A_1 = B_1 - k - 1) + P(B_1 = A_2 - k) \right. \\ \left. + P(A_2 = B_2 - k - 1) + P(B_2 = A_1 - k - 1)] \right)$$

(this is the reading of [6] under which the local bound is derived and the optimal measurements of [6] give the value (1.2) below; see [24, Section 2]). Local hidden variable models satisfy $I_d \leq 2$ [6]. For $d = 2$, I_2 is the CHSH expression; for $d = 3$ see also the Clauser–Horne form of [18].

A *projective strategy on a maximally entangled state* consists of a local dimension $D \geq 1$, the state $\Phi_D = D^{-1/2} \sum_{i=1}^D |ii\rangle \in \mathbb{C}^D \otimes \mathbb{C}^D$, and projection-valued measures $\{P_a^x\}_{a \in \mathbb{Z}_d}$ (Alice, $x = 1, 2$) and $\{Q_b^y\}_{b \in \mathbb{Z}_d}$ (Bob, $y = 1, 2$) on $\mathbb{C}^D$; it gives $P(a, b|x, y) = \langle \Phi_D | P_a^x \otimes Q_b^y | \Phi_D \rangle$. The measurements found numerically by Kaszlikowski et al. [17] and described analytically by Durt, Kaszlikowski and Żukowski [8] (diagonal phase shifts followed by a discrete Fourier transform, then a measurement in the computational basis; we call this the *DKZ strategy*, $D = d$; the explicit bases are recalled in Section 2.2) give

$$(1.2) \quad I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d},$$

for instance $I_{\text{ME}}(3) = 2.8729\dots$, $I_{\text{ME}}(4) = 2.8962\dots$, $I_{\text{ME}}(20) = 2.9547\dots$, and $I_{\text{ME}}(d) \rightarrow 32G/\pi^2 = 2.9698\dots$ as $d \rightarrow \infty$, where G is Catalan's constant (Corollary 8.2). For $d \geq 3$ the maximally entangled state is not the optimal state for I_d [1].

Open Quantum Problem 27 of IQOQI Vienna, “The power of CGLMP inequalities” [29], goes back to questions of Gill [10]. Part A asks whether every facet of the local polytope of the scenario with two settings and d outcomes per party, other than the positivity facets, is of CGLMP type; this was answered negatively by Bancal, Gisin and Pironio [2]. Part B records that the observables which numerically maximise the CGLMP violation on a maximally entangled state are measurements in the computational basis transformed by a discrete Fourier transform and diagonal unitaries, and asks to show that this is necessarily the case; it also asks to show that these measurements give the highest noise resistance and the best Kullback–Leibler discrimination from local models.

1.2. Main results. Our first two results answer the first clause of Part B for projective measurements.

Theorem A (optimality; computer-assisted). *Let $d \geq 2$ and $D \geq 1$. For every projective strategy on the maximally entangled state Φ_D ,*

$$I_d \leq I_{\text{ME}}(d),$$

and the DKZ strategy attains $I_{\text{ME}}(d)$.

Theorem B (rigidity; computer-assisted). *Let $d \geq 2$, $D \geq 1$, and suppose that a projective strategy on Φ_D attains $I_d = I_{\text{ME}}(d)$. Then d divides D , and there is a unitary $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$ such that $uP_a^x u^* = P_a^{x, \text{DKZ}} \otimes \mathbf{1}$ and $\bar{u}Q_b^y \bar{u}^* = Q_b^{y, \text{DKZ}} \otimes \mathbf{1}$ for all x, y, a, b ; moreover $(u \otimes \bar{u})\Phi_D = \Phi_d \otimes \Phi_{D/d}$, and u is unique up to $u \mapsto (\mathbf{1}_d \otimes u')u$ with u' unitary. Consequently, if $d \nmid D$ the maximum of I_d over projective strategies on Φ_D is strictly smaller than $I_{\text{ME}}(d)$.*

The proofs of Theorems A and B are computer-assisted only through Theorem E below; all other steps are proved by hand. The argument has also been formalised in the Lean 4 theorem prover [27, 28] (Section 9.4): for $2 \leq d \leq 20$ the formal proof of Theorems A and B is complete, including Theorem E, and for $d \geq 21$ it is complete except for the input Theorem E. Thus Theorems A and B are proved for every d , with a computer-assisted step (interval arithmetic) for $d \geq 21$, and are fully formalised for $d \leq 20$. The analytic heart of the argument is a matrix inequality, which we now state. Let $S = \{z \in \mathbb{C} : 0 < \text{Re } z < 1\}$ and, for $\lambda \in \mathbb{R}$ and $z = x + iy \in S$,

$$(1.3) \quad h_\lambda(x + iy) = -\frac{1}{\pi^2} \text{Im Li}_2(e^{\pi(y-\lambda)} e^{-i\pi x}),$$

where Li_2 is the dilogarithm (principal branch, holomorphic on $\mathbb{C} \setminus [1, \infty)$). The function h_λ is harmonic on S and extends continuously to the closed strip, with boundary values $(y - \lambda)_+$ on the line $\text{Re } z = 0$ and 0 on the line $\text{Re } z = 1$ (Lemma 3.2). For $0 < x < 1$ let

$$(1.4) \quad K_x(u) = \frac{\sin \pi x}{2(\cosh \pi u - \cos \pi x)} \quad (u \in \mathbb{R}),$$

the Poisson kernel of the strip S for its left edge. For an orthogonal projection E and a Hermitian matrix X we write $\text{Tr}_E f(EXE)$ for the trace over $\text{ran } E$ of f applied to the compression of X to $\text{ran } E$, and X_+ for the positive part of a Hermitian matrix X .

Theorem C (strip inequality). *Let $M \geq 1$, let B be an orthogonal projection on $\mathbb{C}^M$, $P = \mathbf{1} - B$, and let g be a Hermitian $M \times M$ matrix. For every $\lambda \in \mathbb{R}$,*

$$(1.5) \quad \sum_{\nu \in \text{spec}(B+ig)} h_\lambda(\nu) - \text{Tr}[(P(g - \lambda)P)_+] = \int_0^1 \int_{\mathbb{R}} K_{1-\tau}(s - \lambda) \rho(s, \tau) \, ds \, d\tau \geq 0,$$

where the eigenvalues are counted with algebraic multiplicity and $\rho \geq 0$ is the density of Theorem D. Equality holds for one λ if and only if it holds for all λ , if and only if $Bg = gB$.

Theorem D (two-variable BMV positivity for projections). *In the situation of Theorem C let $r = \text{rank } B$, let $\lambda_{\min}, \lambda_{\max}$ be the extreme eigenvalues of g , and for $s \in \mathbb{R}$ and $0 < \tau < 1$ let $\xi_1(s, \tau), \dots, \xi_M(s, \tau)$ be the roots, with multiplicity, of the polynomial $\xi \mapsto \det(g - s - \xi(P - \tau))$ of degree M . Put*

$$\rho(s, \tau) = \frac{1}{2\pi} \sum_{i=1}^M |\text{Im } \xi_i(s, \tau)| \quad (0 < \tau < 1), \quad \rho(s, \tau) = 0 \quad (\tau \notin (0, 1)).$$

Then ρ is continuous on $\mathbb{R} \times (0, 1)$, vanishes unless $\lambda_{\min} < s < \lambda_{\max}$, satisfies $\rho(s, \tau) \leq \frac{M}{2\pi} \|g - s\|(\tau(1 - \tau))^{-1/2}$, hence is integrable, and for all $(a, t) \in \mathbb{C}^2$

$$(1.6) \quad \mathcal{D}(a, t) := \operatorname{Tr} e^{ag-tP} - e^{-t} \operatorname{Tr}_P e^{aPgP} - \operatorname{Tr}_B e^{aBgB} = a^2 \int_0^1 \int_{\mathbb{R}} e^{as-t\tau} \rho(s, \tau) \, ds \, d\tau.$$

Moreover $\int_{\mathbb{R}} \rho(s, \tau) \, ds = \|BgP\|_F^2$ for every $\tau \in (0, 1)$.

Taking $a = 1$ in (1.6) shows that $t \mapsto \operatorname{Tr} e^{g-tP}$ is the Laplace transform of the positive measure $\operatorname{Tr}_B(e^{BgB}) \delta_0 + \operatorname{Tr}_P(e^{PgP}) \delta_1 + (\int_{\mathbb{R}} e^s \rho(s, \tau) \, ds) \, d\tau$ on $[0, 1]$, which is Stahl's theorem [32] (the former Bessis–Moussa–Villani conjecture [3]) for a projection, with an explicit density. Theorem D is stronger: it asserts positivity of a measure in two variables, after the two pinched terms are subtracted and the result is divided by a^2 . Neither modification can be omitted: $\mathcal{D}(0, t) = 0$ for all t , so $\mathcal{D}$ itself is the Laplace transform of a positive measure only if $\mathcal{D} \equiv 0$, and $a^{-2} \operatorname{Tr} e^{ag-tP}$ is not entire.

The last ingredient is a condition on explicit Clausen kernels. Let $N = 4d$ and let $\operatorname{Cl}_2(\theta) = \sum_{n \geq 1} n^{-2} \sin n\theta$ be the Clausen function. Put

$$\Xi(u) = -\frac{N^2}{2\pi^2} \operatorname{Cl}_2\left(\frac{2\pi u}{N}\right), \quad \Xi^\sharp(u) = \Xi(u) + \Xi(2d - u) \quad (0 \leq u \leq 2d),$$

so that $(\Xi^\sharp)''(u) = 2 \csc \frac{\pi u}{2d}$ on $(0, 2d)$. Let $\Delta_d = \{\ell \in \mathbb{R}^d : \ell_r \geq 0, \sum_r \ell_r = d\}$ and $\Delta_d^+ = \{\ell \in \Delta_d : \ell_r > 0 \text{ for all } r\}$. For $\ell \in \Delta_d$, extended d -periodically, let $S_r(m) = \ell_r + \ell_{r+1} + \cdots + \ell_{r+m-1}$ and define the *cell vector* $u^\ell \in \mathbb{R}^{d-1}$ by

$$(1.7) \quad u_m^\ell = \Delta_m^2 \left[\frac{1}{d} \sum_{r=0}^{d-1} \Xi^\sharp(S_r(m)) \right] \quad (1 \leq m \leq d-1),$$

where $\Delta_m^2 f = f(m+1) - 2f(m) + f(m-1)$. Let $v \in \mathbb{R}^{d-1}$, $v_m = 2 \csc \frac{\pi m}{2d}$, and $t_s = e_s + e_{d-s}$ ($1 \leq s \leq d-1$), where (e_m) is the standard basis.

Theorem E (cone condition; computer-assisted). *For every $d \geq 2$ there are a finite positive Borel measure μ on Δ_d with $\mu(\Delta_d^+) = \mu(\Delta_d) > 0$ and numbers $w_s \geq 0$ such that*

$$(1.8) \quad v = \int_{\Delta_d} u^\ell \mu(d\ell) + \sum_{s=1}^{\lfloor d/2 \rfloor} w_s t_s.$$

For $2 \leq d \leq 200$ the measure μ is a finite sum of point masses at rational cell vectors and $w = 0$; the certificates are verified in interval arithmetic, and for $d \leq 20$ also by the Lean kernel in exact rational interval arithmetic. For $d \geq 201$, μ is the law of an explicit random cell vector (a Dirichlet vector with Gaussian-modulated parameters), and the verification is by interval arithmetic for each $201 \leq d \leq 2000$ and by a single interval Taylor-model computation, uniform in d , for $d \geq 2001$ (Section 6).

1.3. Status of Problem 27. Table 1 summarises the status of the clauses of Problem 27 after this work.

1.4. Strategy of the proof. The proof proceeds in six steps; Sections 2–7 follow this order.

(1) *The clock model.* By an exact reduction from [24], recalled in Section 2, $I_d = 4F(V)/(d(d-1))$ for a d -tuple V of unitaries of order four on some $\mathbb{C}^M$, and $I_{\text{ME}}(d) = 4F_{\text{DKZ}}/(d(d-1))$. The spectral projections of the V_k form a family $(Q_x)_{x \in \mathbb{Z}_{4d}}$ of projections, each residue class modulo d being a four-outcome projection-valued measure, and $F(V) - F_{\text{DKZ}} = \frac{1}{2} \langle v, \delta \rangle$, where $\delta \in \mathbb{R}^{d-1}$ collects the deviations of $d-1$ “net pair counts” from their values at the DKZ configuration (Lemma 2.5).

Question	Status	References
27A: all non-positivity facets of the local polytope are of CGLMP type	false ($d = 4$)	Bancal–Gisin–Pironio [2]
27B: DKZ measurements are optimal for I_d on maximally entangled states	proved for every d (projective measurements; computer-assisted for $d \geq 21$; formally verified in Lean for $d \leq 20$)	Theorem A; earlier: $d = 2$ [5], $d \leq 20$ [25], commuting (equal-link) strategies for all d [24]
27B: they are <i>necessarily</i> the optimal ones	proved for every d (projective; computer-assisted for $d \geq 21$; formally verified in Lean for $d \leq 20$)	Theorem B; earlier: $d \leq 20$ [25], equal-link strategies [24]
27B: highest noise resistance	CGLMP witness, white noise: every d ; against all Bell inequalities: $d = 3$ proved, $d = 4, 5$ numerical	Corollary 8.1; [25], [11]
27B: best Kullback–Leibler discrimination	false for $d = 4, \dots, 9$	Zhang [35] ($d = 4$), [25] ($d = 4, \dots, 9$)
Related: maximal value of I_d over all states	exact for $d = 3, \dots, 8$	Ioannou–Rosset [15] ($d = 3, 4$), [25] ($d = 5, \dots, 8$)

TABLE 1. The clauses of Open Quantum Problem 27. “Projective” means that POVMs are not covered.

(2) *Two-variable BMV positivity and the strip inequality* (Section 3): Theorems D and C, proved by hand. Appendix B gives a second proof of both, without distribution theory or the Paley–Wiener–Schwartz theorem, through a Radon identity for matrix pencils.

(3) *A noncommutative continuum theorem* (Section 4): for projection-valued step fields B on the circle, fields $0 \leq A \leq \mathbf{1} - B$, and the conjugate function g of B , the pairing $\int \tau(Ag)$ is at most the value of the classical (commutative) problem. This is a matrix version of the circle Stein–Weiss rearrangement theorem [33, 20]; the strip inequality enters pointwise, through a harmonic function on the disc.

(4) *Cell embeddings* (Section 5): placing the projections Q_x on cells of lengths $\ell_{x \bmod d}$ on a circle of length $4d$ produces a step field to which (3) applies; averaging over rotations gives the linear inequalities $\langle u^\ell, \delta \rangle \leq 0$. Consequently Theorem E implies $\langle v, \delta \rangle \leq 0$, which is Theorem A.

(5) *The cone condition* (Section 6): Theorem E, computer-assisted.

(6) *Rigidity* (Section 7): in the equality case every inequality of the chain is an equality. Through the equality clause of Theorem C this forces $[B(\theta), g(\theta)] = 0$ almost everywhere on a cell embedding with all cells of positive length; a residue argument then shows that all Q_x commute, and the commutative case [24] identifies the strategy.

The continuum problem in (3) is degenerate: its maximum is attained by many configurations, including non-commuting ones (Remark 4.7), and the discrete problem differs from it by terms of order d located at the junctions between blocks. Step (4) avoids any explicit estimate of these junction terms: the cell inequalities are exact consequences of the continuum theorem, and the

discrete kernel is recovered from them by the cone condition. The mechanism behind the cone condition is that the discrete cotangent kernel is the continuum kernel averaged over Poisson-like random cells (Remark 5.6).

1.5. Relation to previous work. For $d = 2$, Theorem A is Tsirelson’s bound [5] on maximally entangled states and Theorem B is the corresponding rigidity. For $d = 3$, Lang, Vértesi and Navascués [21] showed with a semidefinite relaxation that the maximal value on maximally entangled states agrees with $I_{\text{ME}}(3)$ to within 10^{-10} (this is the partial result listed in [29]). In [25] we gave exact sum-of-squares certificates proving Theorems A and B for $3 \leq d \leq 20$, and in [24] we proved both theorems for every d for strategies whose reduced family commutes (equivalently, whose four “link” operators coincide). Ioannou and Rosset [15] obtained exact sum-of-squares bounds for the maximal value of I_3 and I_4 over all states. The SATWAP expressions of Salavrakos et al. [30] are maximally violated by maximally entangled states with measurements of CGLMP type and underlie device-independent self-testing results [31]; they are different Bell expressions. Zhang [35] found the first counterexample to the Kullback–Leibler clause ($d = 4$).

The mechanism of our proof of Theorem D—inverting a trace function of a pencil of Hermitian matrices by a two-dimensional Fourier transform computed along lines, and identifying the density as $\frac{1}{2\pi} \sum |\text{Im}|$ of the roots of an auxiliary pencil by an integral identity for $\log |\det|$ —is the one introduced by Heinävaara [12] for tracial joint spectral measures, which yields, among other results, a new proof of Stahl’s theorem [32] (see also [9, 23]). Theorem D is a variant adapted to a projection and to an affine chart: the subtraction of the two pinched terms and the division by a^2 produce a measure without singular part, which is what the strip inequality requires. We found Theorem D independently, but we do not claim novelty for the mechanism, and we give complete proofs. We have not found Theorem C, or an equivalent statement, in the literature. Its commutative case is an identity, and its continuum consequence (Theorem 4.2) contains the circle version of the theorem of Stein and Weiss [33] on the distribution of the conjugate function of an indicator function, in the form of Laeng [20] (see also [4, 7]).

Appendix B gives a second proof of Theorem D that uses no distributions, no Paley–Wiener–Schwartz theorem and no Fourier inversion in two variables. Its key step is a Radon identity (Theorem B.1): the integral over $\tau \in (0, 1)$ of $\frac{1}{2\pi} \sum_i |\text{Im } y_i(\tau)|$, where the $y_i(\tau)$ are the eigenvalues of $(P - \tau)^{-1}H$, equals $\text{Tr } H_+ - \text{Tr}(BHB + PHP)_+$ for every Hermitian H . We prove it with a complex inviscid Burgers equation satisfied by the pencil roots. The same argument applies to an arbitrary Hermitian matrix in place of the projection, and then gives a proof of Stahl’s theorem without distribution theory or Fourier analysis (Remark B.8); we make no priority claim about this by-product, and refer to [32, 9, 12] for the history of the theorem and of the mechanism. This elementary route is the one followed by the Lean formalisation.

1.6. Conventions. Tr is the trace on $M_M(\mathbb{C})$ and $\tau = \frac{1}{M} \text{Tr}$. Spectra are multisets, counted with algebraic multiplicity. $\|X\|$ is the operator norm and $\|X\|_F$ the Frobenius norm. For $f \in L^1(\mathbb{R})$, $\hat{f}(\sigma) = \int_{\mathbb{R}} f(w) e^{-i\sigma w} dw$; the Hilbert transform is $\mathcal{H}f(w) = \frac{1}{\pi} \text{p.v.} \int_{\mathbb{R}} \frac{f(u)}{w-u} du$, which acts on $L^2(\mathbb{R})$ as the Fourier multiplier $-i \text{sgn } \sigma$. On the circle $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ we use the normalised measure $dm = d\theta/2\pi$, and the conjugate function of $f \in L^2(\mathbb{T})$ is $\text{p.v.} \int \cot \frac{\theta-\varphi}{2} f(\varphi) dm(\varphi)$, the multiplier $-i \text{sgn } n$. Catalan’s constant is $G = \text{Cl}_2(\pi/2) = \sum_{n \geq 0} (-1)^n (2n+1)^{-2}$, and $\text{Cl}_2(\theta) = -\int_0^\theta \log |2 \sin \frac{t}{2}| dt$. For $\alpha, \beta \geq 0$ with $\alpha + \beta \leq 1$ we put

$$(1.9) \quad \Phi_c(\alpha, \beta) = \frac{1}{2\pi^2} \left[\text{Cl}_2(2\pi\alpha) + \text{Cl}_2(2\pi\beta) + \text{Cl}_2(2\pi(1-\alpha-\beta)) \right], \quad \Phi_c\left(\frac{1}{4}, \frac{1}{4}\right) = \frac{G}{\pi^2}.$$

Results labelled *computer-assisted* depend on computations in rigorous interval arithmetic with outward rounding; everything else is proved by hand. Numerical observations that are not part

of any proof are labelled as such. The Lean formalisation (Section 9.4) is an independent check; no proof in this paper depends on it.

2. THE CLOCK MODEL AND THE Q-CONFIGURATION

Throughout, $d \geq 2$, $N = 4d$, $w = e^{2\pi i/d}$, $z = e^{2\pi i/N}$ (so $z^4 = w$ and $z^d = i$), and $\kappa(u) = \cot(\pi u/N)$ for $u \in \mathbb{Z}_N \setminus \{0\}$, $\kappa(0) = 0$.

2.1. Strategies and the clock model. Encode a projective strategy on Φ_D by the unitaries $U_x = \sum_a w^a P_a^x$ and $U'_y = \sum_b w^b Q_b^y$ and put

$$(2.1) \quad R_1 = U_2, \quad R_2 = (U'_2)^T, \quad R_3 = U_1, \quad R_4 = (U'_1)^T,$$

the transpose being taken in the Schmidt basis of Φ_D ; then $R_i^d = \mathbf{1}$, and conversely every quadruple of unitaries on $\mathbb{C}^D$ with $R_i^d = \mathbf{1}$ comes from a projective strategy on Φ_D , through spectral projections. A simultaneous conjugation $R_i \mapsto u R_i u^*$ is the same as applying the local unitary $u \otimes \bar{u}$, which fixes Φ_D . For unitaries $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = \mathbf{1}$ let

$$(2.2) \quad F(V) = \sum_{0 \leq j < k \leq d-1} \operatorname{Re} \left[h_{k-j} \tau(V_j V_k^*) \right], \quad h_m = \sec \frac{\pi m}{2d} - i \csc \frac{\pi m}{2d},$$

and $F_{\text{DKZ}} = F(\mathbf{1}, \dots, \mathbf{1}) = \sum_{m=1}^{d-1} (d-m) \sec \frac{\pi m}{2d}$. Let X be the cyclic shift $X|k\rangle = |k+1\rangle$ on $\mathbb{C}^d$ and define the *covariant strategy* R^V on $\mathbb{C}^d \otimes \mathbb{C}^M$ by

$$(2.3) \quad W = \sum_{k=0}^{d-1} |k\rangle\langle k| \otimes z^k V_k, \quad R_1^V = X \otimes \mathbf{1}_M, \quad R_{i+1}^V = W R_i^V W^* \quad (i = 1, 2, 3).$$

Theorem 2.1 (reduction to the clock model [24, Theorem 2.3]). *Let $d \geq 2$.*

- (1) *For every covariant strategy, $I_d(R^V) = 4F(V)/(d(d-1))$.*
- (2) *For every projective strategy R on Φ_D there are M (one can take $M = 4D$) and unitaries $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = \mathbf{1}$, unique up to a simultaneous unitary conjugation, such that R is unitarily equivalent (by a simultaneous conjugation of $R_1, \dots, R_4$) to a direct summand of R^V and $I_d(R) = I_d(R^V) = 4F(V)/(d(d-1))$.*

In particular $I_d \leq I_{\text{ME}}(d)$ holds for all projective strategies on maximally entangled states if and only if $F(V) \leq F_{\text{DKZ}}$ for all M and all $V_0, \dots, V_{d-1} \in U(M)$ with $V_k^4 = \mathbf{1}$.

The family V in (2) is the *reduced family* of R ; it is obtained from a twirl over a cyclic symmetry of order $4d$ followed by the Stone–von Neumann theorem for $\mathbb{Z}_d$.

2.2. The DKZ strategy and the commutative sector. Let $W_0 = \operatorname{diag}(z^k)_{k=0}^{d-1}$ and $R^{(0)} = R^V$ for $M = 1$, $V \equiv \mathbf{1}$. The DKZ measurements are the measurements in the orthonormal bases $|k\rangle_{A,x} = d^{-1/2} \sum_j e^{-2\pi i j(k+\alpha_x)/d} |j\rangle$ and $|l\rangle_{B,y} = d^{-1/2} \sum_j e^{2\pi i j(l-\beta_y)/d} |j\rangle$ with $\alpha = (0, \frac{1}{2})$ and $\beta = (\frac{1}{4}, -\frac{1}{4})$ (the measurements of [6, eqs. (12)–(15)]), and their chain (2.1) is $GR^{(0)}G^*$ with $G = W_0^{-2}$ [24, Section 2.4]. Hence $I_d(\text{DKZ}) = 4F_{\text{DKZ}}/(d(d-1)) = I_{\text{ME}}(d)$.

Theorem 2.2 (commutative sector [24, Theorem B, Corollary C, Proposition 2.4]). *Let $d \geq 2$.*

- (1) *If $V_0, \dots, V_{d-1} \in U(M)$ commute pairwise and $V_k^4 = \mathbf{1}$, then $F(V) \leq F_{\text{DKZ}}$, with equality if and only if every joint eigenvector v of the V_k satisfies $V_k v = i^{-a_k} v$ with a one-step configuration $a_k = c + [k \geq r]$ ($c \in \mathbb{Z}_4$, $0 \leq r \leq d-1$; c and r may depend on v).*
- (2) *A projective strategy on Φ_D has commuting reduced family if and only if its four link operators coincide, $R_1^n R_2^{-n} = R_2^n R_3^{-n} = R_3^n R_4^{-n} = w^{-n} R_4^n R_1^{-n}$ for all n .*
- (3) *If a projective strategy on Φ_D with commuting reduced family attains $I_d = I_{\text{ME}}(d)$, then $d \mid D$ and there is a unitary $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$ with $u R_i u^* = R_i^{\text{DKZ}} \otimes \mathbf{1}$ for $i = 1, \dots, 4$.*

2.3. Q-configurations.

Definition 2.3. A Q -configuration on $\mathbb{C}^M$ is a family $(Q_x)_{x \in \mathbb{Z}_N}$ of orthogonal projections on $\mathbb{C}^M$ such that for every $k \in \{0, \dots, d-1\}$ the four projections Q_{k-da} , $a \in \mathbb{Z}_4$, are pairwise orthogonal with sum $\mathbf{1}$. We put $A_x = Q_x$ and $B_x = Q_{x+d}$, and for $j \in \mathbb{Z}_N$ we let $(\text{rot}_j Q)_x = Q_{x-j}$.

Every element of $\mathbb{Z}_N$ is uniquely of the form $k - da$ with $0 \leq k \leq d-1$ and $a \in \mathbb{Z}_4$. Given a family V as above, let Q_{k-da} be the spectral projection of $E_k := V_k^*$ for the eigenvalue i^a ; this is a Q -configuration, $E_k = \sum_{a \in \mathbb{Z}_4} i^a Q_{k-da}$, and conversely every Q -configuration arises from a unique family V in this way. The following properties are immediate: $A_x B_x = 0$ (two outcomes of one measurement); for every d -periodic function ℓ on $\mathbb{Z}_N$,

$$(2.4) \quad \sum_{x \in \mathbb{Z}_N} \ell_x B_x = \sum_{x \in \mathbb{Z}_N} \ell_x A_x = \left(\sum_{r=0}^{d-1} \ell_r \right) \mathbf{1};$$

and every rotation $\text{rot}_j Q$ is again a Q -configuration. The Q -configuration of $V \equiv \mathbf{1}$ ($M = 1$) is the *window* $Q_x = \mathbf{1}_{[0,d)}(x)$; its rotations are the windows $\mathbf{1}_{[c,c+d)}$.

Lemma 2.4 (projection form). *For every family V and its Q -configuration,*

$$F(V) = \frac{1}{2} \langle A, B \rangle_\tau - \frac{d}{2}, \quad \langle A, B \rangle_\tau := \sum_{x, y \in \mathbb{Z}_N} \kappa(x - y) \tau(A_x B_y).$$

Proof. This is [24, eq. (13)]; we include the short proof. For $j < k$, $m = k - j$ and $a, b \in \mathbb{Z}_4$ put $x = j - da$, $y = k - db$. Then $\tau(V_j V_k^*) = \tau(E_j^* E_k) = \sum_{a, b} i^{b-a} \tau(Q_x Q_y)$ and $\text{Re}[h_m i^s] = \sec(\frac{\pi m}{2d} - \frac{\pi s}{2})$ for $s \in \mathbb{Z}_4$ (check $s = 0, 1, 2, 3$), which equals $\sec \frac{\pi(y-x)}{2d}$ because $y - x = m - d(b - a)$ and $u \mapsto \sec \frac{\pi u}{2d}$ is even with period $4d$. Since $\tau(Q_x Q_y) = \tau(Q_y Q_x)$,

$$F(V) = \frac{1}{2} \sum_{x \not\equiv y \pmod{d}} \tau(Q_x Q_y) \sec \frac{\pi(x - y)}{2d}.$$

The identity $\sec t = \frac{1}{2} [\cot(\frac{t}{2} + \frac{\pi}{4}) - \cot(\frac{t}{2} - \frac{\pi}{4})]$ with $t = 2\pi(x - y)/N$ gives $\sec \frac{\pi(x - y)}{2d} = \frac{1}{2} [\kappa(x - y + d) - \kappa(x - y - d)]$ for $x \not\equiv y \pmod{d}$, and the two halves give equal sums (exchange x and y and use that κ is odd). Hence $F(V) = \frac{1}{2} \sum_{x \not\equiv y \pmod{d}} \tau(Q_x Q_y) \kappa(x - y + d) = \frac{1}{2} \sum_{x \not\equiv y' \pmod{d}} \tau(A_x B_{y'}) \kappa(x - y')$ with $y' = y - d$. In the full sum $\langle A, B \rangle_\tau$ the pairs with $x \equiv y' \pmod{d}$ contribute only for $x = y' + d$ (otherwise $Q_x Q_{y'+d} = 0$, or $\kappa(0) = 0$ when $x = y'$), and these contribute $\kappa(d) \sum_x \tau(Q_x) = d$. So $\langle A, B \rangle_\tau = 2F(V) + d$. $\square$

2.4. Net pair counts. For a Q -configuration put

$$T(m) = \sum_{y \in \mathbb{Z}_N} \tau(A_{y+m} B_y), \quad \mathcal{N}(m) = T(m) - T(-m), \quad \delta_m = \mathcal{N}(m) - m \quad (1 \leq m \leq d-1),$$

and $\delta = (\delta_m)_{m=1}^{d-1}$. For sites $0 \leq j < k \leq d-1$ let $\pi_{jk}(t) = \sum_{b \in \mathbb{Z}_4} \tau(Q_{j-db} Q_{k-d(b+t)})$, $t \in \mathbb{Z}_4$; this is a probability vector on $\mathbb{Z}_4$ (nonnegative, since $\tau(QQ') \geq 0$ for projections, with sum $\sum_b \tau(Q_{j-db}) = 1$).

Lemma 2.5 (linear form). *For every Q -configuration:*

- (1) $\mathcal{N}(2d - m) = \mathcal{N}(m)$ for $1 \leq m \leq 2d - 1$, and $\mathcal{N}(d) = d$;
- (2) $F(V) = \sum_{m=1}^{d-1} \csc \frac{\pi m}{2d} \mathcal{N}(m)$, the window has $\mathcal{N}(m) = m$, and

$$F(V) - F_{\text{DKZ}} = \frac{1}{2} \langle v, \delta \rangle, \quad v_m = 2 \csc \frac{\pi m}{2d};$$

- (3) for $1 \leq m \leq d-1$, $\mathcal{N}(m) = \sum_{k-j=m} [\pi_{jk}(1) - \pi_{jk}(3)] + \sum_{k-j=d-m} [\pi_{jk}(0) - \pi_{jk}(2)]$; consequently $\langle t_s, \delta \rangle = \mathcal{N}(s) + \mathcal{N}(d-s) - d \leq 0$ for $1 \leq s \leq d-1$.

Proof. (1) Put $C(n) = \sum_u \tau(Q_u Q_{u+n})$; then $C(-n) = C(n)$ by cyclicity of the trace, and $T(m) = \sum_y \tau(Q_{y+m} Q_{y+d}) = C(d-m)$, so $\mathcal{N}(m) = C(m-d) - C(m+d)$. Hence $\mathcal{N}(2d-m) = C(d-m) - C(3d-m) = C(m-d) - C(m+d) = \mathcal{N}(m)$ (indices modulo N), and $\mathcal{N}(d) = C(0) - C(2d) = \sum_u \tau(Q_u) - 0 = d$, because Q_u and Q_{u+2d} are distinct outcomes of one measurement.

(2) Since κ is odd and $\kappa(2d) = 0$, $\langle A, B \rangle_\tau = \sum_{m \in \mathbb{Z}_N} \kappa(m) T(m) = \sum_{m=1}^{2d-1} \kappa(m) \mathcal{N}(m)$, and by (1) this is $\kappa(d)d + \sum_{m=1}^{d-1} (\kappa(m) + \kappa(2d-m)) \mathcal{N}(m) = d + \sum_m v_m \mathcal{N}(m)$, because $\cot \frac{\pi m}{4d} + \tan \frac{\pi m}{4d} = 2 \csc \frac{\pi m}{2d}$. Lemma 2.4 gives the formula for $F(V)$. For the window, $T(m) = \#\{y \in [-d, 0) : y+m \in [0, d)\} = m$ and $T(-m) = 0$ for $1 \leq m \leq d-1$, so $\mathcal{N}(m) = m$; thus $F_{\text{DKZ}} = \sum_m m \csc \frac{\pi m}{2d}$ (which agrees with $\sum_m (d-m) \sec \frac{\pi m}{2d}$ after $m \mapsto d-m$), and the formula for $F(V) - F_{\text{DKZ}}$ follows.

(3) Write $y+d = j-db$ with $0 \leq j \leq d-1$. If $j+m < d$, then $y+m = (j+m) - d(b+1)$ is the outcome $b+1$ of site $k = j+m$; if $j+m \geq d$, then $y+m = (j+m-d) - db$ is the outcome b of site $k' = j+m-d < j$. Summing over b gives $T(m) = \sum_{k-j=m} \pi_{jk}(1) + \sum_{k-j=d-m} \pi_{jk}(0)$; in the same way $T(-m) = \sum_{k-j=m} \pi_{jk}(3) + \sum_{k-j=d-m} \pi_{jk}(2)$. Hence $\mathcal{N}(s) + \mathcal{N}(d-s) = \sum [\pi_{jk}(0) + \pi_{jk}(1) - \pi_{jk}(2) - \pi_{jk}(3)]$, the sum running over the d pairs at distance s or $d-s$ (counted twice if $s = d/2$), and each term is at most 1. $\square$

By Theorem 2.1 and Lemma 2.5, Theorem A is equivalent to

$$(2.5) \quad \langle v, \delta \rangle \leq 0 \quad \text{for every Q-configuration, on every } \mathbb{C}^M.$$

The inequalities $\langle t_s, \delta \rangle \leq 0$ of Lemma 2.5(3) hold pairwise and carry no information about $\langle v, \delta \rangle$ by themselves; they are the “trivial directions” in Theorem E.

3. TWO-VARIABLE BMV POSITIVITY AND THE STRIP INEQUALITY

Throughout this section $M \geq 1$, B is an orthogonal projection on $\mathbb{C}^M$ of rank r , $P = \mathbf{1} - B$, g is Hermitian, $g_d = BgB + PgP$ is the pinching of g , $\lambda_{\min} \leq \lambda_{\max}$ are the extreme eigenvalues of g , and $K = [\lambda_{\min}, \lambda_{\max}] \times [0, 1] \subseteq \mathbb{R}^2$.

3.1. The strip, its Poisson kernel and the functions h_λ .

Lemma 3.1 (strip Poisson kernel). *Let $0 < x < 1$ and let K_x be as in (1.4).*

- (1) $K_x > 0$ and $\int_{\mathbb{R}} K_x = 1 - x$. For $|u| \geq 1$, $K_x(u) \leq 2 \sin(\pi x) e^{-\pi|u|}$; if $x \geq \frac{1}{2}$, then $K_x(u) \leq \sin(\pi x) e^{-\pi|u|}$ for all u .
- (2) For $u \neq 0$, $K_x(u) = \sum_{n \geq 1} \sin(n\pi x) e^{-n\pi|u|}$.
- (3) For $a \in \mathbb{C}$ with $|\operatorname{Re} a| < \pi$, $\int_{\mathbb{R}} e^{-au} K_x(u) du = \frac{\sin(a(1-x))}{\sin a}$ (interpreted as $1-x$ at $a=0$).

Proof. (1) $\cosh \pi u \geq 1 > \cos \pi x$. For $|u| \geq 1$, $\cosh \pi u - \cos \pi x \geq \cosh \pi u - 1 \geq \frac{1}{2} e^{\pi|u|} - 1 \geq \frac{1}{4} e^{\pi|u|}$; for $x \geq \frac{1}{2}$, $\cos \pi x \leq 0$ and $\cosh \pi u - \cos \pi x \geq \frac{1}{2} e^{\pi|u|}$. The mass is (3) at $a=0$.

(2) With $q = e^{-\pi|u|} \in (0, 1)$ and $\theta = \pi x$, $\sum_n q^n \sin n\theta = \operatorname{Im} \frac{qe^{i\theta}}{1-qe^{i\theta}} = \frac{q \sin \theta}{1-2q \cos \theta + q^2} = \frac{\sin \theta}{2 \cosh \pi u - 2 \cos \theta}$.

(3) First let $a = i\kappa$ with $\kappa \in \mathbb{R} \setminus \{0\}$, and $f(u) = e^{-i\kappa u} / (\cosh \pi u - \cos \pi x)$. Integrate f over the boundary of the rectangle $[-L, L] \times [0, 2]$; the vertical sides tend to 0 as $L \rightarrow \infty$, and $f(u+2i) = e^{2\kappa} f(u)$, so the contour integral tends to $(1 - e^{2\kappa}) \int_{\mathbb{R}} f$. Inside the rectangle

the denominator vanishes at $u = ix$ and $u = i(2 - x)$, simply, with residues of f equal to $e^{\kappa x}/(i\pi \sin \pi x)$ and $-e^{\kappa(2-x)}/(i\pi \sin \pi x)$. Hence $\int_{\mathbb{R}} f = \frac{2(e^{\kappa x} - e^{\kappa(2-x)})}{(1 - e^{2\kappa}) \sin \pi x} = \frac{2 \sinh(\kappa(1-x))}{\sinh(\kappa) \sin \pi x}$, that is, $\int e^{-i\kappa u} K_x(u) du = \frac{\sinh(\kappa(1-x))}{\sinh \kappa} = \frac{\sin(a(1-x))}{\sin a}$. Both sides of (3) are holomorphic on $|\operatorname{Re} a| < \pi$ (the left side by the exponential decay in (1); on the right the only zero of $\sin a$ in this strip, $a = 0$, is removable), and they agree on $i\mathbb{R} \setminus \{0\}$, hence everywhere. $\square$

For $\lambda \in \mathbb{R}$ let

$$(3.1) \quad \phi_\lambda(z) = \frac{i}{\pi^2} \operatorname{Li}_2(e^{-i\pi z - \pi\lambda}) \quad (z \in S).$$

For $z = x + iy \in S$ the argument $e^{-i\pi z - \pi\lambda} = e^{\pi(y-\lambda)} e^{-i\pi x}$ has argument $-\pi x \in (-\pi, 0)$, so it avoids the cut $[1, \infty)$; thus ϕ_λ is holomorphic on S , and $\operatorname{Re} \phi_\lambda = h_\lambda$ by (1.3). Moreover $h_\lambda(z) = h_0(z - i\lambda)$.

Lemma 3.2 (the strip functions). *Let $\lambda \in \mathbb{R}$.*

- (1) *For $z = x + iy \in S$, $h_\lambda(z) = \int_{\mathbb{R}} (w - \lambda)_+ K_x(y - w) dw$.*
- (2) *h_λ extends continuously to the closed strip $\bar{S}$, with $h_\lambda(iy) = (y - \lambda)_+$ and $h_\lambda(1 + iy) = 0$. (For $y > \lambda$ the point iy corresponds to an argument of Li_2 on the cut $[1, \infty)$; the boundary value is the limit from inside S , $\operatorname{Im} \operatorname{Li}_2(\varrho - i0) = -\pi \log \varrho$ for $\varrho > 1$.)*
- (3) *$0 \leq h_\lambda(x + iy) \leq (1 - x)(y - \lambda)_+ + 2$ on $\bar{S}$.*
- (4) *For fixed $x \in (0, 1)$, $\partial_y^2 h_\lambda(x + iy) = K_x(y - \lambda) > 0$. For fixed y , $x \mapsto h_\lambda(x + iy)$ is concave on $[0, 1]$.*
- (5) *For $\nu = x + iy \in \bar{S}$ let ω_ν be the measure $K_x(y - w) dw$ if $0 < x < 1$, δ_y if $x = 0$, and 0 if $x = 1$. Then $h_\lambda(\nu) = \int (w - \lambda)_+ \omega_\nu(dw)$, and for $a \in \mathbb{C}$ with $|\operatorname{Re} a| < \pi$, $\int e^{aw} \omega_\nu(dw) = e^{ay} \sin(a(1 - x))/\sin a$.*

Proof. (1) Let $\Phi(z)$ denote the right side. Since $\operatorname{Re} \cot(A + iB) = \sin 2A/(\cosh 2B - \cos 2A)$, $K_x(y - w) = \frac{1}{2} \operatorname{Re} \cot \frac{\pi(z - iw)}{2}$, which is harmonic in $z \in S$; differentiation under the integral is justified by Lemma 3.1(1), so Φ is harmonic on S . So is $h_\lambda = \operatorname{Re} \phi_\lambda$. For $y < \lambda$, the integrand vanishes unless $w > \lambda > y$, where by Lemma 3.1(2) $K_x(y - w) = \sum_n \sin(n\pi x) e^{-n\pi(w-y)}$, uniformly convergent for $w \geq \lambda$; integrating term by term,

$$\Phi(z) = \sum_{n \geq 1} \sin(n\pi x) \frac{e^{n\pi(y-\lambda)}}{(n\pi)^2} = -\frac{1}{\pi^2} \operatorname{Im} \sum_{n \geq 1} \frac{(e^{\pi(y-\lambda)} e^{-i\pi x})^n}{n^2} = h_\lambda(z).$$

Two harmonic functions on the connected set S that agree on the open set $\{y < \lambda\}$ agree everywhere.

(2) As $x \rightarrow 0$, $K_x(y - \cdot)$ is an approximate identity at y : it is positive, has mass $1 - x$, and $\int |u| K_x(u) du \rightarrow 0$. Indeed $\cosh \pi u - \cos \pi x = 2 \sinh^2 \frac{\pi u}{2} + 2 \sin^2 \frac{\pi x}{2} \geq 2(u^2 + x^2)$ (as $\sinh t \geq t$ and $\sin \frac{\pi x}{2} \geq x$ on $[0, 1]$), so $K_x(u) \leq \frac{\pi x}{4(u^2 + x^2)}$ and $\int_{|u| \leq 1} |u| K_x(u) du \leq \frac{\pi x}{4} \log(1 + x^{-2}) \rightarrow 0$, while for $|u| \geq 1$ we use Lemma 3.1(1). Since $|(w - \lambda)_+ - (y - \lambda)_+| \leq |w - y|$, $h_\lambda(x + iy') \rightarrow (y - \lambda)_+$ as $(x, y') \rightarrow (0, y)$. As $x \rightarrow 1$, Lemma 3.1(1) gives $h_\lambda(x + iy) \leq \sin(\pi x) \int (w - \lambda)_+ e^{-\pi|y-w|} dw \rightarrow 0$, locally uniformly in y . The statement about Li_2 on the cut is the classical $\operatorname{Im} \operatorname{Li}_2(\varrho \pm i0) = \pm \pi \log \varrho$ for $\varrho > 1$.

(3) $(w - \lambda)_+ \leq (y - \lambda)_+ + |w - y|$, $\int K_x = 1 - x$, and $\int |u| K_x(u) du \leq \int_{|u| \leq 1} K_x + 4 \int_1^\infty u e^{-\pi u} du \leq 1 + 1$.

(4) For fixed x , $y \mapsto h_\lambda(x + iy)$ is the convolution of the convex function $(\cdot - \lambda)_+$ with the integrable, smooth, exponentially decaying kernel K_x ; its second derivative is the convolution of δ_λ with K_x , that is, $K_x(y - \lambda)$. As h_λ is harmonic, $\partial_x^2 h_\lambda = -K_x(y - \lambda) < 0$ on $(0, 1)$, and $x \mapsto h_\lambda(x + iy)$ is continuous on $[0, 1]$ by (2).

(5) The first formula is (1) and (2); the second follows from Lemma 3.1(3) after the substitution $u = y - w$. $\square$

3.2. The pencil. For $s \in \mathbb{R}$ and $0 < \tau < 1$ put

$$p_{s,\tau}(\xi) = \det(g - s - \xi(P - \tau)), \quad q_{s,\tau}(\xi) = \det(g_d - s - \xi(P - \tau)) \quad (\xi \in \mathbb{C}).$$

Lemma 3.3 (the pencil). *Let $s \in \mathbb{R}$ and $0 < \tau < 1$.*

- (1) $p_{s,\tau}$ and $q_{s,\tau}$ have degree M , the same leading coefficient $c_\tau = \det(\tau - P) = (\tau - 1)^{M-r} \tau^r \neq 0$, and the same sum of roots.
- (2) Every root ξ of $p_{s,\tau}$ or of $q_{s,\tau}$ satisfies $|\xi| \leq \|g - s\| / \min(\tau, 1 - \tau)$.
- (3) Every non-real root ξ of $p_{s,\tau}$ satisfies $|\xi| \leq \|g - s\| / \sqrt{\tau(1 - \tau)}$.
- (4) If $s \leq \lambda_{\min}$ or $s \geq \lambda_{\max}$, all roots of $p_{s,\tau}$ are real.
- (5) All roots of $q_{s,\tau}$ are real.
- (6) The roots of $p_{s,\tau}$, as an unordered M -tuple, depend continuously on $(s, \tau) \in \mathbb{R} \times (0, 1)$.

Consequently ρ is continuous on $\mathbb{R} \times (0, 1)$, vanishes unless $\lambda_{\min} < s < \lambda_{\max}$, satisfies $\rho(s, \tau) \leq \frac{M}{2\pi} \|g - s\| (\tau(1 - \tau))^{-1/2}$, has support in K , and is integrable on $\mathbb{R}^2$.

Proof. (1) $g - s - \xi(P - \tau) = (\tau - P)[\xi - (P - \tau)^{-1}(g - s)]$, so $p_{s,\tau}(\xi) = c_\tau \det(\xi - (P - \tau)^{-1}(g - s))$ and the roots are the eigenvalues of $(P - \tau)^{-1}(g - s)$, with sum $\text{Tr}[(P - \tau)^{-1}(g - s)]$; likewise for $q_{s,\tau}$ with g_d . Since $(P - \tau)^{-1} = \frac{P}{1 - \tau} - \frac{B}{\tau}$ commutes with B , $\text{Tr}[(P - \tau)^{-1}(BgP + PgB)] = 0$, and $g - g_d = BgP + PgB$.

(2) A root ξ comes with a unit vector v such that $(g - s)v = \xi(P - \tau)v$. As $|(P - \tau)v|^2 = (1 - \tau)^2 |Pv|^2 + \tau^2 |Bv|^2 \geq \min(\tau, 1 - \tau)^2$, $|\xi| \min(\tau, 1 - \tau) \leq \|g - s\|$. For $q_{s,\tau}$ use $\|g_d - s\| \leq \|g - s\|$.

(3) With v as in (2), $\langle v, (g - s)v \rangle = \xi \langle v, (P - \tau)v \rangle$, and both inner products are real. If $\text{Im} \xi \neq 0$, then $\langle v, (P - \tau)v \rangle = 0$ and $\langle v, (g - s)v \rangle = 0$; the first gives $|Pv|^2 = \tau$, $|Bv|^2 = 1 - \tau$, hence $|(P - \tau)v|^2 = \tau(1 - \tau)$, and the bound follows as in (2).

(4) If $s \leq \lambda_{\min}$ then $g - s \geq 0$, and a non-real root would give $\langle v, (g - s)v \rangle = 0$, hence $(g - s)v = 0$, hence $\xi(P - \tau)v = 0$, which is impossible. The case $s \geq \lambda_{\max}$ is the same.

(5) Relative to $\mathbb{C}^M = \text{ran } B \oplus \text{ran } P$, $g_d - s - \xi(P - \tau) = (BgB - s + \xi\tau) \oplus (PgP - s - \xi(1 - \tau))$; it is singular if and only if $\xi = (s - \beta)/\tau$ for an eigenvalue β of $BgB|_{\text{ran } B}$ or $\xi = (\mu - s)/(1 - \tau)$ for an eigenvalue μ of $PgP|_{\text{ran } P}$.

(6) The coefficients of $p_{s,\tau}$ are polynomials in (s, τ) , and the leading coefficient c_τ does not vanish on $(0, 1)$; the roots of a polynomial of fixed degree with non-vanishing leading coefficient depend continuously on the coefficients. The final assertions follow from (3), (4), (6), and $\int_0^1 (\tau(1 - \tau))^{-1/2} d\tau = \pi$. $\square$

3.3. A Jensen-type identity.

Lemma 3.4. *Let $p(\xi) = c \prod_{i=1}^M (\xi - \xi_i)$ and $q(\xi) = c \prod_{i=1}^M (\xi - y_i)$ with $c \neq 0$, all y_i real, and $\sum_i \xi_i = \sum_i y_i$. Then $\log |p/q| \in L^1(\mathbb{R})$ and*

$$\int_{\mathbb{R}} \log \left| \frac{p(\xi)}{q(\xi)} \right| d\xi = \pi \sum_{i=1}^M |\text{Im} \xi_i|.$$

Proof. Write $\xi_i = \alpha_i + i\beta_i$. Then $\log |p/q| = A + C$ with $A(\xi) = \frac{1}{2} \sum_i \log(1 + \beta_i^2 / (\xi - \alpha_i)^2) \geq 0$ and $C(\xi) = \sum_i [\log |\xi - \alpha_i| - \log |\xi - y_i|]$. Each summand of A is integrable, with $\int_{\mathbb{R}} \frac{1}{2} \log(1 + b^2/u^2) du = |b| \int_0^\infty \log(1 + v^{-2}) dv = \pi|b|$, since $v \log(1 + v^{-2}) + 2 \arctan v$ is an antiderivative of $\log(1 + v^{-2})$ with limits 0 at 0 and π at ∞ . The function C has only logarithmic singularities, and $C(\xi) = \sum_i [\log |1 - \alpha_i/\xi| - \log |1 - y_i/\xi|] = O(\xi^{-2})$ as $|\xi| \rightarrow \infty$ because $\sum \alpha_i = \text{Re} \sum \xi_i = \sum y_i$; so $C \in L^1(\mathbb{R})$. For $L > \max_i (|\alpha_i|, |y_i|)$ and real a with $|a| < L$, $m_L(a) := \int_{-L}^L \log |\xi - a| d\xi =$

$(L - a) \log(L - a) + (L + a) \log(L + a) - 2L$ is even in a with $m_L''(a) = 2L/(L^2 - a^2)$, so $m_L(a) = m_L(0) + a^2/L + O(a^4/L^3)$. Hence $\int_{-L}^L C = \sum_i (\alpha_i^2 - y_i^2)/L + O(L^{-3}) \rightarrow 0$, and $\int_{\mathbb{R}} C = 0$. $\square$

3.4. Real slices and limits. For $\xi \in \mathbb{R}$ let $\lambda_1(\xi), \dots, \lambda_M(\xi)$ be the eigenvalues of $g - \xi P$ and $\lambda_1^0(\xi), \dots, \lambda_M^0(\xi)$ those of $g_d - \xi P$, that is, the eigenvalues of $BgB|_{\text{ran } B}$ together with those of $PgP|_{\text{ran } P}$ shifted by $-\xi$. Put

$$U_\xi(w) = \sum_{j=1}^M [(w - \lambda_j(\xi))_+ - (w - \lambda_j^0(\xi))_+], \quad n_\xi(w) = \sum_{j=1}^M [\mathbf{1}_{w > \lambda_j(\xi)} - \mathbf{1}_{w > \lambda_j^0(\xi)}].$$

Lemma 3.5 (real slices). *Let $\xi \in \mathbb{R}$.*

- (1) U_ξ is continuous, piecewise linear and compactly supported, and $U'_\xi = n_\xi$ almost everywhere; n_ξ is bounded with compact support.
- (2) For every $a \in \mathbb{C} \setminus \{0\}$, $\int_{\mathbb{R}} e^{aw} U_\xi(w) dw = a^{-2} \mathcal{D}(a, a\xi)$.
- (3) For $w \notin \text{spec}(g - \xi P) \cup \text{spec}(g_d - \xi P)$, $(\mathcal{H}n_\xi)(w) = \frac{1}{\pi} \log |\det(g - \xi P - w) / \det(g_d - \xi P - w)|$.

Proof. (1) U_ξ vanishes below all eigenvalues, and above all of them it equals $\sum_j (\lambda_j^0 - \lambda_j) = \text{Tr}(g_d - \xi P) - \text{Tr}(g - \xi P) = 0$. (2) As distributions, $U'_\xi = \sum_j (\delta_{\lambda_j(\xi)} - \delta_{\lambda_j^0(\xi)})$, and U_ξ has compact support, so two integrations by parts give $\int e^{aw} U_\xi = a^{-2} \sum_j (e^{a\lambda_j(\xi)} - e^{a\lambda_j^0(\xi)}) = a^{-2} [\text{Tr } e^{a(g - \xi P)} - \text{Tr}_B e^{aBgB} - e^{-a\xi} \text{Tr}_P e^{aPgP}] = a^{-2} \mathcal{D}(a, a\xi)$. (3) Pair λ_j with λ_j^0 : the function $\mathbf{1}_{w > \lambda_j} - \mathbf{1}_{w > \lambda_j^0}$ is $\pm$ the indicator of the interval between them, and $\frac{1}{\pi} \int_{\alpha}^{\beta} \frac{du}{w - u} = \frac{1}{\pi} \log \left| \frac{w - \alpha}{w - \beta} \right|$; summing over j gives $\frac{1}{\pi} \log |\prod_j (w - \lambda_j) / \prod_j (w - \lambda_j^0)|$. $\square$

Lemma 3.6 (limits). *For every Hermitian matrix A ,*

$$\lim_{t \rightarrow +\infty} \text{Tr } e^{A - tP} = \text{Tr}_B e^{BAB} \quad \text{and} \quad \lim_{t \rightarrow -\infty} e^t \text{Tr } e^{A - tP} = \text{Tr}_P e^{PAP}.$$

Proof. Both sides of the first statement are multiplied by e^c if A is replaced by $A + c$, so we may assume $A \geq \mathbf{1}$. Let $\mu_1 \geq \dots \geq \mu_M$ be the eigenvalues of $H_t = A - tP$ and $\beta_1 \geq \dots \geq \beta_r$ those of $BAB|_{\text{ran } B}$. The min-max principle with subspaces of $\text{ran } B$, on which $H_t = A$, gives $\mu_k \geq \beta_k$ for $k \leq r$. For $v = b + p$ ($b \in \text{ran } B$, $p \in \text{ran } P$) and $t > \|A\|$, $\langle v, H_t v \rangle \leq \langle b, Ab \rangle + 2\|A\|\|b\|\|p\| - (t - \|A\|)\|p\|^2 \leq \langle b, (A + \varepsilon_t)b \rangle$ with $\varepsilon_t = \|A\|^2/(t - \|A\|)$. The form $v \mapsto \langle Bv, (A + \varepsilon_t)Bv \rangle$ has the eigenvalues $\beta_k + \varepsilon_t > 0$ and 0, so $\mu_k \leq \beta_k + \varepsilon_t$ for $k \leq r$. Finally $\mu_{r+1} \leq \|A\| - t$ by the min-max principle with the subspace $\text{ran } P$ of codimension r . Hence $\text{Tr } e^{H_t} \rightarrow \sum_k e^{\beta_k}$. The second statement is the first one with B and P exchanged, since $e^t \text{Tr } e^{A - tP} = \text{Tr } e^{A - |t|B}$ for $t < 0$. $\square$

3.5. Proof of Theorem D. The properties of ρ are Lemma 3.3. We prove (1.6) in four steps.

Step 0: $\mathcal{D}/a^2$ is entire. $\mathcal{D}$ is entire on $\mathbb{C}^2$. Since $e^{-tP} = B + e^{-t}P$, $\mathcal{D}(0, t) = \text{Tr } e^{-tP} - (M - r)e^{-t} - r = 0$ and $\partial_a \mathcal{D}(0, t) = \text{Tr}(ge^{-tP}) - e^{-t} \text{Tr}(PgP) - \text{Tr}(BgB) = 0$. Hence $\mathcal{G}(a, t) := \mathcal{D}(a, t)/a^2 = \int_0^1 (1 - u) \partial_a^2 \mathcal{D}(ua, t) du$ is entire on $\mathbb{C}^2$.

Step 1: a distribution with support in K . For $\zeta = (\zeta_1, \zeta_2) \in \mathbb{C}^2$ put $\hat{\mathcal{G}}(\zeta) = \mathcal{G}(-i\zeta_1, i\zeta_2)$, so that $e^{-i(\zeta_1 s + \zeta_2 \tau)} = e^{as - t\tau}$ when $(a, t) = (-i\zeta_1, i\zeta_2)$. Let $\eta = \text{Im } \zeta$ and $H_K(\eta) = \max_{(s, \tau) \in K} (s\eta_1 + \tau\eta_2)$, the supporting function of K . For $X = ag - tP$ the Hermitian part of X is $\eta_1 g + \eta_2 P$, because $\text{Re } a = \eta_1$ and $\text{Re } t = -\eta_2$. If $u(\theta) = e^{\theta X} v$ then $\frac{d}{d\theta} |u|^2 = 2\langle u, (\eta_1 g + \eta_2 P)u \rangle \leq 2\lambda_{\max}(\eta_1 g + \eta_2 P)|u|^2$, so $|\text{Tr } e^X| \leq Me^{\lambda_{\max}(\eta_1 g + \eta_2 P)} \leq Me^{H_K(\eta)}$, the last step because $\lambda_{\max}(\eta_1 g + \eta_2 P) = \max_{|v|=1} (\eta_1 \langle v, gv \rangle + \eta_2 \langle v, Pv \rangle)$ and $(\langle v, gv \rangle, \langle v, Pv \rangle) \in K$. Similarly $|e^{-t} \text{Tr}_P e^{aPgP}| \leq (M - r)e^{H_K(\eta)}$ and $|\text{Tr}_B e^{aBgB}| \leq re^{H_K(\eta)}$, since the eigenvalues of the compressions lie in

$[\lambda_{\min}, \lambda_{\max}]$ and $(\mu, 1), (\beta, 0) \in K$. Hence $|\mathcal{D}| \leq 2Me^{H_K(\eta)}$, and $|\hat{\mathcal{G}}(\zeta)| \leq 2Me^{H_K(\eta)}$ for $|\zeta_1| \geq 1$. For $|\zeta_1| < 1$ the maximum principle for $z \mapsto \hat{\mathcal{G}}(z, \zeta_2)$ on the disc $|z| \leq 2$ gives $|\hat{\mathcal{G}}(\zeta)| \leq \frac{M}{2} \max_{|z|=2} e^{H_K(\operatorname{Im} z, \eta_2)} \leq \frac{M}{2} e^{H_K(\eta) + 3c_K}$, where $c_K = \max(|\lambda_{\min}|, |\lambda_{\max}|)$ is a Lipschitz constant of H_K in η_1 and $|\operatorname{Im} z - \eta_1| \leq 3$. So $|\hat{\mathcal{G}}(\zeta)| \leq Ce^{H_K(\operatorname{Im} \zeta)}$ on $\mathbb{C}^2$. By the Paley–Wiener–Schwartz theorem [13, Theorem 7.3.1] there is a distribution Θ with support in the convex compact set K whose Fourier–Laplace transform $\Theta(e^{-i\langle \cdot, \zeta \rangle})$ is $\hat{\mathcal{G}}(\zeta)$; equivalently,

$$(3.2) \quad \Theta(e^{as-t\tau}) = \mathcal{G}(a, t) = \frac{\mathcal{D}(a, t)}{a^2} \quad \text{for all } (a, t) \in \mathbb{C}^2.$$

On $\mathbb{R}^2$, $|\hat{\mathcal{G}}| \leq C$, since $H_K(0) = 0$.

Step 2: $\Theta = \rho$ on the open strip $\mathbb{R} \times (0, 1)$. Let $\varphi \in C_c^\infty(\mathbb{R}^2)$ and $\hat{\varphi}(k) = \int \varphi(z) e^{-ik \cdot z} dz$. The Fourier transform of Θ is the bounded function $\hat{\mathcal{G}}$ on $\mathbb{R}^2$ [13, Theorem 7.1.14], so

$$(3.3) \quad \Theta(\varphi) = (2\pi)^{-2} \int_{\mathbb{R}^2} \hat{\mathcal{G}}(k) \hat{\varphi}(-k) dk,$$

with an integrable integrand. The map $(\sigma, \xi) \mapsto k = (\sigma, -\sigma\xi)$ is a diffeomorphism of $(\mathbb{R} \setminus \{0\}) \times \mathbb{R}$ onto the complement of the null set $\{k_1 = 0\}$, with Jacobian $|\sigma|$; by Fubini,

$$\Theta(\varphi) = (2\pi)^{-2} \int_{\mathbb{R}} d\xi \int_{\mathbb{R}} d\sigma |\sigma| \hat{\mathcal{G}}(\sigma, -\sigma\xi) \hat{\varphi}(-\sigma, \sigma\xi).$$

Fix ξ . (i) By Lemma 3.5(2) with $a = -i\sigma$, $\hat{\mathcal{G}}(\sigma, -\sigma\xi) = \mathcal{D}(a, a\xi)/a^2 = \hat{U}_\xi(\sigma)$. (ii) $\hat{\varphi}(-\sigma, \sigma\xi) = \iint \varphi(s, \tau) e^{i\sigma(s-\xi\tau)} ds d\tau = \hat{\Psi}_\xi(-\sigma)$, where $\Psi_\xi(w) = \int_{\mathbb{R}} \varphi(w + \xi\tau, \tau) d\tau$ belongs to $C_c^\infty(\mathbb{R})$. (iii) By Lemma 3.5(1), $i\sigma\hat{U}_\xi = \hat{n}_\xi$, so $|\sigma|\hat{U}_\xi(\sigma) = -i \operatorname{sgn}(\sigma) \hat{n}_\xi(\sigma) = (\mathcal{H}n_\xi)^\wedge(\sigma)$, with $\mathcal{H}n_\xi \in L^2(\mathbb{R})$. By Parseval's identity $\int fh = (2\pi)^{-1} \int \hat{f}(\sigma) \hat{h}(-\sigma) d\sigma$ ($f \in L^2$, h Schwartz),

$$\frac{1}{2\pi} \int_{\mathbb{R}} |\sigma| \hat{U}_\xi(\sigma) \hat{\Psi}_\xi(-\sigma) d\sigma = \int_{\mathbb{R}} (\mathcal{H}n_\xi)(w) \Psi_\xi(w) dw = \iint (\mathcal{H}n_\xi)(s - \xi\tau) \varphi(s, \tau) ds d\tau.$$

(iv) Since $g - \xi P - (s - \xi\tau) = g - s - \xi(P - \tau)$, Lemma 3.5(3) gives $(\mathcal{H}n_\xi)(s - \xi\tau) = \frac{1}{\pi} \log |p_{s,\tau}(\xi)/q_{s,\tau}(\xi)|$ for $\tau \notin \{0, 1\}$ and almost every s . Therefore

$$(3.4) \quad \Theta(\varphi) = \frac{1}{2\pi^2} \int_{\mathbb{R}} d\xi \iint \varphi(s, \tau) \log \left| \frac{p_{s,\tau}(\xi)}{q_{s,\tau}(\xi)} \right| ds d\tau.$$

Now assume $\operatorname{supp} \varphi \subseteq [-S_0, S_0] \times [\epsilon, 1 - \epsilon]$ with $0 < \epsilon < \frac{1}{2}$. For (s, τ) in this set, Lemma 3.3(2) puts all roots of $p_{s,\tau}$ and $q_{s,\tau}$ in the disc $|\xi| \leq R := (\|g\| + S_0)/\epsilon$, and by Lemma 3.3(1) the two polynomials have the same leading coefficient and the same sum of roots. For $|\xi| \leq 2R$, $|\log |p/q|| \leq \sum_i (|\log |\xi - \xi_i|| + |\log |\xi - y_i||)$, whose integral over $[-2R, 2R]$ is bounded uniformly. For $|\xi| > 2R$, using $|\log |1 - z| + \operatorname{Re} z| \leq |z|^2$ for $|z| \leq \frac{1}{2}$ and $\operatorname{Re} \sum_i (\xi_i - y_i) = 0$, $|\log |p/q||(\xi) = |\sum_i (\log |1 - \xi_i/\xi| - \log |1 - y_i/\xi|)| \leq 2MR^2/\xi^2$. So the integrand of (3.4) is absolutely integrable on $\mathbb{R} \times \operatorname{supp} \varphi$, and Fubini and Lemma 3.4 (applicable by Lemma 3.3(1),(5)) give

$$\Theta(\varphi) = \iint \varphi(s, \tau) \left[\frac{1}{2\pi^2} \int_{\mathbb{R}} \log \left| \frac{p_{s,\tau}(\xi)}{q_{s,\tau}(\xi)} \right| d\xi \right] ds d\tau = \iint \varphi(s, \tau) \rho(s, \tau) ds d\tau.$$

Step 3: no singular part on the edges. By Lemma 3.3, ρ defines a distribution with support in K , and $\Theta - \rho$ vanishes on $\mathbb{R} \times (0, 1)$ and outside K , so its support lies in the union of the disjoint segments $L_0 = [\lambda_{\min}, \lambda_{\max}] \times \{0\}$ and $L_1 = [\lambda_{\min}, \lambda_{\max}] \times \{1\}$. Splitting $\Theta - \rho$ with smooth cutoffs and applying the structure theorem for distributions supported in a hyperplane [13,

Theorem 2.3.5] to each piece, there are $N_0 \geq 0$ and compactly supported distributions α_k, β_k on $\mathbb{R}$ with

$$\Theta - \rho = \sum_{k=0}^{N_0} [\alpha_k \otimes \delta^{(k)}(\tau) + \beta_k \otimes \delta^{(k)}(\tau - 1)].$$

Evaluating on $e^{as-t\tau}$, and using $\langle \delta^{(k)}, f \rangle = (-1)^k f^{(k)}(0)$,

$$\mathcal{G}(a, t) - \iint e^{as-t\tau} \rho(s, \tau) ds d\tau = \sum_{k=0}^{N_0} t^k [A_k(a) + e^{-t} B_k(a)],$$

where $A_k(a) = \alpha_k(e^{as})$ and $B_k(a) = \beta_k(e^{as})$. Fix a real $a \neq 0$. As $t \rightarrow +\infty$: $\mathcal{G}(a, t) \rightarrow 0$ by Lemma 3.6 (with $A = ag$), the integral tends to 0 by dominated convergence ($\rho \in L^1$ with compact support, $e^{-t\tau} \leq 1$, and $e^{-t\tau} \rightarrow 0$ for $\tau > 0$), and $t^k e^{-t} B_k(a) \rightarrow 0$; so the polynomial $\sum_k A_k(a) t^k$ tends to 0, and every $A_k(a) = 0$. As $t \rightarrow -\infty$: $e^t \mathcal{G}(a, t) = a^{-2} [e^t \text{Tr} e^{ag-tP} - \text{Tr}_P e^{aPgP} - e^t \text{Tr}_B e^{aBgB}] \rightarrow 0$ by Lemma 3.6, and $e^t \iint e^{as-t\tau} \rho = \iint e^{as+t(1-\tau)} \rho \rightarrow 0$; so $\sum_k B_k(a) t^k \rightarrow 0$ and every $B_k(a) = 0$. The functions A_k, B_k are entire (Fourier–Laplace transforms of compactly supported distributions) and vanish on $\mathbb{R} \setminus \{0\}$, hence identically, so $\alpha_k = \beta_k = 0$. Thus $\Theta = \rho$, and (3.2) is (1.6).

Step 4: the τ -marginals. By Duhamel's formula,

$$\partial_a^2 \text{Tr} e^{ag-tP} \big|_{a=0} = \int_0^1 \text{Tr}(g e^{-\theta t P} g e^{-(1-\theta)t P}) d\theta,$$

and with $e^{-\theta t P} = B + e^{-\theta t} P$ this equals $\text{Tr}(BgB)^2 + e^{-t} \text{Tr}(PgP)^2 + 2\|BgP\|_F^2 (1 - e^{-t})/t$. The first two terms are removed by the subtractions in $\mathcal{D}$, so

$$\mathcal{G}(0, t) = \frac{1}{2} \partial_a^2 \mathcal{D}(0, t) = \|BgP\|_F^2 \frac{1 - e^{-t}}{t} = \|BgP\|_F^2 \int_0^1 e^{-t\tau} d\tau.$$

By (3.2) at $a = 0$, where now $\Theta = \rho$, and uniqueness of Laplace transforms, $(\int \rho(s, \tau) ds) d\tau = \|BgP\|_F^2 d\tau$ on $[0, 1]$; the marginal is continuous in $\tau \in (0, 1)$ (dominated convergence, Lemma 3.3), so the identity holds for every τ . This completes the proof of Theorem D. $\square$

3.6. Proof of Theorem C. Let $\nu_k = x_k + iy_k$ ($k = 1, \dots, M$) be the eigenvalues of $X = B + ig$. Since $\text{Re}\langle v, Xv \rangle = \langle v, Bv \rangle \in [0, 1]$ for unit v , $x_k \in [0, 1]$. Let $\omega = \sum_k \omega_{\nu_k}$ (Lemma 3.2(5)) and $\sigma = \sum_\mu \delta_\mu$, the sum over the eigenvalues μ of $PgP|_{\text{ran } P}$. By Lemma 3.2(5), the left side of (1.5) is $\int (w - \lambda)_+ d(\omega - \sigma)(w)$; both measures are finite, and ω is the sum of finitely many atoms and a measure with a density that is $O(e^{-\pi|w|})$ as $|w| \rightarrow \infty$ (Lemma 3.1(1)).

(a) *Laplace transforms.* For real a with $0 < |a| < \pi$, Lemma 3.2(5) gives

$$\hat{\omega}(a) := \int e^{aw} \omega(dw) = \sum_k e^{ay_k} \frac{\sin(a(1-x_k))}{\sin a} = \frac{\text{Tr} e^{ag+iaP} - \text{Tr} e^{ag-iaP}}{2i \sin a},$$

because $e^{-ia(X-1)} = e^{ag+iaP}$ has the eigenvalues $e^{ay_k} e^{ia(1-x_k)}$ and $e^{ia(X^*-1)} = e^{ag-iaP}$ has the eigenvalues $e^{ay_k} e^{-ia(1-x_k)}$. Also $\hat{\sigma}(a) = \text{Tr}_P e^{aPgP}$. In $\mathcal{D}(a, -ia) - \mathcal{D}(a, ia)$ the B -terms cancel, and

$$\mathcal{D}(a, -ia) - \mathcal{D}(a, ia) = \text{Tr} e^{ag+iaP} - \text{Tr} e^{ag-iaP} - (e^{ia} - e^{-ia}) \text{Tr}_P e^{aPgP} = 2i \sin a [\hat{\omega}(a) - \hat{\sigma}(a)].$$

By Theorem D at $t = \mp ia$, the left side is $2ia^2 \iint e^{as} \sin(a\tau) \rho(s, \tau) ds d\tau$. With Lemma 3.1(3) for $x = 1 - \tau$, $\sin(a\tau)/\sin a = \int e^{-au} K_{1-\tau}(u) du$, and Tonelli's theorem (all integrands are

nonnegative for real a),

$$(3.5) \quad \hat{\omega}(a) - \hat{\sigma}(a) = a^2 \int_{\mathbb{R}} e^{aw} V(w) dw, \quad V(w) := \int_0^1 \int_{\mathbb{R}} K_{1-\tau}(s-w) \rho(s, \tau) ds d\tau.$$

V is finite, continuous and $O(e^{-\pi|w|})$: $\int K_{1-\tau} = \tau \leq 1$, $\sup_s \rho(s, \tau) \leq C(\tau(1-\tau))^{-1/2}$, ρ has compact support, and $K_{1-\tau}(u) \leq 2 \sin(\pi\tau) e^{-\pi|u|}$ for $|u| \geq 1$.

(b) *Moments.* By (3.5), $\hat{\omega} - \hat{\sigma} = O(a^2)$ as $a \rightarrow 0$ (both sides extend holomorphically to $|\operatorname{Re} a| < \pi$), so ω and σ have the same mass and the same first moment. Put $W(w) = \int (w-u)_+ d(\omega - \sigma)(u)$. Then W is continuous; for w below all atoms of ω and σ , $W(w) = \int_{u < w} (w-u) d\omega(u) = O(e^{-\pi|w|})$, and for w above all atoms, by the moment identities, $W(w) = \int_{u > w} (u-w) d\omega(u) = O(e^{-\pi w})$; the same holds for W' . As $W'' = \omega - \sigma$ in the sense of distributions, two integrations by parts give $\int e^{aw} W(w) dw = a^{-2}(\hat{\omega}(a) - \hat{\sigma}(a)) = \int e^{aw} V(w) dw$ for real $0 < |a| < \pi$. Both sides are holomorphic in $|\operatorname{Re} a| < \pi$, so they agree on $i\mathbb{R}$: the integrable continuous functions W and V have the same Fourier transform, and $W = V$.

(c) *Conclusion.* Since $(u-\lambda)_+ = (\lambda-u)_+ + (u-\lambda)$ and $\omega - \sigma$ has zero mass and zero first moment, $\int (u-\lambda)_+ d(\omega - \sigma) = W(\lambda) = V(\lambda)$, which is (1.5).

(d) *Equality.* $K_{1-\tau}(s-\lambda) > 0$ for $0 < \tau < 1$, and $\rho \geq 0$ is continuous on $\mathbb{R} \times (0, 1)$; so $V(\lambda) = 0$ for one λ forces $\rho = 0$, hence $\mathcal{D} \equiv 0$ by Theorem D, hence $\|BgP\|_F^2 = 0$ by Step 4 of Section 3.5, that is, $BgP = 0 = PgB$ and $[B, g] = BgP - PgB = 0$. Conversely, if $[B, g] = 0$ then $g = g_d$, $p_{s,\tau} = q_{s,\tau}$, $\rho = 0$ and $V \equiv 0$. $\square$

3.7. Remarks.

Remark 3.7 ($M = 2$). Let $B = \operatorname{diag}(1, 0)$ and $g = \begin{pmatrix} \alpha & c \\ \bar{c} & \beta \end{pmatrix}$. Then $p_{s,\tau}(\xi)$ has discriminant $(m(\tau) - s)^2 - 4\tau(1-\tau)|c|^2$ with $m(\tau) = (1-\tau)\alpha + \tau\beta$, so

$$\rho(s, \tau) = \frac{(4\tau(1-\tau)|c|^2 - (s - m(\tau))^2)_+^{1/2}}{2\pi\tau(1-\tau)} :$$

each slice $\tau = \text{const}$ is a semicircle law of radius $2|c|\sqrt{\tau(1-\tau)}$ and mass $|c|^2 = \|BgP\|_F^2$, centred on the segment from α (the eigenvalue of BgB , at $\tau = 0$) to β (that of PgP , at $\tau = 1$).

Remark 3.8 (convex order). In the proof of Theorem C, ω (the sweep of the eigenvalues of $B + ig$ onto the left edge of $\bar{S}$ by harmonic measure) and σ (the eigenvalue measure of $PgP|_{\operatorname{ran} P}$) have equal masses $M - r$ and equal means, and $\int (\lambda - u)_+ d(\omega - \sigma) \geq 0$ for all λ . Hence Theorem C says exactly that σ is dominated by ω in the convex order: $\operatorname{Tr}_P f(PgP) \leq \int f d\omega$ for every convex f of at most linear growth, with equality for all such f if and only if $[B, g] = 0$. Applying this to (P, g) , whose matrix $P + ig = \mathbf{1} - (B + ig)^*$ has the eigenvalues $1 - \bar{v}$, gives the mirror statement: the eigenvalue measure of $BgB|_{\operatorname{ran} B}$ is dominated in convex order by the sweep of $\operatorname{spec}(B + ig)$ onto the right edge. When B and g commute both statements are identities. In this sense Theorem C is a Schur–Horn type majorisation for the non-normal matrix $B + ig$; first-order (distribution-function) versions are false whenever $[B, g] \neq 0$, because ω has a density as soon as one eigenvalue lies in the open strip.

Remark 3.9 (relation to Heinävaara’s tracial joint spectral measures). For Hermitian A, B , Heinävaara [12] constructs a positive measure $\mu_{A,B}$ on $\mathbb{R}^2$ such that $\operatorname{tr} \mathcal{T}(f)(xA + yB) = \int f(ax + by) d\mu_{A,B}(a, b)$ for a large class of f , where $\mathcal{T}(f)(x) = \int_0^1 \frac{1-t}{t} f(xt) dt$ (so $\mathcal{T}$ maps x^k to $x^k/(k(k+1))$), computes the absolutely continuous part of $\mu_{A,B}$ explicitly as $\frac{1}{2\pi} \sum_i |\operatorname{Im} \lambda_i(C_{a,b})|$ for an auxiliary pencil $C_{a,b}$ ([12, Theorem 3.1]), and deduces Stahl’s theorem. That proof computes a Fourier transform in polar coordinates, one line through the origin at a time, and uses

an integral identity for $\log |\det|$ of a pencil. Our proof of Theorem D follows the same route in an affine chart adapted to the projection P : the lines are the real slices of Lemma 3.5, the division by a^2 plays the role of $\mathcal{T}$, the pinched subtractions remove the contributions of the edges $\tau \in \{0, 1\}$, and Lemma 3.4 is the corresponding $\log |\det|$ identity. We use only the statement of Theorem D in the form given above, and we have given a complete proof of it. Appendix B gives a second proof, in which the Fourier inversion along lines is replaced by a Radon identity for the pencil (Theorem B.1), proved with a complex Burgers equation for its roots.

Remark 3.10 (an independent proof for rank one, and an automorphism of order three). For rank $B \leq 1$ there is a second proof of Theorem C (inequality part), by a different method; we sketch it, the details being in the accompanying notes [26]. Let $B = vv^*$, and let $\nu_k = x_k + iy_k$ be the eigenvalues of $B + ig$, so that $x_k \in [0, 1]$ and $\sum_k x_k = 1$. If all $x_k > 0$, the matrix determinant lemma gives $\det(g - iB - t)/\det(g + iB - t) = \exp(-2\pi i F(t))$ for real t , where $F(t) = \sum_k [\frac{1}{2} + \frac{1}{\pi} \arctan \frac{t - y_k}{x_k}]$ is the distribution function of the sum of the Cauchy laws with centres y_k and scales x_k , and the left side equals $(1 - iG(t))/(1 + iG(t))$ with $G(t) = \langle v, (g - t)^{-1}v \rangle$; the zeros of G are the eigenvalues of $PgP|_{\text{ran } P}$. Hence these eigenvalues are the integer quantiles $\gamma_j = \min\{t : F(t) \geq j\}$, $1 \leq j \leq M - 1$, and conversely every configuration with $x_k > 0$ and $\sum x_k = 1$ arises. Theorem C for rank one thus becomes the inequality $\sum_k h_0(\nu_k) \geq \sum_j (\gamma_j)_+$ for M points in the strip. It is proved by induction on M , for the more general constraint $\sum_k x_k = s \leq 1$ with a correction term: a penalised minimiser is either reducible (a point on or near ∂S , handled by the induction hypothesis) or an interior critical point; at an interior critical point all ν_k are zeros, counted with multiplicity, of a holomorphic function on S of the form $\phi'_0 + \frac{i}{\pi} \sum_j \frac{c_j}{z - i\gamma_j} - m_0 + ia z + \varepsilon \pi \cos \pi z$ ($c_j, \varepsilon > 0, a \geq 0$), and an argument-principle count shows that such a function has at most $J + 1 \leq M - 1$ zeros in S , where J is the number of positive quantiles. Combined with a duality argument it also gives Theorem C whenever rank $P \leq 3$, hence for all $M \leq 5$.

The derivative $\phi'_0(z) = -\frac{1}{\pi} \log(1 - e^{-i\pi z})$ has a curious structure. Let $\chi = i\phi'_0$. In the coordinate $w = e^{-i\pi z}$, which maps S onto the lower half-plane, χ acts as the Möbius map $w \mapsto 1/(1 - w)$; hence χ is an automorphism of S of order three, with fixed point $\frac{1}{3}$, and $z + \chi(z) + \chi(\chi(z)) = 1$ on S (the three values of w have product -1 and arguments in $(-\pi, 0)$). By the holomorphic functional calculus, $X + \chi(X) + \chi(\chi(X)) = \mathbf{1}$ for every matrix X with spectrum in S . If $B + ig$ has no eigenvalue on ∂S , the derivative of $g \mapsto \sum_{\nu \in \text{spec}(B+ig)} h_0(\nu)$ in a Hermitian direction δg is $\text{Tr}[\text{Re } \chi(B + ig) \delta g]$, where $\text{Re } Y = \frac{1}{2}(Y + Y^*)$; so the critical points of the defect in (1.5) (at $\lambda = 0$, where $PgP|_{\text{ran } P}$ is invertible) are the solutions of $\text{Re } \chi(B + ig) = Q$, with Q the spectral projection of PgP for $(0, \infty)$.

4. A NONCOMMUTATIVE CONTINUUM THEOREM

Let $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ with $dm = d\theta/2\pi$, and let $\mathbb{D}$ be the open unit disc. A *projection step field* on $\mathbb{C}^M$ is given by points $t_0 < t_1 < \dots < t_{n-1} < t_0 + 2\pi$ (indices modulo n , $t_n = t_0 + 2\pi$), the open arcs $J_j = (t_j, t_{j+1})$, and orthogonal projections B_j on $\mathbb{C}^M$; it is the function $B(\theta) = B_j$ for $\theta \in J_j$ (values at the points t_j play no role). Its *conjugate function* is

$$(4.1) \quad g(\theta) = \text{p.v.} \int_{\mathbb{T}} \cot \frac{\theta - \varphi}{2} B(\varphi) dm(\varphi) = \frac{1}{\pi} \sum_{j=0}^{n-1} (B_j - B_{j-1}) \log \left| \sin \frac{\theta - t_j}{2} \right| \quad (\theta \notin \{t_j\}),$$

where the second expression follows from $\frac{d}{d\varphi} \log \left| \sin \frac{\theta - \varphi}{2} \right| = -\frac{1}{2} \cot \frac{\theta - \varphi}{2}$ and summation by parts. Thus g is Hermitian, real-analytic on each arc J_j , and in $L^p(\mathbb{T})$ for every $p < \infty$. We write $P(\theta) = \mathbf{1} - B(\theta)$ and $\bar{B} = \int_{\mathbb{T}} B dm$.

Lemma 4.1 (Herglotz function of a step field). *For $z \in \mathbb{D}$ let $\mathcal{F}(z) = \int_{\mathbb{T}} \frac{e^{i\varphi} + z}{e^{i\varphi} - z} B(\varphi) dm(\varphi)$. Then:*

- (1) $\mathcal{F}(z) = \bar{B} + \frac{i}{\pi} \sum_j (B_j - B_{j-1}) \operatorname{Log}(1 - ze^{-it_j})$ (principal logarithm);
- (2) $\mathcal{F}$ is holomorphic on $\mathbb{D}$ and $\mathcal{F}(0) = \bar{B}$; its Hermitian part $\operatorname{Re} \mathcal{F}(z) := \frac{1}{2}(\mathcal{F}(z) + \mathcal{F}(z)^*)$ is $\int_{\mathbb{T}} P_z(\varphi) B(\varphi) dm(\varphi)$, where $P_z > 0$ is the Poisson kernel;
- (3) $\mathcal{F}$ extends continuously to $\bar{\mathbb{D}} \setminus \{e^{it_j}\}$, with $\mathcal{F}(e^{i\theta}) = B(\theta) + ig(\theta)$ for $\theta \notin \{t_j\}$;
- (4) for $\frac{1}{2} \leq r < 1$, $\|\mathcal{F}(re^{i\theta})\| \leq C_0 + \frac{1}{\pi} \sum_j |\log |\sin \frac{\theta - t_j}{2}||$, with $C_0 = 1 + n(\log 2 + \frac{\pi}{2})/\pi$.

Proof. (1) Since $\frac{e^{i\varphi} + z}{e^{i\varphi} - z} = 1 + \frac{2}{i} \frac{d}{d\varphi} \operatorname{Log}(1 - ze^{-i\varphi})$ and $\operatorname{Re}(1 - ze^{-i\varphi}) > 0$, $\frac{1}{2\pi} \int_a^b \frac{e^{i\varphi} + z}{e^{i\varphi} - z} d\varphi = \frac{b-a}{2\pi} - \frac{i}{\pi} [\operatorname{Log}(1 - ze^{-ib}) - \operatorname{Log}(1 - ze^{-ia})]$. Multiply by B_j for the arc J_j , sum over j , and regroup the logarithms by endpoints. (2) is clear. (3) The right side of (1) is continuous on $\bar{\mathbb{D}} \setminus \{e^{it_j}\}$. For $u \in (0, 2\pi)$, $1 - e^{iu} = 2 \sin \frac{u}{2} e^{i(u-\pi)/2}$, so $\operatorname{Log}(1 - e^{iu}) = \log |2 \sin \frac{u}{2}| + \frac{i}{2}(u - \pi)$. The imaginary part of $\mathcal{F}(e^{i\theta})$ is therefore $\frac{1}{\pi} \sum_j (B_j - B_{j-1}) \log |2 \sin \frac{\theta - t_j}{2}| = g(\theta)$ (the $\log 2$ terms cancel since $\sum_j (B_j - B_{j-1}) = 0$). For the real part let $\theta \in J_x$ and $u_j = (\theta - t_j) \bmod 2\pi \in (0, 2\pi)$, so $u_j = \theta - t_j$ for $j \leq x$ and $u_j = \theta - t_j + 2\pi$ for $j > x$ (with $0 \leq j \leq n-1$). Then $\operatorname{Re} \mathcal{F}(e^{i\theta}) = \bar{B} - \frac{1}{2\pi} \sum_j (B_j - B_{j-1}) u_j$, and summation by parts gives $\sum_j (B_j - B_{j-1}) u_j = -\sum_j (B_j - B_{j-1}) t_j + 2\pi \sum_{j>x} (B_j - B_{j-1}) = 2\pi(\bar{B} - B_{n-1}) + 2\pi(B_{n-1} - B_x)$, so $\operatorname{Re} \mathcal{F}(e^{i\theta}) = B_x$. (4) $\|B_j - B_{j-1}\| \leq 1$, and for $\frac{1}{2} \leq r < 1$, $\sqrt{2} |\sin \frac{u}{2}| \leq |1 - re^{iu}| \leq 2$, hence $|\operatorname{Log}(1 - re^{iu})| \leq \log 2 + \frac{\pi}{2} + |\log |\sin \frac{u}{2}||$. $\square$

Theorem 4.2 (continuum theorem). *Let B be a projection step field on $\mathbb{C}^M$ with conjugate function g , and let A be a measurable field of Hermitian matrices with $0 \leq A(\theta) \leq P(\theta)$. Put $\alpha = \int \tau(A) dm$, $\beta = \int \tau(B) dm$, and let $\beta_1, \dots, \beta_M \in [0, 1]$ be the eigenvalues of $\bar{B}$. For $\lambda \in \mathbb{R}$ and $\theta \notin \{t_j\}$ put*

$$D_\lambda(\theta) = \sum_{\nu \in \operatorname{spec}(B(\theta) + ig(\theta))} h_\lambda(\nu) - \operatorname{Tr}[(P(\theta)(g(\theta) - \lambda)P(\theta))_+],$$

$$S_\lambda(\theta) = \tau[(P(\theta)(g(\theta) - \lambda)P(\theta))_+] - \tau[A(\theta)(g(\theta) - \lambda)].$$

Then $D_\lambda \geq 0$, $S_\lambda \geq 0$, both are integrable, and

$$(4.2) \quad \lambda\alpha + h_\lambda(\beta) - \int_{\mathbb{T}} \tau(Ag) dm = \left[h_\lambda(\beta) - \frac{1}{M} \sum_{i=1}^M h_\lambda(\beta_i) \right] + \frac{1}{M} \int_{\mathbb{T}} D_\lambda dm + \int_{\mathbb{T}} S_\lambda dm,$$

where the bracket is also nonnegative. In particular $\int \tau(Ag) dm \leq \lambda\alpha + h_\lambda(\beta)$ for every λ , and if $0 < \alpha, \beta$ and $\alpha + \beta < 1$, then $\int \tau(Ag) dm \leq \Phi_c(\alpha, \beta)$.

Proof. Signs. $D_\lambda \geq 0$ is Theorem C for the pair $(B(\theta), g(\theta))$ (note that $\operatorname{Tr}[(P(g - \lambda)P)_+]$ is the same whether $P(g - \lambda)P$ is regarded on $\mathbb{C}^M$ or on $\operatorname{ran} P$). For S_λ : $0 \leq A \leq P$ gives $BAB = 0$, hence $AB = 0$ and $A = PAP$; with $Y = P(g - \lambda)P$, $\tau(A(g - \lambda)) = \tau(AY) \leq \tau(AY_+) \leq \tau(PY_+) = \tau(Y_+)$. The bracket is nonnegative because $x \mapsto h_\lambda(x)$ is concave on $[0, 1]$ (Lemma 3.2(4)) and $\frac{1}{M} \sum_i \beta_i = \tau(\bar{B}) = \beta$.

A harmonic function. Let $K_0 = \{v : B_j v = 0 \ \forall j\}$ and $K_1 = \{v : B_j v = v \ \forall j\}$; these subspaces and $K' = (K_0 \oplus K_1)^\perp$ reduce every B_j , and $\mathcal{F}(z) = 0 \oplus \mathbf{1} \oplus \mathcal{F}'(z)$ accordingly. For a unit vector $v \in K'$ and $z \in \mathbb{D}$, $\langle v, \operatorname{Re} \mathcal{F}(z)v \rangle = \sum_j \omega_j(z) \langle v, B_j v \rangle$ with harmonic measures $\omega_j(z) > 0$, $\sum_j \omega_j(z) = 1$; this is 0 only if $v \in K_0$ and 1 only if $v \in K_1$, which is impossible. By compactness the numerical range of $\operatorname{Re} \mathcal{F}'(z)$ is a compact subset of $(0, 1)$, so $\operatorname{spec} \mathcal{F}'(z) \subseteq S$. By the holomorphic functional calculus (a Riesz–Dunford integral over a contour in S , locally

constant in z), $z \mapsto \operatorname{Re} \operatorname{Tr} \phi_\lambda(\mathcal{F}'(z))$ is harmonic on $\mathbb{D}$. Hence

$$u_\lambda(z) := \sum_{\nu \in \operatorname{spec} \mathcal{F}(z)} h_\lambda(\nu) = \dim K_0 h_\lambda(0) + \dim K_1 h_\lambda(1) + \operatorname{Re} \operatorname{Tr} \phi_\lambda(\mathcal{F}'(z))$$

is harmonic on $\mathbb{D}$, and $u_\lambda(0) = \sum_i h_\lambda(\beta_i)$ because $\mathcal{F}(0) = \bar{B}$.

Boundary passage. Every eigenvalue ν of $\mathcal{F}(z)$, $z \in \mathbb{D} \setminus \{e^{it_j}\}$, lies in $\bar{S}$ and satisfies $|\operatorname{Im} \nu| \leq \|\mathcal{F}(z)\|$; by Lemma 3.2(3), $0 \leq u_\lambda(re^{i\theta}) \leq M(\|\mathcal{F}(re^{i\theta})\| + |\lambda| + 2)$, which by Lemma 4.1(4) is bounded, for $\frac{1}{2} \leq r < 1$, by an integrable function of θ . For $\theta \notin \{t_j\}$, $\mathcal{F}(re^{i\theta}) \rightarrow B(\theta) + ig(\theta)$ as $r \rightarrow 1$ (Lemma 4.1(3)), the eigenvalues converge, and h_λ is continuous on $\bar{S}$; so $u_\lambda(re^{i\theta}) \rightarrow u_\lambda^*(\theta) := \sum_{\nu \in \operatorname{spec}(B(\theta) + ig(\theta))} h_\lambda(\nu)$. The mean value property on $|z| = r$ and dominated convergence give $\int_{\mathbb{T}} u_\lambda^* dm = \sum_i h_\lambda(\beta_i)$.

Conclusion. By definition, $\tau(A(g - \lambda)) = \frac{1}{M} u_\lambda^* - \frac{1}{M} D_\lambda - S_\lambda$ off $\{t_j\}$, and every term is integrable ($g \in L^p$, A bounded, u_λ^* dominated as above). Integrating, $\int \tau(Ag) dm - \lambda \alpha = \frac{1}{M} \sum_i h_\lambda(\beta_i) - \frac{1}{M} \int D_\lambda - \int S_\lambda$, which is (4.2). The last assertion follows from (4.2) and Lemma 4.4 below, whose proof uses only the identity (4.2) for $M = 1$. $\square$

Lemma 4.3 (two adjacent arcs). *Let $\alpha, \beta \geq 0$ with $\alpha + \beta \leq 1$, and let g_β be the conjugate function of the scalar step field $\mathbf{1}_{(-2\pi\beta, 0)}$. Then $\int_0^{2\pi\alpha} g_\beta dm = \Phi_c(\alpha, \beta)$. Equivalently, for $a, b \geq 0$ with $a + b \leq N$, $\int_0^a \int_{-b}^0 \cot \frac{\pi(\xi - \eta)}{N} d\eta d\xi = N^2 \Phi_c(a/N, b/N)$.*

Proof. The function $\Xi(u) = -\frac{N^2}{2\pi^2} \operatorname{Cl}_2(2\pi u/N)$ is continuous, odd and N -periodic, with $\Xi''(u) = \cot \frac{\pi u}{N}$ for $u \notin N\mathbb{Z}$ (since $\operatorname{Cl}_2''(\theta) = -\frac{1}{2} \cot \frac{\theta}{2}$) and Ξ' locally integrable. Integrating twice, $\int_0^a \int_{-b}^0 \Xi''(\xi - \eta) d\eta d\xi = \Xi(a + b) - \Xi(a) - \Xi(b) + \Xi(0)$, which equals $\frac{N^2}{2\pi^2} [\operatorname{Cl}_2(2\pi \frac{a}{N}) + \operatorname{Cl}_2(2\pi \frac{b}{N}) - \operatorname{Cl}_2(2\pi \frac{a+b}{N})] = N^2 \Phi_c(\frac{a}{N}, \frac{b}{N})$, because $-\operatorname{Cl}_2(\theta) = \operatorname{Cl}_2(2\pi - \theta)$. The first form follows by the substitution $\theta = 2\pi\xi/N$, $\varphi = 2\pi\eta/N$. $\square$

Lemma 4.4 (the classical value). *Let $\alpha, \beta > 0$ with $\alpha + \beta < 1$. The function $f(\lambda) = \lambda\alpha + h_\lambda(\beta)$ is strictly convex on $\mathbb{R}$, and*

$$\min_{\lambda \in \mathbb{R}} f(\lambda) = \Phi_c(\alpha, \beta), \quad \text{attained only at } \lambda_0 = \frac{1}{\pi} \log \frac{\sin \pi(\alpha + \beta)}{\sin \pi\alpha}.$$

In particular $\min_\lambda [\frac{\lambda}{4} + h_\lambda(\frac{1}{4})] = \Phi_c(\frac{1}{4}, \frac{1}{4}) = G/\pi^2$, attained only at $\lambda^ = \frac{\log 2}{2\pi}$.*

Proof. Since $h_\lambda(\beta) = h_0(\beta - i\lambda)$, Lemma 3.2(4) gives $f''(\lambda) = K_\beta(\lambda) > 0$. Apply Theorem 4.2 with $M = 1$ to $B = \mathbf{1}_{(-2\pi\beta, 0)}$ and $A = \mathbf{1}_{(0, 2\pi\alpha)}$. The conjugate function $g_\beta(\theta) = \frac{1}{\pi} \log |\sin \frac{\theta + 2\pi\beta}{2} / \sin \frac{\theta}{2}|$ has derivative $\frac{1}{2\pi} [\cot \frac{\theta + 2\pi\beta}{2} - \cot \frac{\theta}{2}] < 0$ on $(0, 2\pi(1 - \beta))$, where it decreases from $+\infty$ to $-\infty$. With $\lambda_0 = g_\beta(2\pi\alpha)$, which is the stated value, $S_{\lambda_0} = 0$ off finitely many points: on $(0, 2\pi\alpha)$ both terms equal $g_\beta - \lambda_0 > 0$, on $(2\pi\alpha, 2\pi(1 - \beta))$ both vanish, and on the arc of B both vanish. For $M = 1$, $D_\lambda \equiv 0$ (if $B(\theta) = 1$ both terms of D_λ vanish, if $B(\theta) = 0$ both equal $(g - \lambda)_+$ and the bracket in (4.2) vanishes). Hence $f(\lambda) - \int Ag_\beta dm = \int S_\lambda \geq 0$ for all λ , with equality at λ_0 , and $\int Ag_\beta dm = \Phi_c(\alpha, \beta)$ by Lemma 4.3. For $\alpha = \beta = \frac{1}{4}$, $\lambda_0 = \frac{1}{\pi} \log \frac{1}{\sin(\pi/4)} = \frac{\log 2}{2\pi}$ and $\Phi_c = \frac{1}{2\pi^2} [2 \operatorname{Cl}_2(\frac{\pi}{2}) + \operatorname{Cl}_2(\pi)] = G/\pi^2$. $\square$

The identity $\min_\lambda f = \Phi_c$ is a Legendre-type identity, $\inf_y [h_0(x + iy) - py] = \Phi_c(x, p)$; it can also be derived from the Bloch–Wigner evaluation of $\operatorname{Im} \operatorname{Li}_2$ at the vertex of a triangle with angles $\pi(\alpha, \beta, 1 - \alpha - \beta)$ [34]. At $\alpha = \beta = \frac{1}{4}$ it is the classical value $\operatorname{Im} \operatorname{Li}_2(\frac{1-i}{2}) = \frac{\pi}{8} \log 2 - G$ [22].

Corollary 4.5 (equality). *Let B be a projection step field on $\mathbb{C}^M$ with $\int B \, dm = \frac{1}{4}\mathbf{1}$, and let $0 \leq A \leq \mathbf{1} - B$ be measurable with $\int \tau(A) \, dm = \frac{1}{4}$. Then*

$$\frac{G}{\pi^2} - \int_{\mathbb{T}} \tau(Ag) \, dm = \frac{1}{M} \int_{\mathbb{T}} D_{\lambda^*} \, dm + \int_{\mathbb{T}} S_{\lambda^*} \, dm \geq 0.$$

If equality holds, then $[B(\theta), g(\theta)] = 0$ for almost every θ .

Proof. All β_i equal $\frac{1}{4}$, so the bracket in (4.2) vanishes; take $\lambda = \lambda^*$ and use Lemma 4.4. In the equality case $D_{\lambda^*} = 0$ almost everywhere, and the equality clause of Theorem C gives $[B(\theta), g(\theta)] = 0$. $\square$

Remark 4.6 (scope). For $M = 1$, Theorem 4.2 contains the circle version of the bathtub bound obtained from the Stein–Weiss theorem [33, 20] (cf. [24, Proposition 4.6]); the proof above uses only the mean value property of u_{λ} . The theorem extends to measurable projection fields, by the standard boundary theory of Herglotz functions and the exponential integrability of conjugate functions of bounded functions [36]; we only need step fields.

Remark 4.7 (non-commuting maximisers in the continuum). The bound $\Phi_c(\alpha, \beta)$ is attained by many fields, including non-commuting ones. For instance, if Θ is a rational $M \times M$ inner function with $\Theta(0) = 0$ (a finite Blaschke–Potapov product), the spectral projections $B = \mathbf{1}_{(-2\pi\beta, 0)}(\Theta(e^{i\theta}))$ and $A = \mathbf{1}_{(0, 2\pi\alpha)}(\Theta(e^{i\theta}))$ satisfy $\int B \, dm = \beta\mathbf{1}$ and $\int A \, dm = \alpha\mathbf{1}$ (because $\int \Theta(e^{i\theta})^k \, dm = \Theta(0)^k = 0$ for $k \geq 1$), the conjugate function of B is $g_{\beta}(\Theta(e^{i\theta}))$, and in the measurable version of Theorem 4.2 (Remark 4.6) all inequalities of (4.2) are equalities at λ_0 , since A, B and g commute pointwise; the field B does not commute with itself at different points unless the factors of Θ commute. Thus the continuum problem alone cannot detect the structure of the optimal discrete strategies. In the proof of Theorem B the needed rigidity comes instead from the cell structure (Proposition 7.3). We do not use this remark.

5. CELL EMBEDDINGS

5.1. Cells and kernels. Let $\ell \in \Delta_d$, extended to $\mathbb{Z}$ by $\ell_x = \ell_{x \bmod d}$. Put $L_0 = 0$ and $L_{x+1} = L_x + \ell_x$, so that $L_{x+d} = L_x + d$ and $L_{x+N} = L_x + N$, and define the cells $I_x = [L_x, L_{x+1}) \subseteq \mathbb{T}_N := \mathbb{R}/N\mathbb{Z}$, $x \in \mathbb{Z}_N$; their lengths are ℓ_x , and $S_r(m) = L_{r+m} - L_r$. We write $\kappa(u) = \cot(\pi u/N)$ also for real $u \notin N\mathbb{Z}$. For $x \neq y$ in $\mathbb{Z}_N$ let

$$K^{\ell}(x, y) = \int_{I_x} \int_{I_y} \kappa(\xi - \eta) \, d\eta \, d\xi, \quad K^{\ell}(x, x) = 0,$$

and for a Q -configuration Q put $Q^{\ell}(Q) = \sum_{x, y \in \mathbb{Z}_N} K^{\ell}(x, y) \tau(A_x B_y)$.

Lemma 5.1. *For $x \neq y$ the integral defining $K^{\ell}(x, y)$ converges absolutely, and*

$$K^{\ell}(x, y) = \Xi(L_{x+1} - L_y) - \Xi(L_x - L_y) - \Xi(L_{x+1} - L_{y+1}) + \Xi(L_x - L_{y+1}).$$

Moreover $K^{\ell}(x + d, y + d) = K^{\ell}(x, y) = -K^{\ell}(y, x)$, and $K^{\ell}(x, y) = 0$ if $\ell_x = 0$ or $\ell_y = 0$.

Proof. The cells are disjoint; $\kappa(\xi - \eta)$ is singular on $\bar{I}_x \times \bar{I}_y$ only at a common endpoint, where $|\kappa(\xi - \eta)| \leq C/|\xi - \eta|$ with ξ, η on opposite sides of it, and $\int_0^1 \int_0^1 \frac{ds \, dt}{s+t} < \infty$. The formula follows by integrating $\Xi'' = \kappa$ twice, as in Lemma 4.3. The symmetries follow from $I_{x+d} = I_x + d$ and the oddness of κ . $\square$

Proposition 5.2 (cell inequalities). *For every $\ell \in \Delta_d$ and every Q -configuration Q on $\mathbb{C}^M$,*

$$Q^{\ell}(Q) \leq \frac{N^2 G}{\pi^2},$$

with equality for every window. More precisely, let B and A be the step fields on $\mathbb{T}$ equal to B_x and A_x on the arc $J_x = (2\pi L_x/N, 2\pi L_{x+1}/N)$, for every x with $\ell_x > 0$, and let g be the conjugate function of B ; then $\int B \, dm = \int A \, dm = \frac{1}{4}\mathbf{1}$, $Q^\ell(Q) = N^2 \int_{\mathbb{T}} \tau(Ag) \, dm$, and Corollary 4.5 applies.

Proof. The arcs J_x with $\ell_x > 0$ are disjoint, nonempty and cover $\mathbb{T}$ up to finitely many points, so B is a projection step field; $0 \leq A \leq \mathbf{1} - B$ because $A_x B_x = 0$; and by (2.4), $\int B \, dm = \frac{1}{N} \sum_x \ell_x B_x = \frac{d}{N} \mathbf{1} = \frac{1}{4} \mathbf{1}$, and likewise $\int A \, dm = \frac{1}{4} \mathbf{1}$, so $\int \tau(A) \, dm = \frac{1}{4}$. For $\theta \in J_x$, $\tau(A(\theta)g(\theta)) = \sum_{y \neq x} \tau(A_x B_y) \int_{J_y} \cot \frac{\theta - \varphi}{2} \, dm(\varphi)$, because $\tau(A_x B_x) = 0$ (no principal value is needed). Integrating over J_x , summing over x , and substituting $\theta = 2\pi\xi/N$, $\varphi = 2\pi\eta/N$ gives $\int \tau(Ag) \, dm = N^{-2} \sum_{x \neq y} K^\ell(x, y) \tau(A_x B_y) = N^{-2} Q^\ell(Q)$, all integrals being absolutely convergent by Lemma 5.1. Corollary 4.5 gives the inequality. For the window $\mathbf{1}_{[c, c+d)}$ ($M = 1$), $A = 1$ exactly on $[L_c, L_c + d)$ and $B = 1$ exactly on $[L_c - d, L_c)$ (each window contains every residue once), two adjacent arcs of length d ; Lemma 4.3 gives $Q^\ell = N^2 \Phi_c(\frac{1}{4}, \frac{1}{4}) = N^2 G/\pi^2$. $\square$

5.2. Rotation averages.

Proposition 5.3 (averaged cell inequalities). *For every $\ell \in \Delta_d$ and every Q -configuration,*

$$(5.1) \quad \frac{1}{d} \sum_{j=0}^{d-1} \left[Q^\ell(\text{rot}_j Q) - \frac{N^2 G}{\pi^2} \right] = \langle u^\ell, \delta \rangle,$$

with u^ℓ as in (1.7). Equivalently $u_m^\ell = \tilde{K}(m) + \tilde{K}(2d - m)$ with $\tilde{K}(m) = \frac{1}{d} \sum_{r=0}^{d-1} K^\ell(r + m, r)$. Consequently $\langle u^\ell, \delta \rangle \leq 0$.

Proof. By definition $Q^\ell(\text{rot}_j Q) = \sum_{x,y} K^\ell(x + j, y + j) \tau(A_x B_y)$. Since K^ℓ is invariant under $(x, y) \mapsto (x + d, y + d)$, the average over $0 \leq j \leq d - 1$ equals the average over $j \in \mathbb{Z}_N$, which depends only on $x - y$ and equals $\tilde{K}(x - y)$. Hence the left side of (5.1), plus $N^2 G/\pi^2$, is $\sum_{m \in \mathbb{Z}_N} \tilde{K}(m) T(m)$. The function $\tilde{K}$ is odd on $\mathbb{Z}_N$ (Lemma 5.1), so $\tilde{K}(0) = \tilde{K}(2d) = 0$ and, by Lemma 2.5(1),

$$\sum_m \tilde{K}(m) T(m) = \sum_{m=1}^{2d-1} \tilde{K}(m) \mathcal{N}(m) = d \tilde{K}(d) + \sum_{m=1}^{d-1} [\tilde{K}(m) + \tilde{K}(2d - m)] \mathcal{N}(m).$$

For the window, $\mathcal{N}(m) = m$ and every rotation is a window, so Proposition 5.2 gives $N^2 G/\pi^2 = d \tilde{K}(d) + \sum_m [\tilde{K}(m) + \tilde{K}(2d - m)] m$. Subtracting gives (5.1) with $u_m^\ell = \tilde{K}(m) + \tilde{K}(2d - m)$. It remains to identify this with (1.7). By Lemma 5.1, $K^\ell(r + m, r) = \Xi(S_r(m + 1)) - \Xi(S_r(m)) - \Xi(S_{r+1}(m)) + \Xi(S_{r+1}(m - 1))$, and $S_{r+d} = S_r$, so $\tilde{K}(m) = \Delta_m^2 \Pi(m)$ with $\Pi(k) = \frac{1}{d} \sum_r \Xi(S_r(k))$. For $0 \leq j \leq d$, $S_r(2d - j) = d + S_r(d - j) = 2d - S_{r+d-j}(j)$, so $\Pi(2d - j) = \frac{1}{d} \sum_r \Xi(2d - S_r(j))$, and $\tilde{K}(2d - m) = \Delta_m^2 [\frac{1}{d} \sum_r \Xi(2d - S_r(m))]$ for $1 \leq m \leq d - 1$. Adding gives (1.7). The last assertion follows from Proposition 5.2. $\square$

Proposition 5.4 (the cone condition implies optimality). *Let $d \geq 2$, and suppose that $v = \int u^\ell \mu(d\ell) + \sum_s w_s t_s$ for a finite positive Borel measure μ on Δ_d and $w_s \geq 0$. Then $\langle v, \delta \rangle \leq 0$ for every Q -configuration on every $\mathbb{C}^M$; hence $F(V) \leq F_{\text{DKZ}}$ for every family V of order-four unitaries, and $I_d \leq I_{\text{ME}}(d)$ for every projective strategy on a maximally entangled state.*

Proof. The map $\ell \mapsto u^\ell$ is continuous on the compact set Δ_d (as $\Xi^\#$ is continuous on $[0, 2d]$), so the integral is defined, and $\langle v, \delta \rangle = \int \langle u^\ell, \delta \rangle \mu(d\ell) + \sum_s w_s \langle t_s, \delta \rangle \leq 0$ by Proposition 5.3 and Lemma 2.5(3). The rest is Lemma 2.5(2) and Theorem 2.1. $\square$

Remark 5.5 (uniform cells and junction terms). For uniform cells $\ell \equiv 1$, $K^1(x, y) = \bar{\kappa}(x - y)$ with $\bar{\kappa}(m) = \int_{-1}^1 (1 - |s|) \kappa(m + s) ds$, and $u^1 = v + j$, where $j_m = |e_J(m)| + |e_J(2d - m)| \geq 0$ and $e_J(m) = \kappa(m) - \bar{\kappa}(m)$ is the junction function of [24, Lemma 4.9]. So the uniform embedding gives $\langle v, \delta \rangle \leq -\langle j, \delta \rangle$, a bound with an error made of junction terms; this is how the commutative case is treated in [24], where those terms are estimated using the cyclic order of optimal colourings, an argument with no known operator analogue. The cone condition replaces this estimate: since u^1 is not proportional to v , non-uniform cells are needed.

Remark 5.6 (Poisson cells). Let $\ell = d \cdot (Y_0, \dots, Y_{d-1})$ with (Y_r) uniformly distributed on the probability simplex (the Dirichlet law with all parameters 1; equivalently, the cell boundaries are d independent uniform points on a circle of length d). Then $S_r(m)/d$ has the Beta($m, d - m$) law, and the Bernstein derivative formula gives $\mathbb{E}u_m^\ell = \int_0^d (\Xi^\#)''(s) b_{m,d}(s/d) ds$ with $b_{m,d}(x) = \binom{d}{m} x^m (1 - x)^{d-m}$. As $(\Xi^\#)''(s) = \frac{4d}{\pi s} + O(1)$ and $\int_0^1 b_{m,d}(x) \frac{dx}{x} = \frac{1}{m}$, the averaged cell vector reproduces *exactly* the singular part $\frac{4d}{\pi m}$ of v_m : the discrete kernel is the continuum kernel averaged over Poisson-like cells. The residual $\mathbb{E}u^\ell - v$ is smooth and of relative size $O(1/d)$; the constructions of Section 6 correct it by modulating the Dirichlet parameters. (This remark explains the construction and is not used in the proofs; Lemma 6.3 below is the precise statement that is used.)

6. THE CONE CONDITION FOR EVERY d (COMPUTER-ASSISTED)

This section proves Theorem E. The proof is computer-assisted: it depends on computations in rigorous interval arithmetic (the interval type of the Python library `mpmath` [16], with outward rounding, at 160 bits for $d \leq 2000$ and at 120–240 bits for $d \geq 2001$). Below we state the mathematical reductions, with proofs, and then exactly what is verified, by which program, and with which margins. Programs, certificates and logs are listed in Section 9. There are three regimes: $2 \leq d \leq 200$ (Section 6.2), $201 \leq d \leq 2000$ (Section 6.5) and $d \geq 2001$ (Section 6.6).

6.1. Potentials. For a probability measure μ on Δ_d and $0 \leq m \leq d$ let $P_\mu(m) = \int \frac{1}{d} \sum_{r=0}^{d-1} \Xi^\#(S_r(m)) \mu(d\ell)$. Since $S_r(0) = 0$ and $S_r(d) = d$, $P_\mu(0) = \Xi^\#(0)$ and $P_\mu(d) = \Xi^\#(d)$ for every μ , and $\int u^\ell \mu(d\ell) = \Delta^2 P_\mu$ on $\{1, \dots, d-1\}$. Let P_v be the solution of $\Delta^2 P_v = v$ on $\{1, \dots, d-1\}$ with $P_v(0) = \Xi^\#(0)$ and $P_v(d) = \Xi^\#(d)$. For $X: \{0, \dots, d\} \rightarrow \mathbb{R}$ write $X_A(m) = \frac{1}{2}(X(m) - X(d - m))$ and $X_S(m) = \frac{1}{2}(X(m) + X(d - m))$.

Lemma 6.1 (reduction to potentials). *Suppose that μ satisfies (i) $(P_\mu)_A = (P_v)_A$, and (ii) $W := (P_v - P_\mu)_S$ has $\Delta^2 W(m) \geq 0$ for $1 \leq m \leq d-1$. Then (1.8) holds with this μ and suitable $w_s \geq 0$.*

Proof. W is symmetric with $W(0) = W(d) = 0$, and by (i) $P_\mu + W = P_v$; applying Δ^2 gives $\int u^\ell d\mu + \Delta^2 W = v$. The vector $\Delta^2 W$ is symmetric under $m \mapsto d - m$ and nonnegative, hence a nonnegative combination of the t_s . $\square$

6.2. Regime I: $2 \leq d \leq 200$.

Proposition 6.2 (computer-assisted). *For every $2 \leq d \leq 200$ there are $K_d \leq 6.3d$ cell vectors $\ell^{(k)} = n^{(k)}/16$ with integer entries $n_r^{(k)} \geq 1$ and $\sum_r n_r^{(k)} = 16d$, and weights $c_k > 0$ with $d \cdot \min_k c_k \geq 0.0816$, such that $\sum_k c_k u^{\ell^{(k)}} = v$. In particular Theorem E holds with $\mu = \sum_k c_k \delta_{\ell^{(k)}}$, which is supported in Δ_d^+ , and $w = 0$.*

What is verified. Since all window sums of $\ell = n/16$ are multiples of $1/16$, only the values $\Xi^\#(j/16)$, $0 \leq j \leq 16d$, are needed, and $\Xi^\#(j/q) = -\frac{N^2}{2\pi^2} [\text{Cl}_2(\pi j/K_q) + \text{Cl}_2(\pi(K_q - j)/K_q)]$ with

$q = 16$, $K_q = 2dq$. Each value $\text{Cl}_2(t)$, $0 < t \leq \pi$, is enclosed using

$$\text{Cl}_2(t) = t - t \log t + \sum_{n=1}^{70} \frac{|B_{2n}| t^{2n+1}}{2n(2n+1)!} + R_{70}(t), \quad 0 \leq R_{70}(t) \leq \frac{t \pi^2/6}{\frac{3}{4} \cdot 71 \cdot 143 \cdot 471},$$

with exact rational Bernoulli numbers; the tail bound follows from $|B_{2n}| = 2(2n)!\zeta(2n)/(2\pi)^{2n}$, $\zeta(2n) \leq \zeta(2)$ and $(t/2\pi)^{2n} \leq 4^{-n}$. The vectors $u^{\ell(k)}$ are then enclosed through the window-sum form (1.7). The cell vectors and approximate weights $\tilde{c}$ were found by floating-point linear programming (HiGHS [14]); this search is not part of the verification. The exact weights are $c = \tilde{c} + U^\top y$, where U is the matrix with columns $u^{\ell(k)}$ and $(UU^\top)y = v - U\tilde{c}$; then $Uc = v$ exactly. With an approximate inverse R of $UU^\top$, the bound $\|y\|_\infty \leq \|R(v - U\tilde{c})\|_\infty / (1 - \|\mathbf{1} - RUU^\top\|_\infty)$ holds as soon as $\|\mathbf{1} - RUU^\top\|_\infty < 1$; all quantities are computed in interval arithmetic (160 bits), so the bound holds for the exact U . The program `verify_cone.py` (driver `run_verify.py`) certifies, for every $2 \leq d \leq 200$: $\|\mathbf{1} - RUU^\top\|_\infty \leq 9.4 \cdot 10^{-34}$, $\|y\|_\infty \leq 2.1 \cdot 10^{-11}$, and $\min_k c_k > 0$. Every d was certified at the first attempt; $3.5d \leq K_d \leq 6.3d$; the smallest certified weight is $4.25 \cdot 10^{-4}$ (at $d = 198$), and $0.0816 \leq d \cdot \min_k c_k \leq 0.2572$. The certificates contain the integer vectors $n^{(k)}$ and the weights. A complete re-verification of all 199 stored certificates (program `lp_reverify.py`, which reruns `verify_cone.py`) passed and reproduced every certified bound. For $2 \leq d \leq 20$ the same certificates are also re-evaluated by the Lean kernel in exact rational interval arithmetic (Section 9.4).

Independent check. The program `crosscheck_certs.py` recomputes $\sum_k c_k u^{\ell(k)} - v$ without interval arithmetic, with `mpmath.clsin` at 50 digits and, for $d \leq 10$, with the cell-pair kernel of Lemma 5.1 instead of the window-sum form: the maximal deviation is between $3 \cdot 10^{-15}$ and $2.5 \cdot 10^{-13}$ for $d \in \{3, 5, 8, 10, 20, 37\}$, and at most $1.5 \cdot 10^{-10}$ for $d \in \{150, 200\}$ (weights stored in double precision).

6.3. Dirichlet cells. For $0 < b < d$ let X_b have the Beta($b, d-b$) law, and put $\Xi(b) = \mathbb{E} \Xi^\sharp(dX_b)$, $\Xi(0) = \Xi^\sharp(0)$, $\Xi(d) = \Xi^\sharp(d)$. If $\beta_0, \dots, \beta_{d-1} > 0$ with $\sum_r \beta_r = d$ and ℓ/d has the Dirichlet law with parameters (β_r) , then by the aggregation property of the Dirichlet law $S_r(m)/d$ has the Beta($B_r(m), d - B_r(m)$) law, $B_r(m) = \beta_r + \dots + \beta_{r+m-1}$, so that

$$(6.1) \quad \mathbb{E} \left[\frac{1}{d} \sum_r \Xi^\sharp(S_r(m)) \right] = \frac{1}{d} \sum_r \Xi(B_r(m)).$$

Lemma 6.3 (the singular part is reproduced exactly). *Write $\Xi^\sharp = \Xi_s + \Xi_r$ with $\Xi_s(u) = \frac{4d}{\pi} u \log \frac{u}{d}$ and $\Xi_r(u) = d^2 g_r(u/d)$, where*

$$g_r(x) = -\frac{8}{\pi^2} \left[\text{Cl}_2\left(\frac{\pi x}{2}\right) + \text{Cl}_2\left(\pi - \frac{\pi x}{2}\right) \right] - \frac{4}{\pi} x \log x$$

extends to an odd analytic function on the disc $|x| < 2$. Let Ξ_s, Ξ_r be the corresponding parts of Ξ , and let $P_{v,s}, P_{v,r}$ be the solutions of $\Delta^2 P = \Xi_s''$, respectively Ξ_r'' , on $\{1, \dots, d-1\}$ with the boundary values of Ξ_s , respectively Ξ_r , at 0 and d (so $P_v = P_{v,s} + P_{v,r}$). Then for every integer $0 \leq m \leq d$,

$$\Xi_s(m) = P_{v,s}(m) = \frac{4d}{\pi} \left[m\psi(m) + 1 - \frac{m}{d} - m\psi(d) \right],$$

where ψ is the digamma function and $m\psi(m)$ is read as -1 at $m = 0$. Consequently $P_v - \Xi = P_{v,r} - \Xi_r$ involves only the analytic function g_r .

Proof. Near $x = 0$, $\text{Cl}_2(\theta) = \theta - \theta \log \theta + (\text{odd power series})$ shows that the logarithms cancel; g_r is analytic for $|x| < 2$ because Cl_2 is analytic on $(0, 2\pi)$. For $X \sim \text{Beta}(b, d-b)$, $\mathbb{E}[X \log X] = \frac{b}{d} [\psi(b+1) - \psi(d+1)]$, so $\Xi_s(b) = \frac{4d}{\pi} b [\psi(b+1) - \psi(d+1)]$; at integers use $m\psi(m+1) = m\psi(m) + 1$

and $\psi(d+1) = \psi(d) + \frac{1}{d}$. The right side vanishes at $m = 0$ and $m = d$, as do $\Xi_s(0)$ and $\Xi_s(d)$, and $\Delta^2[m\psi(m)] = \frac{1}{m}$ for $m \geq 1$, while $\Xi_s''(m) = \frac{4d}{\pi m}$; so both sides solve the same discrete Dirichlet problem. $\square$

6.4. Gaussian modulation. Call $\Upsilon: \{0, \dots, d\} \rightarrow [0, \infty)$ a *variogram* if $\Upsilon(m) = \Upsilon(d-m)$, $\Upsilon(0) = \Upsilon(d) = 0$, and, extending Υ d -periodically,

$$\hat{a}_k(\Upsilon) := -\frac{2}{d} \sum_{m=0}^{d-1} \Upsilon(m) \cos \frac{2\pi km}{d} \quad (1 \leq k < \frac{d}{2}), \quad \hat{a}_{d/2}(\Upsilon) := -\frac{1}{d} \sum_{m=0}^{d-1} (-1)^m \Upsilon(m) \quad (d \text{ even})$$

are all nonnegative. If all $\hat{a}_k(\Upsilon) > 0$, the function $c(j) = \frac{1}{2}(\Upsilon(j+1) - 2\Upsilon(j) + \Upsilon(j-1))$ on $\mathbb{Z}_d$ has discrete Fourier transform $\hat{c}(k) = d \sin^2(\pi k/d) \hat{a}_k(\Upsilon)$ for $1 \leq k < d/2$ (and $2d \hat{a}_{d/2}$ at $k = d/2$), and $\hat{c}(0) = 0$; so there is a centred Gaussian vector $(\eta_r)_{r \in \mathbb{Z}_d}$, invariant in law under rotations, with covariance c , with $\sum_r \eta_r = 0$, with $\text{Var}(\eta_r + \dots + \eta_{r+m-1}) = \Upsilon(m)$, and with a positive density on the hyperplane $\{\sum_r \eta_r = 0\}$. Let $E = \{\max_r |\eta_r| < \frac{3}{4}\}$ and let μ_Υ be the law of the random cell vector ℓ which, given η , is d times a Dirichlet vector with parameters $(1 + \eta_r)$ on E and with all parameters 1 on the complement E^c .

For $1 \leq m \leq d-1$ let $L_m = \frac{3}{4} \min(m, d-m)$, and for $t \geq 0$ and $\zeta \sim N(0, 1)$ let

$$H(m, t) = \mathbb{E}[\Xi(m + \sqrt{t}\zeta); |\sqrt{t}\zeta| < L_m] + \Xi(m) \mathbb{P}(|\sqrt{t}\zeta| \geq L_m).$$

For $1 \leq m < d/2$ put $\text{Hc}_m(t) = [H(m, t) - \Xi(m)] - [H(d-m, t) - \Xi(d-m)]$ and $\tau_m = \text{Dl}(m) - \text{Dl}(d-m)$ with $\text{Dl} = P_v - \Xi$. The *decoupled equations* are

$$(6.2) \quad \text{Hc}_m(\Upsilon(m)) = \tau_m \quad (1 \leq m < d/2).$$

Lemma 6.4 (decoupling). *Let Υ be a variogram with all $\hat{a}_k(\Upsilon) > 0$, and put*

$$\varepsilon_t = \frac{3d}{8} \sup_{0 < b < d} |\Xi'(b)| \cdot 2d \exp\left(-\frac{9}{32\Upsilon(1)}\right),$$

$$e_m(\Upsilon) = [P_{\mu_\Upsilon}(m) - \Xi(m)] - [P_{\mu_\Upsilon}(d-m) - \Xi(d-m)] - \text{Hc}_m(\Upsilon(m)).$$

Then $|P_{\mu_\Upsilon}(m) - H(m, \Upsilon(m))| \leq \varepsilon_t$ for $1 \leq m \leq d-1$, and $|e_m(\Upsilon)| \leq 2\varepsilon_t$ for $1 \leq m < d/2$.

Proof. By (6.1), rotation invariance of (η, E) , and $B_r(m) = m + Z_r(m)$ with $Z_r(m) = \eta_r + \dots + \eta_{r+m-1}$, $P_{\mu_\Upsilon}(m) = \mathbb{E}[\Xi(m+Z); E] + \mathbb{P}(E^c)\Xi(m)$ with $Z = Z_0(m) \sim N(0, \Upsilon(m))$. On E , $|Z| < \frac{3}{4}m$, and since $\sum_r \eta_r = 0$ also $|Z| < \frac{3}{4}(d-m)$; so $\{|Z| \geq L_m\} \subseteq E^c$, and $P_{\mu_\Upsilon}(m) - H(m, \Upsilon(m)) = -\mathbb{E}[(\Xi(m+Z) - \Xi(m))\mathbf{1}_{|Z| < L_m} \mathbf{1}_{E^c}]$, which is bounded by $L_m \sup |\Xi'| \mathbb{P}(E^c)$. Finally $\mathbb{P}(E^c) \leq \sum_r \mathbb{P}(|\eta_r| \geq \frac{3}{4}) \leq 2d \exp(-\frac{9}{32\Upsilon(1)})$ since $\text{Var} \eta_r = \Upsilon(1)$, and $L_m \leq \frac{3d}{8}$. The bound on e_m follows since $\Upsilon(d-m) = \Upsilon(m)$. $\square$

Lemma 6.5 (from the decoupled to the exact equations). *Let $\text{lo}_m < \text{hi}_m$ for $1 \leq m < d/2$, and let $\mathcal{Q}$ be the set of variograms with $\Upsilon(m) \in [\text{lo}_m, \text{hi}_m]$ for $1 \leq m < d/2$ (and a fixed value $\Upsilon(d/2)$ if d is even). Suppose that*

- (B1) $\hat{a}_k(\Upsilon) > 0$ for all k and all $\Upsilon \in \mathcal{Q}$;
- (B2) $\varepsilon_t \leq \bar{\varepsilon}$ for all $\Upsilon \in \mathcal{Q}$;
- (B3) $\text{Hc}_m(\text{lo}_m) < \tau_m - 2\bar{\varepsilon}$ and $\text{Hc}_m(\text{hi}_m) > \tau_m + 2\bar{\varepsilon}$ for all m .

Then some $\Upsilon^\# \in \mathcal{Q}$ satisfies $(P_{\mu_{\Upsilon^\#}})_A = (P_v)_A$, that is, condition (i) of Lemma 6.1.

Proof. Condition (i) for μ_Υ is the system $F_m(\Upsilon) := \text{Hc}_m(\Upsilon(m)) + e_m(\Upsilon) - \tau_m = 0$, $1 \leq m < d/2$. By (B1) the Gaussian law depends continuously on $\Upsilon \in \mathcal{Q}$ and has a positive density on the hyperplane $\{\sum \eta_r = 0\}$, so the boundary of E is a null set and $\Upsilon \mapsto P_{\mu_\Upsilon}$ is continuous; hence F is continuous on the box $\mathcal{Q}$. By Lemma 6.4, (B2) and (B3), $F_m < 0$ on the face $\Upsilon(m) = \text{lo}_m$

and $F_m > 0$ on the face $\Upsilon(m) = \text{hi}_m$. The Poincaré–Miranda theorem [19] gives a zero of F in $\mathcal{Q}$. $\square$

Condition (B3) is obtained from a sign change and a derivative bound: if $\text{Hc}_m(a_m) < \tau_m < \text{Hc}_m(b_m)$, $\text{lo}_m \leq a_m - \varrho$, $\text{hi}_m \geq b_m + \varrho$, and $\text{Hc}'_m \geq c_* > 0$ on $[\text{lo}_m, \text{hi}_m]$ with $c_*\varrho \geq 2\bar{\varepsilon}$, then $\text{Hc}_m(\text{lo}_m) \leq \text{Hc}_m(a_m) - c_*\varrho < \tau_m - 2\bar{\varepsilon}$, and similarly at hi_m , so (B3) holds. For condition (ii) of Lemma 6.1 at $\Upsilon^\#$, note that $W = (P_v - P_{\mu_{\Upsilon^\#}})_S$ and that $|P_{\mu_\Upsilon}(m) - H(m, \Upsilon(m))| \leq \bar{\varepsilon}$ by Lemma 6.4, while both functions take the same values at $m = 0$ and $m = d$. So it suffices that $\Delta^2(P_v - H(\cdot, \Upsilon))_S > 4\bar{\varepsilon}$ on $\{1, \dots, d-1\}$ for every $\Upsilon \in \mathcal{Q}$. Since moreover $(P_v - P_{\mu_{\Upsilon^\#}})_A = 0$, one can instead use the one-sided slack $X_\Upsilon(m) := P_v(m) - H(m, \Upsilon(m))$ ($1 \leq m \leq d/2$), extended by $X_\Upsilon(d-m) = X_\Upsilon(m)$ and $X_\Upsilon(0) = 0$: then $W = X_{\Upsilon^\#} + e$ with $|e| \leq \bar{\varepsilon}$, and it suffices that $\Delta^2 X_\Upsilon > 4\bar{\varepsilon}$ for every $\Upsilon \in \mathcal{Q}$.

6.5. Regime II: $201 \leq d \leq 2000$.

Proposition 6.6 (computer-assisted). *For every $201 \leq d \leq 2000$ the hypotheses of Lemma 6.5 and condition (ii) of Lemma 6.1 hold for explicit intervals $[\text{lo}_m, \text{hi}_m]$. Hence Theorem E holds with $\mu = \mu_{\Upsilon^\#}$, a probability measure with $\mu(\Delta_d^+) = 1$.*

What is verified. For each $201 \leq d \leq 2000$, one run of the program `verify_cert.py` certifies all hypotheses of Lemma 6.5 and condition (ii) of Lemma 6.1. Every quantity is computed in one process, at one precision (160-bit interval arithmetic throughout) and by one version of the code, whose SHA-256 hash is stored in every certificate (Section 9). The run computes:

- (1) enclosures of $\Xi^{(2j)}(m)$ for $j \leq 7$ and all integers $1 \leq m \leq d-1$ (and bounds for $\Xi^{(16)}$ on discs, used in (3)): the singular part through $\Xi_s(b) = \frac{4d}{\pi}b[\psi(b+1) - \psi(d+1)]$ and the polygamma values $\psi^{(n)}(m+1) = (-1)^{n+1}n![\zeta(n+1) - H_m^{(n+1)}]$, where $\zeta(s)$, $2 \leq s \leq 17$, is enclosed by the Euler–Maclaurin formula in exact rational arithmetic with its remainder bound; the regular part through the series $\Xi_r(b) = d^2 \sum_{i \text{ odd}} g_i \frac{(b)_i}{(d)_i}$ (g_i the Taylor coefficients of g_r , $(b)_i$ the rising factorial, $\mathbb{E}X_b^i = (b)_i/(d)_i$), whose Taylor coefficients at the integers are computed by exact truncated products for $i \leq 141$, with a tail bound from $0 \leq \frac{d^k}{db^k} \frac{(b)_i}{(d)_i} \leq k! \binom{i}{k} / (d)_k$ on $[0, d]$;
- (2) the targets τ_m , through Lemma 6.3 and the discrete potential $P_{v,r}$ (prefix sums), with $g_r(1)$ enclosed including the tail of its series;
- (3) root boxes $[a_m, b_m]$ for (6.2): the Gaussian average is expanded exactly to order 14, $H(m, t) - \Xi(m) = \sum_{j=1}^7 t^j \Xi^{(2j)}(m) / (2^j j!) + R(m, t)$, where R contains the order-16 Lagrange remainder (bounded through $\sup |\Xi^{(16)}|$ on $|z| \leq \frac{1}{2} \min(m, d-m)$ and a Gaussian tail estimate on the rest of the window, using $|\psi^{(n)}(z)| \leq (n-1)!z^{-n} + n!z^{-n-1}$) and the truncation $|\sqrt{t}\zeta| < L_m$ (truncated Gaussian moments); an untrusted search proposes $[a_m, b_m]$, and a certified sign change of $\text{Hc}_m - \tau_m$ at its ends is checked;
- (4) the tail quantities, in logarithms: $\sup |\Xi'| \leq \frac{4d}{\pi}(H_d + 1) + S_1 d$ (H_d the harmonic number, $S_1 = \sum_{i \text{ odd}} i|g_i| \leq 1.8884$), $\mathbb{P}(E^c)$ and $\bar{\varepsilon}$ from Lemma 6.4, and a radius $\varrho = 2^e$, $e \in \mathbb{Z}$, with $\varrho \geq 8\bar{\varepsilon}/c_{1,\min}$, where $c_{1,\min} = \min_m \frac{1}{2}(\Xi''(m) - \Xi''(d-m))$;
- (5) the box $\mathcal{Q} = \prod_m [\text{lo}_m, \text{hi}_m]$ with $\text{lo}_m \leq a_m - \varrho$ and $\text{hi}_m \geq b_m + \varrho$ (outward rounding), and, on the whole of $\mathcal{Q}$: condition (B1); condition (B2), using $\Upsilon(1) \leq \text{hi}_1$; the bound $c_D := \min_m \inf\{\text{Hc}'_m(t) : t \in [\text{lo}_m, \text{hi}_m]\} > 0$ with $c_D \varrho > 2\bar{\varepsilon}$, which gives (B3) by the remark after Lemma 6.5 (the derivative of the order-14 expansion and of the remainder is bounded with the heat equation $\partial_t \phi_t = \frac{1}{2} \phi_t''$ for the Gaussian density and two integrations by parts on $[-L_m, L_m]$); condition (B3) once more, directly, by interval evaluation of

$Hc_m - \tau_m$ at lo_m and hi_m ; and $\Delta^2(P_v - H(\cdot, \Upsilon))_S > 4\bar{\varepsilon}$ on $\{1, \dots, d-1\}$ for all $\Upsilon \in \mathcal{Q}$, which gives condition (ii) by the same remark.

All factorials, binomial coefficients, Bernoulli numbers and rational constants enter as exact integers or rationals with outward rounding, the results of the elementary interval functions of `mpmath` are widened by a further 16 units in the last place, every decision is a comparison of interval endpoints, and every certified bound is stored as the exact decimal expansion of the corresponding endpoint. The program `audit_cert_g.py` re-checks every condition from these stored values, together with the code hash: all 1800 certificates pass. The smallest certified values over $201 \leq d \leq 2000$ are: $d^2 \min_k \hat{a}_k \geq 0.06319$ for even d (at $d = 2000$; the minimum over k is at $k = \lfloor d/2 \rfloor$) and $d^2 \min_k \hat{a}_k \geq 0.12638$ for odd d ; $d \cdot \min_m \Delta^2(P_v - H)_S \geq 0.660$; $c_D/(c_{1,\min}/4) \geq 4.0009$ (a ratio above 1 is needed) and $c_D \varrho/(2\bar{\varepsilon}) \geq 10^{0.61}$; the face values of (B3) exceed $2\bar{\varepsilon}$ by a factor of at least $10^{16.7}$; $\mathbb{P}(E^c) \leq 4.2 \cdot 10^{-37}$, $\bar{\varepsilon} \leq 10^{-31.16}$ and $\varrho \leq 2^{-94}$. (Earlier runs, which used separate programs for the Gaussian and the fixed-point conditions and 100-bit arithmetic for $d \leq 600$, gave lower apparent margins, 0.0285 and 0.0798; they are superseded by the certificates described here.)

6.6. Regime III: $d \geq 2001$. For large d the variogram coefficients of the decoupled solution are of size d^{-2} at high frequencies and are controlled by a local criterion. Let $\varepsilon = 1/d$, $A_0 = \frac{\pi}{2} - 1$,

$$B(b) = b[2\psi'(b+1) + b\psi''(b+1)], \quad f(b) = b^2 \left(\frac{1}{B(b)} - 1 \right) \quad (b > 0), \quad f(0) = 0,$$

$a(m) = \frac{1}{6} - f(m)$ for $m \geq 0$, and $F_A(m) = A_0\varepsilon[f(m) + f(d-m) - f(d)]$ for $0 \leq m \leq d$. For a symmetric function X on $\mathbb{Z}_d$ let $\Delta^4 X$ be its centred periodic fourth difference.

Lemma 6.7 (positivity criterion). (1) (computer-assisted) *The sequence $(a(m))_{m \geq 0}$ is positive and convex, and $m^2 a(m) \leq \frac{13}{180}$ for $m \geq 1$. Consequently, for all θ ,*

$$\mathcal{A}(\theta) := a(0) + 2 \sum_{m \geq 1} a(m) \cos m\theta \geq \Delta^2 a(1) = 2f(1) - f(2) \geq 0.105487,$$

where $f(1) = (\pi^2/3 - 2\zeta(3))^{-1} - 1$ and $f(2) = 4(2\pi^2/3 + 4 - 8\zeta(3))^{-1} - 4$.

(2) *Let $d \geq 5$, let Υ_e be symmetric on $\mathbb{Z}_d$ with $\Upsilon_e(0) = 0$, and let $D_B = \Delta^4(\Upsilon_e - F_A)$. If $D_B(1) \geq -\gamma\varepsilon^2$ and $D_B(m) \geq \beta'\varepsilon^3$ for $2 \leq m \leq d/2$, with $\gamma \geq 0$ and $\beta' > 0$, then for $1 \leq k \leq d/2$*

$$\hat{a}_k(\Upsilon_e) \geq w_k \frac{\varepsilon^2}{8} \left[P_d - \frac{4\gamma^2}{\beta'd(1-4/d)} \right], \quad P_d = 16A_0 \left(\inf_{\theta} \mathcal{A}(\theta) - \frac{0.1446}{d-1} \right),$$

where $w_k = 1$ for $k < d/2$ and $w_{d/2} = \frac{1}{2}$.

(3) *If $|\Upsilon(m) - \Upsilon_e(m)| \leq \delta_{\max}$ for all m , then $\hat{a}_k(\Upsilon) \geq \hat{a}_k(\Upsilon_e) - 2w_k \delta_{\max}$.*

Proof. (1) The positivity, convexity and the bound $m^2 a(m) \leq \frac{13}{180}$ are verified in interval arithmetic for $m \leq 3000$, and for $m > 3000$ from the asymptotic expansion $f(b) = \frac{1}{6} - \frac{13}{180}b^{-2} + \frac{683}{7560}b^{-4} + E(b)$, $|E(b)| \leq 0.409b^{-6}$, obtained from the Binet expansion of the polygamma functions with its explicit remainder (program `t35_polya.py`). Since a is convex with $a(m) \rightarrow 0$, Pólya's argument writes $\mathcal{A} = \sum_{L \geq 1} \Delta^2 a(L) L \mathcal{K}_L$ with the nonnegative Fejér kernels $\mathcal{K}_L(\theta) = L^{-2} (\sin \frac{L\theta}{2} / \sin \frac{\theta}{2})^2$, and $L \mathcal{K}_L \equiv 1$ for $L = 1$. The values of $f(1), f(2)$ follow from $\psi'(2) = \frac{\pi^2}{6} - 1$, $\psi''(2) = 2 - 2\zeta(3)$, $\psi'(3) = \frac{\pi^2}{6} - \frac{5}{4}$, $\psi''(3) = \frac{9}{4} - 2\zeta(3)$.

(2) For symmetric X on $\mathbb{Z}_d$ with $X(0) = 0$, $\theta = 2\pi k/d$ and $s = \sin \frac{\theta}{2}$, summation by parts gives $8ds^4 \hat{a}_k(X) = w_k \sum_{m=1}^{d-1} \Delta^4 X(m) (1 - \cos m\theta)$, because the symbol of Δ^4 is $16s^4$ and $\sum_m \Delta^4 X(m) = 0$. For $X = F_A$, using $F_A(m) = A_0\varepsilon[\frac{1}{6} + a(d) - a(m) - a(d-m)]$ (valid

also at $m = 0$), $\sum_m \Delta^4 F_A(m)(1 - \cos m\theta) = 16A_0\varepsilon s^4[a(0) + 2\sum_{m=1}^{d-1} a(m)\cos m\theta + a(d)] \geq 16A_0\varepsilon s^4[\mathcal{A}(\theta) - 2\sum_{m \geq d} a(m)] \geq P_d\varepsilon s^4$, since $2\sum_{m \geq d} a(m) \leq \frac{26}{180(d-1)} < \frac{0.1446}{d-1}$. For the remainder, $D_B(d-m) = D_B(m)$, $1 - \cos \theta = 2s^2$, and $\sum_{m=2}^{d-2} (1 - \cos m\theta) = d - 2 + 2\cos \theta \geq d - 4$, so $\sum_m D_B(m)(1 - \cos m\theta) \geq -4\gamma\varepsilon^2 s^2 + \beta'\varepsilon^2(1 - 4\varepsilon)$. Dividing by $8ds^4$ and putting $y = s^{-2} \geq 1$, $\hat{a}_k(\Upsilon_e) \geq w_k \frac{\varepsilon^2}{8} [P_d - 4\gamma\varepsilon y + \beta'\varepsilon(1 - 4\varepsilon)y^2]$, and the minimum over $y \in \mathbb{R}$ of the bracket is the stated bound.

(3) $|\hat{a}_k(\delta)| \leq \frac{2}{d} \sum_m |\delta(m)| \leq 2\delta_{\max}$, and half of this for $k = d/2$. $\square$

For example, $\gamma = 3$, $\beta' = \frac{1}{4}$ and $\delta_{\max} = \varepsilon^2/20$ in Lemma 6.7(2),(3) give $\hat{a}_k(\Upsilon) \geq w_k \varepsilon^2 [\frac{1}{8}(0.962733 - \frac{36}{0.25 \cdot 2001 \cdot (1-4/2001)}) - \frac{1}{10}] \geq 0.011 w_k \varepsilon^2$ for every $d \geq 2001$; the bounds certified below are much better.

Proposition 6.8 (computer-assisted, uniform in d). *Let Υ_e be the explicit model of Appendix A. For every $d \geq 2001$, with $\varepsilon = 1/d$, let $\bar{\varepsilon}_d$ be the quantity ε_t of Lemma 6.4 evaluated with $\Upsilon(1) \leq 0.6451\varepsilon$ and $\sup |\Xi'| \leq \frac{4d}{\pi}(H_d + 1) + 1.8884d$, let $c_d = 0.1935\varepsilon$ and $\varrho_d = 2\bar{\varepsilon}_d/c_d$. Then:*

- (1) *for each $1 \leq m < d/2$ there are $a_m < b_m$ within $6.6468 \cdot 10^{-4}\varepsilon^2$ of $\Upsilon_e(m)$ with $\text{Hc}_m(a_m) < \tau_m < \text{Hc}_m(b_m)$, and $\text{Hc}'_m \geq c_d$ on $[a_m - \varrho_d, b_m + \varrho_d]$;*
- (2) *$D_B[\Upsilon_e](1) \geq -2.765192\varepsilon^2$ and $D_B[\Upsilon_e](m) \geq 6.022431\varepsilon^3$ for $2 \leq m \leq d/2$;*
- (3) *for every symmetric Υ with $|\Upsilon(m) - \Upsilon_e(m)| \leq 6.6468 \cdot 10^{-4}\varepsilon^2 + \varrho_d$ for $1 \leq m \leq d/2$, the one-sided slack X_Υ of the remark after Lemma 6.5 satisfies $\Delta^2 X_\Upsilon(m) \geq 0.620154\varepsilon - 4\varrho_d d^2$ for $1 \leq m \leq d-1$;*
- (4) *every such Υ has $\Upsilon(1) \leq 0.6451\varepsilon$; consequently, for every $d \geq 2001$ and in natural logarithms, $\log \mathbb{P}(E^c) \leq -864$, $\log \bar{\varepsilon}_d \leq -847$ and $\log \varrho_d \leq -837$ (each bound decreases in d , so $d = 2001$ is the worst case); in particular $2\varrho_d$, $4(\bar{\varepsilon}_d + \varrho_d d^2)$ and ϱ_d are smaller than $0.1186944\varepsilon^2$, 0.620154ε and $10^{-6}\varepsilon$, respectively, by factors larger than e^{800} .*

What is verified. The model Υ_e is real-analytic in $(x, \varepsilon) = (m/d, 1/d)$, and every statement is proved by interval Taylor models over boxes in (x, ε) or (w, ε) , $w = 1/m$, with $\varepsilon \in [0, 1/2001]$, so that each box covers all $d \geq 2001$ at once (Appendix A); the box endpoints lie on exact decimal grids and the ε -ranges are stored exactly. Item (1) is the *pointwise lemma* (program `a10_pointwise.py`, 159 boxes; it also certifies that the model root is unique) together with the derivative bound of `a12_fp_alld.py`: a certified lower bound 0.15204 for the derivative of the scaled decoupled equation on all boxes, which gives $\text{Hc}'_m \geq \frac{4}{\pi} 0.15204\varepsilon \geq c_d$. Item (2) is assembled from `a06_zone0.py` (outer zone $x \in [0.011, 0.501]$: a fourth-derivative density ≥ 7.419), `a07_zoneI.py` (inner zone $x \leq 0.013$, $m \geq 3$: ≥ 6.022431), `a08_A2.py` (the leading inner term $A_2(m) = \Delta^4 \varphi_2(m)$ is ≥ 0 for all $m \geq 2$, and $A_2(1) = -2.7559456\dots$) and `a09_smallm.py` ($m = 1, 2$: $D_B(1) \geq -2.765192\varepsilon^2$, $D_B(2) \geq 206.884\varepsilon^3$); the contribution of the term $A_0\varepsilon f(d-m)$ of F_A is at most $2.03 \cdot 10^{-11}\varepsilon^3$ (`a13_constants.py`). Item (3) is `a11_slack.py` (slack densities of the model ≥ 0.651054 in the outer zone, ≥ 0.8260 in the inner zone, ≥ 0.9100 at $m = 1$) together with a perturbation bound of 0.030900ε : with $G(b, t) = H(b, t) - \Xi(b)$ and G_4 its expansion to order t^4 , the program `a13_constants.py` certifies $|G - G_4| \leq 138.103\varepsilon^3$ (orders 5–7 and the remainder) and $|\partial_t G_4| \leq 11.5703d$, and the Euler–Maclaurin defect of the model target is bounded analytically. Item (4) follows from the scan of `a10_pointwise.py` ($u_e \leq 0.645042$ for $m = 1$), Lemma 6.4 and item (1); every margin is monotone in d , so $d = 2001$ is the worst case. The program `audit_alld.py` checks the coverage of all boxes exactly, with rational arithmetic, and evaluates every margin in interval arithmetic from the logs. A final rigour pass (documented with the certificates) completed the written proofs of the model identities used in Appendix A, among them the first-order identity of the inner zone, which was also checked symbolically and

numerically to $6 \cdot 10^{-25}$, and corrected several code-level enclosure defects; one of them had left a sliver of size about 10^{-20} of the parameter range uncovered at $d = 2001$. No conclusion changed.

Proof of Theorem E for $d \geq 2001$. Let $\text{lo}_m = a_m - \varrho_d$ and $\text{hi}_m = b_m + \varrho_d$. Since $c_d \varrho_d = 2\bar{\varepsilon}_d$, the remark after Lemma 6.5 and Proposition 6.8(1),(4) give (B2) and (B3) with $\bar{\varepsilon} = \bar{\varepsilon}_d$. For Υ in the resulting box, $|\Upsilon(m) - \Upsilon_e(m)| \leq 6.6468 \cdot 10^{-4} \varepsilon^2 + \varrho_d$, so Lemma 6.7, with $\gamma = 2.765192$, $\beta' = 6.022431$, $P_d \geq 0.962733$ (from $\inf \mathcal{A} \geq 0.1054878997577$) and $d \geq 2001$, gives

$$\begin{aligned} \hat{a}_k(\Upsilon) &\geq w_k \left[\frac{\varepsilon^2}{8} \left(0.962733 - \frac{4\gamma^2}{\beta' \cdot 2001 \cdot (1 - 4/2001)} \right) - 2(6.6468 \cdot 10^{-4} \varepsilon^2 + \varrho_d) \right] \\ &\geq w_k (0.1186944 \varepsilon^2 - 2\varrho_d) > 0, \end{aligned}$$

which is (B1). Condition (ii) of Lemma 6.1 at the point $\Upsilon^\#$ provided by Lemma 6.5 follows from the remark after Lemma 6.5 in its one-sided form, since by Proposition 6.8(3),(4) $\Delta^2 X_\Upsilon \geq 0.620154 \varepsilon - 4\varrho_d d^2 > 4\bar{\varepsilon}_d$ on the box. Lemmas 6.5 and 6.1 give Theorem E with $\mu = \mu_{\Upsilon^\#}$, and $\mu(\Delta_d^+) = 1$ because, given η , ℓ/d is a Dirichlet vector with all parameters $\geq \frac{1}{4}$. $\square$

6.7. Summary. Propositions 6.2, 6.6 and 6.8 prove Theorem E for every $d \geq 2$. In all three regimes $\mu(\Delta_d \setminus \Delta_d^+) = 0$: in Regime I every cell vector has $n_r^{(k)} \geq 1$, and in Regimes II and III the cell vector is almost surely in the open simplex. This property is used in the proof of Theorem B.

All values are certified, and a single program checks all three regimes. It verifies the 199 certificates of Regime I (their exact structure, and a matching verification record of `verify_cone.py`), runs `audit_cert_g.py` on the 1800 certificates of Regime II (1800 of 1800 pass, all with the same code hash), checks every box and every margin of Regime III, and prints ALL-D CERTIFIED (every $d \geq 2$). This program, `audit_alld.py`, and its output are included with the certificates.

7. PROOFS OF THEOREMS A AND B

7.1. Proof of Theorem A. By Theorem E and Proposition 5.4, $I_d \leq I_{\text{ME}}(d)$ for every projective strategy on every maximally entangled state, and the DKZ strategy attains $I_{\text{ME}}(d)$ (Section 2.2). $\square$

Remark 7.1 (a deficit identity). The proof gives more than an inequality. For a family V with Q-configuration Q , Lemma 2.5, Theorem E, Proposition 5.3 and Corollary 4.5 give

$$\begin{aligned} F_{\text{DKZ}} - F(V) &= \frac{N^2}{2} \int_{\Delta_d} \frac{1}{d} \sum_{j=0}^{d-1} \left[\frac{1}{M} \int_{\mathbb{T}} D_{\lambda^*}^{(\ell,j)} dm + \int_{\mathbb{T}} S_{\lambda^*}^{(\ell,j)} dm \right] \mu(d\ell) \\ &\quad + \frac{1}{2} \sum_s w_s [d - \mathcal{N}(s) - \mathcal{N}(d-s)], \end{aligned}$$

where $D_{\lambda^*}^{(\ell,j)}$ and $S_{\lambda^*}^{(\ell,j)}$ are the (nonnegative) strip and bathtub defects of Theorem 4.2 for the cell embedding of $\text{rot}_j Q$ with cells ℓ . Thus the CGLMP deficit of any projective strategy on a maximally entangled state is an explicit positive combination of integrated defects of Theorem C, of bathtub slacks, and of pair slacks.

7.2. Proof of Theorem B. We need one lemma about cotangent sums.

Lemma 7.2 (residue lemma). *Let $t_0, \dots, t_{n-1} \in \mathbb{R}$ be pairwise distinct modulo 2π and $J_0, \dots, J_{n-1} \in M_M(\mathbb{C})$. If $\sum_j J_j \cot \frac{\theta - t_j}{2} = 0$ for all θ in a nonempty open interval $I \subseteq \mathbb{R} \setminus \bigcup_j (t_j + 2\pi\mathbb{Z})$, then $J_0 = \dots = J_{n-1} = 0$.*

Proof. Entrywise, $R(z) = \sum_j J_j \cot \frac{z-t_j}{2}$ is holomorphic on the connected open set $\Omega = \mathbb{C} \setminus \bigcup_j (t_j + 2\pi\mathbb{Z})$ and vanishes on I , so $R = 0$ on Ω . Near $z = t_j$ the terms with index $j' \neq j$ are holomorphic, and $\cot \frac{z-t_j}{2} = \frac{2}{z-t_j} + O(z-t_j)$; so the residue of R at t_j is $2J_j$, which must vanish. $\square$

Proposition 7.3 (pointwise commutation forces commutation). *Let $\ell \in \Delta_d^+$, let Q be a Q -configuration, and let B and g be the step field and its conjugate function from Proposition 5.2. If $[B(\theta), g(\theta)] = 0$ for almost every θ , then $Q_x Q_y = Q_y Q_x$ for all $x, y \in \mathbb{Z}_N$.*

Proof. Since all cells have positive length, the points $t_x = 2\pi L_x/N$ ($x \in \mathbb{Z}_N$) are pairwise distinct modulo 2π and the arcs $J_x = (t_x, t_{x+1})$ are nonempty. Fix x . On J_x , $B(\theta) = B_x$ and g is real-analytic by (4.1), so $\theta \mapsto [B_x, g(\theta)]$ is continuous on J_x and vanishes almost everywhere there, hence identically; so does its derivative,

$$0 = \frac{d}{d\theta} [B_x, g(\theta)] = \frac{1}{2\pi} \sum_{j \in \mathbb{Z}_N} [B_x, B_j - B_{j-1}] \cot \frac{\theta - t_j}{2} \quad (\theta \in J_x).$$

By Lemma 7.2, $[B_x, B_j] = [B_x, B_{j-1}]$ for every j ; so $j \mapsto [B_x, B_j]$ is constant on $\mathbb{Z}_N$, with value $[B_x, B_x] = 0$. As x runs over $\mathbb{Z}_N$, $B_x = Q_{x+d}$ runs over all the Q 's. $\square$

Proof of Theorem B. Let a projective strategy R on Φ_D attain $I_{\text{ME}}(d)$, let V be its reduced family (Theorem 2.1) and Q the Q -configuration of V . Then $F(V) = F_{\text{DKZ}}$, so $\langle v, \delta \rangle = 0$ by Lemma 2.5.

Step 1: a tight cell inequality with all cells positive. By Theorem E, $0 = \langle v, \delta \rangle = \int \langle u^\ell, \delta \rangle \mu(d\ell) + \sum_s w_s \langle t_s, \delta \rangle$, and every term is nonpositive (Proposition 5.3 and Lemma 2.5(3)). Hence the continuous nonpositive function $\ell \mapsto \langle u^\ell, \delta \rangle$ vanishes μ -almost everywhere. As $\mu(\Delta_d^+) = \mu(\Delta_d) > 0$, there is a cell vector $\ell^* \in \Delta_d^+$ with $\langle u^{\ell^*}, \delta \rangle = 0$. By Proposition 5.3, the d numbers $Q^{\ell^*}(\text{rot}_j Q) - N^2 G/\pi^2$ are nonpositive with mean 0; so $Q^{\ell^*}(Q) = N^2 G/\pi^2$.

Step 2: commutation almost everywhere. For the step fields of Proposition 5.2 built from ℓ^* , $\int \tau(Ag) dm = N^{-2} Q^{\ell^*}(Q) = G/\pi^2$; by Corollary 4.5, $[B(\theta), g(\theta)] = 0$ for almost every θ .

Step 3: commutation. By Proposition 7.3 all Q_x commute. Hence the unitaries $E_k = \sum_a i^a Q_{k-da}$ commute, and so do the $V_k = E_k^*$.

Step 4: the commutative case. By Theorem 2.2(2),(3), $d \mid D$ and there is a unitary $u: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$ with $u R_i u^* = R_i^{\text{DKZ}} \otimes \mathbf{1}$ for $i = 1, \dots, 4$.

Step 5: measurements, state and uniqueness. From $u U_x u^* = U_x^{\text{DKZ}} \otimes \mathbf{1}$ we get $u P_a^x u^* = P_a^{x, \text{DKZ}} \otimes \mathbf{1}$ (spectral projections). From $u (U_y')^\top u^* = (U_y^{\text{DKZ}})^\top \otimes \mathbf{1}$, by transposition, $\bar{u} U_y' \bar{u}^* = U_y^{\text{DKZ}} \otimes \mathbf{1}$, so $\bar{u} Q_b^y \bar{u}^* = Q_b^{y, \text{DKZ}} \otimes \mathbf{1}$. For any unitary u , $(u \otimes \bar{u}) \sum_i |ii\rangle = \sum_{j,k} (u u^*)_{jk} |jk\rangle = \sum_j |jj\rangle$, so $(u \otimes \bar{u}) \Phi_D = \Phi_d \otimes \Phi_{D/d}$ in the product basis. If u_1, u_2 both have these properties, then $u_2 u_1^*$ commutes with all $R_i^{\text{DKZ}} \otimes \mathbf{1}$; the $R_i^{\text{DKZ}} = G R_i^{(0)} G^*$ generate $M_d(\mathbb{C})$ (the algebra generated by the $R_i^{(0)}$ contains X and $R_1^{(0)}(R_2^{(0)})^{-1} = z^{-1}(\mathbf{1} + (i-1)|0\rangle\langle 0|)$, hence all matrix units [24, proof of Corollary C]), so $u_2 u_1^* \in \mathbf{1}_d \otimes U(D/d)$. Finally, for fixed D the projective strategies on Φ_D form a compact set on which I_d is continuous, so the maximum is attained; if it were $I_{\text{ME}}(d)$, then $d \mid D$. $\square$

Remark 7.4 (why all cells must be positive). If $\ell_r = 0$ for some residue r , then $K^\ell(x, y) = 0$ whenever $x \equiv r$ or $y \equiv r \pmod{d}$, so Q^ℓ does not depend on the measurement of site r , and a tight cell inequality cannot force that measurement to commute with the others. The uniform cell vector lies in Δ_d^+ , but u^1 is not proportional to v (Remark 5.5), so $\langle v, \delta \rangle = 0$ does not make the uniform cell inequality tight. Step 1 needs a cone representation charging Δ_d^+ , which Theorem E provides.

8. CONSEQUENCES

8.1. Noise robustness of the CGLMP witness.

Corollary 8.1. *Let p be the behaviour of a projective strategy on Φ_D , let $p_0(a, b|x, y) = d^{-2}$, and $p_v = vp + (1 - v)p_0$ for $0 \leq v \leq 1$. Then $I_d(p_v) = vI_d(p)$, so the CGLMP inequality detects p_v (that is, $I_d(p_v) > 2$) if and only if $v > 2/I_d(p)$. Consequently the critical visibility $2/I_d(p)$ (defined when $I_d(p) > 2$) satisfies $2/I_d(p) \geq 2/I_{\text{ME}}(d)$, with equality if and only if the strategy is, up to a local unitary $u \otimes \bar{u}$, the DKZ strategy tensored with the identity (in particular $d \mid D$). The same holds for white noise added to the state, $v|\Phi_D\rangle\langle\Phi_D| + (1 - v)\mathbf{1}/D^2$, whenever every outcome of every measurement has $\text{Tr } P_a^x = \text{Tr } Q_b^y = D/d$ (for example rank-one measurements with $D = d$), because then the noise term produces p_0 .*

Proof. I_d is linear in the behaviour, and $I_d(p_0) = 0$ because each of the eight probabilities in each bracket of (1.1) equals $1/d$ under p_0 . Apply Theorems A and B. Under the trace condition, $\text{Tr}((P_a^x \otimes Q_b^y)\mathbf{1}/D^2) = (D/d)^2/D^2 = d^{-2}$. $\square$

For measurements with outcomes of unequal traces, white noise on the state produces a local behaviour whose CGLMP value need not vanish, and Corollary 8.1 says nothing. The question of Gill [10] with respect to *all* Bell inequalities is settled for $d = 3$ (through the complete list of facets of the local polytope with three outcomes) and numerically supported for $d = 4, 5$ [25, 11]; it is open for $d \geq 4$. If the state may also be chosen, the DKZ measurements on Φ_d are not the most robust ones [1, 11].

8.2. The continuum limit.

Corollary 8.2. *As $d \rightarrow \infty$,*

$$I_{\text{ME}}(d) = \frac{32G}{\pi^2} - \frac{c_1}{d} + O(d^{-2}), \quad c_1 = 2 + \frac{4}{\pi} - \frac{32G}{\pi^2} = 0.30342\dots,$$

so the maximum of I_d over projective strategies on maximally entangled states tends to $32G/\pi^2 = 2.96981\dots$

Proof. By Lemma 2.5(2), $F_{\text{DKZ}} = \sum_{m=1}^{d-1} m \csc x_m$ with $x_m = \frac{\pi m}{2d}$. Write $m \csc x_m = \frac{2d}{\pi}(1 + f(x_m))$ with $f(x) = x \csc x - 1$, which is smooth on $[0, \frac{\pi}{2}]$ with $f(0) = 0$, $f(\frac{\pi}{2}) = \frac{\pi}{2} - 1$ and $\int_0^{\pi/2} f = 2G - \frac{\pi}{2}$ (because $\int_0^{\pi/2} x \csc x \, dx = 2G$). The trapezoidal rule gives $\sum_{m=1}^{d-1} f(x_m) = \frac{2d}{\pi}(2G - \frac{\pi}{2}) - \frac{1}{2}(\frac{\pi}{2} - 1) + O(\frac{1}{d})$, hence $F_{\text{DKZ}} = \frac{8G}{\pi^2}d^2 - (\frac{1}{2} + \frac{1}{\pi})d + O(1)$ and $I_{\text{ME}}(d) = \frac{4F_{\text{DKZ}}}{d(d-1)}$ has the stated expansion. $\square$

In the language of Sections 4 and 5, $N^{-2}(2F_{\text{DKZ}} + d) \rightarrow \Phi_c(\frac{1}{4}, \frac{1}{4}) = G/\pi^2$: the discrete value converges to the value of the continuum problem, which by Theorem 4.2 is the same for matrix-valued and for scalar configurations. The continuum theorem alone does not give the exact discrete bound: the discrete and continuum problems differ by terms of order d (Remark 5.5), and the exact bound requires the cell embeddings and Theorem E.

8.3. Open questions. (1) *POVMs.* Our results concern projective measurements; the reduction of Section 2 uses projections in an essential way. The numerical evidence of [21] for $d = 3$ suggests that POVMs do not increase the maximum. (2) *All states.* The maximal value of I_d over all states is known exactly only for $d \leq 8$ ([15, 25]). (3) *Robustness of rigidity.* The deficit identity of Remark 7.1 suggests a quantitative version of Theorem B; it would require a lower bound for the defect in Theorem C in terms of $\|[B, g]\|_F^2$. (4) *An analytic proof of Theorem E.* The mechanism (Remark 5.6) and the continuum limit of the Gaussian modulation are explicit, and the large- d argument of Section 6.6 is an expansion with certified remainders;

a proof without interval arithmetic for all d seems within reach but is not attempted here. Such a proof, once formalised, would also complete the Lean proof of Theorems A and B for $d \geq 21$. (5) *Kullback–Leibler*. The DKZ measurements are not optimal for the Kullback–Leibler clause for $d = 4, \dots, 9$ [35, 25]; whether they are optimal for $d = 3$ is open.

9. VERIFICATION AND REPRODUCIBILITY

9.1. Computer-assisted components. Only Theorem E (and the convexity statement of Lemma 6.7(1)) depend on computer calculations. All verifications use interval arithmetic with outward rounding (`mpmath` [16]); floating-point computations are used only to find candidates (linear-programming columns, approximate roots) and play no role in the verification. Table 2 lists the programs; they are in the accompanying repository [26] (directory `cone-certificates/`), together with their outputs and with written accounts of every bound they use. The program `verify_cert.py` of Regime II has the SHA-256 hash

ff56740d15d40a07c99deb6433fb1b0f40b5a2d5b356b3ec9da0270ff1fab6a7,

which is stored in each of its 1800 certificates and checked by `audit_cert_g.py`. The single command `python audit_alld.py` re-checks all three regimes and prints ALL-D CERTIFIED (every $d \geq 2$).

Statement	Programs	Output and margins
Prop. 6.2 ($d \leq 200$)	<code>verify_cone.py</code> , <code>run_verify.py</code> , <code>lp_reverify.py</code> ; check <code>crosscheck_certs.py</code>	199 certificates <code>cert_d{d}.json</code> , all re-verified; $d \min_k c_k \geq 0.0816$
Prop. 6.6 ($201 \leq d \leq 2000$)	<code>verify_cert.py</code> (tests <code>test_verify_cert.py</code>); audit <code>audit_cert_g.py</code>	1800 certificates <code>cert_g_d{d}.json</code> , all pass; $d^2 \min \hat{a}_k \geq 0.06319$ (even d), ≥ 0.12638 (odd d); $d \min \Delta^2(P_v - H)_S \geq 0.660$; derivative ratio ≥ 4.0009
Prop. 6.8 ($d \geq 2001$)	<code>a06_zone0.py</code> , <code>a07_zoneI.py</code> , <code>a08_A2.py</code> , <code>a09_smallm.py</code> , <code>a10_pointwise.py</code> , <code>a11_slack.py</code> , <code>a12_fp_alld.py</code> , <code>a13_constants.py</code> , <code>a14_identities.py</code> (libraries a02–a05)	$Q_O \geq 7.419$, $\Lambda(R_2) \geq 6.022$, $D_B(1) \geq -2.765192\varepsilon^2$, $ \Upsilon_* - \Upsilon_e \leq 6.6468 \cdot 10^{-4}\varepsilon^2$, $\hat{a}_k \geq 0.1186944 w_k \varepsilon^2$, $\Delta^2 W \geq 0.620154 \varepsilon$, $ \mathbf{G} - \mathbf{G}_4 \leq 138.103 \varepsilon^3$, $ \partial_t \mathbf{G}_4 \leq 11.5703 d$
Lemma 6.7(1) Theorem E, all $d \geq 2$	<code>t35_polya.py</code> <code>audit_alld.py</code>	$\inf \mathcal{A} \geq 0.1054878997577$ ALL-D CERTIFIED

TABLE 2. Programs for the computer-assisted part.

9.2. Independent verification. Each analytic proof was refereed by an independent checker, who re-derived every step and wrote the numerical checks from scratch, without importing any of our programs; the reports and scripts are in the repository [26] (directory `proofs/`). There were four such verifications.

- (1) Theorems D and C, by the proofs of Section 3: verdict correct. The independent re-implementation (adaptive quadrature between exactly located branch points of ρ , and $\mathcal{D}$ by a 30-digit matrix exponential) reproduces (1.6) at 82 complex points (a, t) on 13

instances ($M = 3, \dots, 6$, all ranks for $M \leq 5$), with worst relative error $6.8 \cdot 10^{-13}$, and the defect formula (1.5) at 51 pairs (instance, λ) on 9 instances, including spectra up to ± 150 , clustered spectra and $M = 6$, to $1.3 \cdot 10^{-12}$ (absolute). The marginal identity $\int \rho(s, \tau) ds = \|BgP\|_F^2$ holds to 10^{-13} , and the back-projection integral in Step 2 equals ρ to 10^{-20} (40-digit arithmetic, determinants). Fixed-order quadrature rules between branch points are not sufficient: ρ has complex singularities close to the real s -axis.

- (2) The distribution-free proof of Appendix B, including the general case of Remark B.8: verdict correct. The identities involved, among them the Radon identity of Theorem B.1 on 22 instances with $M \leq 8$, were confirmed to 34–46 digits.
- (3) The reduction chain (Section 2), the continuum theorem (Section 4), the cell inequalities (Section 5), the logic of the cone condition (Section 6) and the rigidity argument (Section 7): verdict correct.
- (4) The rank-one argument of Remark 3.10: verdict correct (no mathematical error; some expository gaps, none of which changes a statement). The check includes the zero count on 1800 extreme instances and the inequality at $M = 3, \dots, 8$ with entries up to 300, re-evaluated at 50 digits.

9.3. Further numerical checks. The following checks are numerical; they are not part of any proof, and were made to guard against errors in formulas, signs and constants.

- Sections 2 and 5: on random non-commuting families ($d = 3, \dots, 12$, $M = 2, 3, 4$) the identities of Lemmas 2.4 and 2.5 and Proposition 5.3 hold to $2 \cdot 10^{-12}$; with the certificates of Proposition 6.2 at $d = 25$ and $d = 40$, the identity $\langle v, \delta \rangle = \sum_k c_k \langle u^{\ell(k)}, \delta \rangle$ holds to $1.6 \cdot 10^{-11}$ on random non-commuting configurations.
- Section 4: the closed form of Lemma 4.1 agrees with the Herglotz integral to $2 \cdot 10^{-16}$; the deficit identity (4.2) holds term by term on cell-embedded fields ($d = 4$, $M = 2, 3$) to 10^{-8} ; the values in Lemma 4.4 at $\alpha = \beta = \frac{1}{4}$ agree to 40 digits.
- Consistency: Theorems A and B were proved for $3 \leq d \leq 20$ by exact sum-of-squares certificates in [25], by a method sharing no step with the present one; and gradient ascents of F over non-commuting families ($d = 3, \dots, 6$, $M = 2, 3$) that reach F_{DKZ} end at commuting families, as Theorem B predicts.

9.4. Formalisation. The argument of this paper has been formalised in the Lean 4 theorem prover [27] with the mathematical library Mathlib [28], in the versions

Lean 4.33.1, Mathlib revision 0df444a360eaa60ab8c11dca51a86af692955474,

in 96 files of about 33,000 lines (directory `lean/` of [26]). No file uses `sorry`, `admit`, an `axiom` declaration or `native_decide`, and `#print axioms` reports, for every main theorem, only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. Table 3 lists the main theorems (file `Main.lean` and the files it imports).

Thus the maximally entangled clause of Problem 27B, Theorems A and B, is fully machine-checked for $2 \leq d \leq 20$. For $d \geq 21$ the formal proof is complete except for the single input CONE_d (`Hyp_ConeCertPos_large`), which is established by the interval-arithmetic computations of Section 6, outside Lean. Theorems C and D are formalised in full, for every matrix size and with no assumption: the identity (1.6) on $\mathbb{C}^2$, with $\rho \geq 0$ measurable, supported in K and bounded by $C(\tau(1 - \tau))^{-1/2}$, and the identity (1.5), that is, the strip inequality with its exact defect, together with its equality case. The formal proof of Theorem D follows the distribution-free route of Appendix B: the Radon identity, then Tonelli’s theorem and the identity theorem. In the uniqueness step (b) of the proof of Theorem C, the formal proof uses the Fourier inversion theorem of Mathlib in place of Lemma B.7. The continuum theorem is formalised for step fields, with the explicit Herglotz functions of Lemma 4.1.

Lean theorem	Statement	Assumptions
QQP27.maxEntClause_le_twenty	for every $2 \leq d \leq 20$: Theorems A and B, and the DKZ strategy attains $I_{\text{ME}}(d)$	none
QQP27.maxEntClause_all	Theorems A and B for every $d \geq 2$	CONE_d for $d \geq 21$ only (Hyp_ConeCertPos_large)
QQP27.stripInequality, QQP27.stripEquality	Theorem C for every matrix size M , and its equality case	none
QQP27.StripL3b.theorem1_bmv2	Theorem D, for all $(a, t) \in \mathbb{C}^2$	none
QQP27.StripL3b.hyp_RI	the Radon identity, Theorem B.1	none
QQP27.Red.hyp_reduction	the exact reduction: strategy $\rightarrow$ clock model $\rightarrow$ Q-configuration $\rightarrow$ linear form (Theorem 2.1, Lemma 2.5)	none
QQP27.Cell.continuumCell	the continuum theorem for step fields: Theorem C implies every cell inequality (Proposition 5.3)	none
QQP27.ClassB. classicalTheoremB_all	the classical (commuting) theorem for every d (Theorem 2.2), including the discrete Hilbert-transform rearrangement ($N = 4d$, $ A = B = d$)	none
QQP27.coneCertPos_le_twenty	Theorem E for $2 \leq d \leq 20$: the certificates of Proposition 6.2 re-evaluated by the Lean kernel in exact rational interval arithmetic	none
QQP27.ConeCertificate. fullCheck_sound	soundness of the certificate checker	none
QQP27.dkz_cg1mp	the DKZ strategy attains $I_{\text{ME}}(d)$, for every d	none

TABLE 3. The main theorems of the Lean formalisation.

A few auxiliary statements of this paper are proved on paper only, because the formal proof does not need them: the continuity of ρ (the formal proof uses its measurability, and a bound in Frobenius norm instead of the operator norm), the marginal identity $\int \rho(s, \tau) ds = \|BgP\|_F^2$ of Theorem D, and the closed formula (1.3) for h_λ (in Lean, h_λ is defined as the Poisson integral of Lemma 3.2(1)). Likewise, the formal rigidity statement is the main assertion of Theorem B: d divides D , and a unitary u maps the measurements to the DKZ measurements tensored with the identity. The uniqueness of u and the strict inequality for $d \nmid D$, which follow from it by the short arguments at the end of Section 7, are not part of the formal statement.

Remark 9.1 (the finite form of the cone condition). In Lean the cone condition is stated in finite form: $v = \sum_j \alpha_j u^{\ell_j}$ for finitely many $\ell_j \in \Delta_d$ and $\alpha_j \geq 0$, with $\alpha_j > 0$ for at least one $\ell_j \in \Delta_d^+$. For $21 \leq d \leq 200$ this is Proposition 6.2. For $d \geq 201$ it follows from Theorem E. Indeed, the measure μ of Propositions 6.6 and `efprop:regimeIII` is a probability measure carried by Δ_d^+ and $\ell \mapsto u^\ell$ is continuous, so the barycentre $\int u^\ell \mu(d\ell)$ lies in the convex hull of $\{u^\ell : \ell \in \Delta_d^+\}$ (in finite dimension the barycentre of a probability measure carried by a set lies in its convex hull), and by Carathéodory's theorem it is a convex combination of at most d such vectors. Finally, each t_s is a positive multiple of u^ℓ for the cell vector ℓ with $\ell_0 = a$, $\ell_s = d - a$ ($0 < a < d$) and all other cells empty. Indeed, by Lemma 5.1, $K^\ell(x, y) = 0$ unless x and y are congruent to 0 or s modulo d , so u^ℓ is supported on $\{s, d - s\}$; its two entries are equal, by the reflections

$\xi \mapsto d + a - \xi$ and $\xi \mapsto 2d + a - \xi$ of $\mathbb{T}_N$, and positive, because they are sums of integrals of $\kappa(\xi - \eta)$ over cells with $\xi - \eta \in (0, 2d)$, where $\kappa > 0$.

Data and code availability. Certificates, verification programs, logs, the Lean formalisation, the reports of the independent verifications and the notes referred to above are available at [26], <https://github.com/anshM123/anshM123-IQ0QI-Vienna-Open-Problem-27>, under the MIT license. The exact certificates for $d \leq 20$ of [25] are available at <https://github.com/anshM123/CGLMP>.

APPENDIX A. THE MODEL FOR LARGE d

This appendix defines the model Υ_e of Proposition 6.8 and records the decomposition used in its verification. Throughout $d \geq 2001$, $\varepsilon = 1/d$, and for real $b \in (0, d)$ we write $x = b/d$ and $w = 1/b$.

Ingredients. Let $\kappa_n(b) = \frac{d^n}{db^n} [b\psi(b+1)]$ and $B_{2j}(b) = b^{2j-1} \kappa_{2j}(b)/(2j-2)!$ (so that $B_{2j}(b) \rightarrow 1$ as $b \rightarrow \infty$), $k_j = 2(2j-2)!/(2^j j!)$. Let $g_r(x) = \sum_{i \text{ odd}} g_i x^i$ (Lemma 6.3), $a_n = (-1)^n g_r^{(n)}(1)/n!$, $P_n(x, \varepsilon) = \prod_{k < n} \frac{x+k\varepsilon}{1+k\varepsilon}$ (so $P_n = \mathbb{E}X_b^n$), $\Gamma(x, \varepsilon) = \sum_{i \text{ odd}} g_i P_i(x, \varepsilon) = \bar{\Xi}_r(b)/d^2$, and $S(x, \varepsilon) = \Gamma(x, \varepsilon) - \sum_n a_n P_n(x, \varepsilon)$ (the second sum is $\bar{\Xi}_r(d-b)/d^2$, since X_{d-b} and $1 - X_b$ have the same law). With $c_j(b) = [\bar{\Xi}^{(2j)}(b) - \bar{\Xi}^{(2j)}(d-b)]/(2^j j!)$, the decoupled equation (6.2) reads $\sum_{j \geq 1} c_j(m) t^j + (\text{remainder}) = \tau_m$, and in the scaled unknown $u = t/(\varepsilon m^2)$, after multiplication by $\frac{\pi}{2m}$, its coefficients are

$$\text{Ch}_j(b) = \frac{\pi}{2} \varepsilon^j b^{2j-1} c_j(b) = \varepsilon^{j-1} \left\{ k_j \left[B_{2j}(b) - \left(\frac{x}{1-x} \right)^{2j-1} B_{2j}(d-b) \right] + \frac{\pi}{2} \frac{x^{2j-1} \partial_x^{2j} S(x, \varepsilon)}{2^j j!} \right\}.$$

Let $p = g_r - \frac{\varepsilon^2}{12} g_r'' + \frac{\varepsilon^4}{240} g_r''''$ (an Euler–Maclaurin interpolant), $\tilde{P}(b) = d^2[p(x) - x(p(1) - g_r(1))]$, $\tau_e(b) = [\tilde{P} - \bar{\Xi}_r](b) - [\tilde{P} - \bar{\Xi}_r](d-b)$ and $\hat{\tau}_e = \frac{\pi}{2} \tau_e/b$.

The model. $u_e(b)$ is the root in $(0, 1)$ of $\sum_{j=1}^4 \text{Ch}_j(b) u^j = \hat{\tau}_e(b)$, and $\Upsilon_e(b) = \varepsilon b^2 u_e(b)$; $\Upsilon_e(0) = 0$ and $\Upsilon_e(d-b) = \Upsilon_e(b)$. The scan of `a10_pointwise.py` certifies that this root exists and is unique, so u_e is real-analytic; with the exact discrete targets and Gaussian order 7 the model reproduces the certified solutions at $d = 1000$ to relative accuracy 10^{-18} (numerical check).

Decomposition. Expanding $u_e = U_0(w) + xU_1(w) + x^2 R_2(x, w)$ in x at fixed w , one has $b^2 U_0 = A_0 b^2/B(b)$ and $b^3 U_1 = \varphi_2(b)$ for an explicit function φ_2 , so that

$$\Upsilon_e(b) = \varepsilon A_0 b^2 + A_0 \varepsilon f(b) + \varepsilon^2 \varphi_2(b) + \varepsilon^3 b^4 R_2(x, w),$$

and, since Δ^4 annihilates quadratic polynomials, $D_B[\Upsilon_e](m) = \varepsilon^2 A_2(m) + \varepsilon^3 \Delta^4[b^4 R_2](m) - A_0 \varepsilon \Delta^4[f(d - \cdot)](m)$ with $A_2 = \Delta^4 \varphi_2$. For $m \geq 3$, $\Delta^4 h(m) = \int_{-2}^2 h''''(m+s) M_4(s) ds$ with the cardinal B-spline $M_4 \geq 0$ of mass 1, and $\frac{d^4}{db^4} [b^4 R_2] = \Lambda(R_2)$ with $\Lambda = (\vartheta+1)(\vartheta+2)(\vartheta+3)(\vartheta+4)$, $\vartheta = x\partial_x - w\partial_w$. In the *inner zone* ($x \leq 0.013$) the certified bounds are $A_2(m) \geq 0$ for $m \geq 2$ and $\Lambda(R_2) \geq 6.022$; in the *outer zone* ($x \in [0.011, 0.501]$) one uses instead $\varepsilon^{-3} \Delta^4[\Upsilon_e - \varepsilon A_0 b^2/B](m) = \int Q_O(x_s, \varepsilon) M_4(s) ds$ with $Q_O = \partial_x^4 [x^2(u_e - A_0/B(1/w))]$ at fixed ε , certified ≥ 7.419 . Since the stencil of Δ^4 has width $4/d \leq 0.002$, every m is covered by one of the two zones; $m = 1, 2$ are treated directly. All enclosures are Taylor models in (x, ε) or (w, ε) with $\varepsilon \in [0, 1/2001]$: thin centre jets of the model root (Krawczyk iteration, then implicit differentiation degree by degree), Lagrange remainders from jets over the full box, special functions (polygamma and Binet functions, Pochhammer sums with rigorous tails) by interval series. For boxes with $x > 0.42$ the exactly antisymmetric coefficients are divided by $1 - 2x$ before enclosing (they vanish at $x = \frac{1}{2}$).

Pointwise lemma and slack. With the exact targets τ_m and the Gaussian expansion to order 7 plus remainder, `a10_pointwise.py` bounds the difference between the decoupled solution and the model by $6.6468 \cdot 10^{-4} \varepsilon^2$, using analytic bounds for the Euler–Maclaurin defect of $\tilde{P}$ (at most $235.6 \varepsilon^6$ for its second differences), for the Gaussian remainder, and for the higher coefficients $\text{Ch}_5, \dots, \text{Ch}_7$. For the slack, let $\text{Dl}_e = \tilde{P} - \Xi_r$ (the model of Dl , by Lemma 6.3) and $W_e(b) = \text{Dl}_e(b) - \sum_{j=1}^4 \Upsilon_e(b)^j \Xi^{(2j)}(b)/(2^j j!)$, which is symmetric under $b \mapsto d - b$ by the model equation. Its leading term is exactly linear, $-(1 - \frac{2}{\pi})b$, so $\Delta^2 W_e = O(\varepsilon)$; the density $\Delta^2 W_e/\varepsilon$ and the perturbation from W_e to W are certified as stated in Section 6.6.

APPENDIX B. A DISTRIBUTION-FREE PROOF OF THEOREM D

The proof of Theorem D in Section 3.5 uses the Paley–Wiener–Schwartz theorem, Fourier inversion along lines and the structure of distributions supported on a line, and the proof of Theorem C in Section 3.6 uses the uniqueness of Fourier transforms. Here we give a second proof of both theorems, which uses no distributions. Its tools are linear algebra, the Cauchy integral formula on a circle and Hurwitz’s theorem, differentiation under the integral sign, dominated convergence, Weierstrass’ theorem on locally uniform limits of holomorphic functions, the theorems of Fubini and Tonelli, the identity theorem in one variable, the Weierstrass approximation theorem, and a Gaussian approximate identity in one variable (Lemma B.7). Its key step is a Radon identity for the pencil (Theorem B.1), which we prove through a complex inviscid Burgers equation satisfied by the roots of the pencil. This is the route followed by the Lean formalisation (Section 9.4).

B.1. The Radon identity. Throughout this appendix H is a Hermitian $M \times M$ matrix, B an orthogonal projection of rank r , $P = \mathbf{1} - B$, and $H_d := BHB + PHP$. Let $\lambda_1, \dots, \lambda_M$ be the eigenvalues of H and $\lambda_1^0, \dots, \lambda_M^0$ those of H_d , which consist of the eigenvalues $\mu_1, \dots, \mu_{M-r}$ of $PHP|_{\text{ran } P}$ and $\beta_1, \dots, \beta_r$ of $BHB|_{\text{ran } B}$; note that $\text{Tr } H = \text{Tr } H_d$. We write $\mathbb{C}_\pm = \{z : \pm \text{Im } z > 0\}$, and Log is the principal logarithm on $\mathbb{C} \setminus (-\infty, 0]$. For $0 < \tau < 1$ and $c \in \mathbb{C}$ put

$$p(y; \tau, c) = \det(H + c - y(P - \tau)), \quad Y(\tau, c) = (P - \tau)^{-1}(H + c) = \left(\frac{P}{1 - \tau} - \frac{B}{\tau}\right)(H + c).$$

Since $H + c - y(P - \tau) = (P - \tau)(Y(\tau, c) - y)$, the roots of $p(\cdot; \tau, c)$, with multiplicity, are the eigenvalues of $Y(\tau, c)$, and the leading coefficient $\det(\tau - P) = (\tau - 1)^{M-r} \tau^r$ does not depend on c . Let $y_1(\tau), \dots, y_M(\tau)$ be the roots for $c = 0$, and put

$$\mathsf{L}(H) = \frac{1}{2\pi} \int_0^1 \sum_{i=1}^M |\text{Im } y_i(\tau)| d\tau, \quad \mathsf{R}(H) = \text{Tr } H_+ - \text{Tr } (H_d)_+.$$

Theorem B.1 (Radon identity). *The function $\tau \mapsto \sum_i |\text{Im } y_i(\tau)|$ is continuous on $(0, 1)$ and bounded by $M\|H\|/\sqrt{\tau(1-\tau)}$, and $\mathsf{L}(H) = \mathsf{R}(H)$.*

If $r \in \{0, M\}$, then $P - \tau$ is a nonzero multiple of $\mathbf{1}$, all roots are real and $H_d = H$, so both sides vanish. We assume $1 \leq r \leq M - 1$ from now on. Applied to $H = g - \xi P - w$, Theorem B.1 says that the integrals of the density ρ of Theorem D along the lines $s = w + \xi\tau$ are the real slices $U_\xi(w)$ of Lemma 3.5 (Section B.4); this explains the name.

B.2. The complexified pencil. In this subsection $\text{Im } c > 0$ and $0 < \tau < 1$, and we put $\Gamma_c = \|H\| + |c|$. For a root y of $p(\cdot; \tau, c)$ we fix a unit vector v with $(H + c)v = y(P - \tau)v$, and we put $a = \langle v, (P - \tau)v \rangle = \|Pv\|^2 - \tau \in [-\tau, 1 - \tau]$.

Lemma B.2 (location and count). (1) $\text{Im } c = a \text{Im } y$. Hence $a \neq 0$, $y \in \mathbb{C}_+$ if and only if $a > 0$, $\text{Im } y \geq \text{Im } c/(1 - \tau)$ if $y \in \mathbb{C}_+$, and $\text{Im } y \leq -\text{Im } c/\tau$ if $y \in \mathbb{C}_-$.

- (2) No root of $p(\cdot; \tau, c)$ is real, and exactly $M - r$ roots, counted with multiplicity, lie in $\mathbb{C}_+$.
 (3) For H_d in place of H the roots are the numbers $(\mu_k + c)/(1 - \tau) \in \mathbb{C}_+$ and $-(\beta_j + c)/\tau \in \mathbb{C}_-$.

Proof. (1) $\langle v, (H + c)v \rangle = \langle v, Hv \rangle + c$ and $\langle v, (H + c)v \rangle = ya$, where $\langle v, Hv \rangle$ and a are real; taking imaginary parts gives $\text{Im } c = a \text{Im } y$. As $\text{Im } c > 0$, $a \neq 0$ and $\text{Im } y = \text{Im } c/a$, with $0 < a \leq 1 - \tau$ or $-\tau \leq a < 0$. (2) No root is real by (1). For $\sigma \in [0, 1]$, part (1) applies to the Hermitian matrix σH , so no root of $\det(\sigma H + c - y(P - \tau))$ is real. The leading coefficient does not depend on σ , so the roots, as an unordered M -tuple, depend continuously on σ , and the number of roots in $\mathbb{C}_+$ is constant on $[0, 1]$. At $\sigma = 0$ the polynomial is $(c - y(1 - \tau))^{M-r}(c + y\tau)^r$, with the root $c/(1 - \tau) \in \mathbb{C}_+$ of multiplicity $M - r$ and the root $-c/\tau \in \mathbb{C}_-$ of multiplicity r . (3) $H_d + c - y(P - \tau)$ is block diagonal, equal to $PHP + c - y(1 - \tau)$ on $\text{ran } P$ and to $BHB + c + y\tau$ on $\text{ran } B$. $\square$

Lemma B.3 (bounds). (1) Every root satisfies $|y| \leq \Gamma_c / \min(\tau, 1 - \tau)$.
 (2) If $y \in \mathbb{C}_+$ and $\tau \leq \frac{1}{2}$, or $y \in \mathbb{C}_-$ and $\tau \geq \frac{1}{2}$, then $|y| \leq \Gamma_c / \sqrt{\tau(1 - \tau)}$.
 (3) Let $\Sigma_{\pm}(\tau, c)$ be the sums of the roots in $\mathbb{C}_{\pm}$, and $\Sigma_+^0(\tau, c) = (\text{Tr}_P(PHP) + (M - r)c)/(1 - \tau)$, $\Sigma_-^0(\tau, c) = -(\text{Tr}_B(BHB) + rc)/\tau$ the corresponding sums for H_d . Then $\Sigma_+ + \Sigma_- = \Sigma_+^0 + \Sigma_-^0$, and

$$|\Sigma_+(\tau, c) - \Sigma_+^0(\tau, c)| \leq M\Gamma_c(2 + (\tau(1 - \tau))^{-1/2}).$$

Proof. We have $|y|(P - \tau)v| = |(H + c)v| \leq \Gamma_c$. (1) $|(P - \tau)v|^2 = (1 - \tau)^2|Pv|^2 + \tau^2|Bv|^2 \geq \min(\tau, 1 - \tau)^2$. (2) With $|Pv|^2 = a + \tau$, $|(P - \tau)v|^2 = (1 - \tau)^2(a + \tau) + \tau^2(1 - \tau - a) = \tau(1 - \tau) + a(1 - 2\tau)$, and $a(1 - 2\tau) \geq 0$ in both cases by Lemma B.2(1). (3) $\Sigma_+ + \Sigma_- = \text{Tr } Y(\tau, c)$, and $\text{Tr}[(\frac{P}{1 - \tau} - \frac{B}{\tau})(H - H_d)] = \text{Tr}[(\frac{P}{1 - \tau} - \frac{B}{\tau})(PHB + BHP)] = 0$, so $\text{Tr } Y(\tau, c)$ is the same for H and for H_d . For $\tau \leq \frac{1}{2}$, part (2) gives $|\Sigma_+| \leq (M - r)\Gamma_c / \sqrt{\tau(1 - \tau)}$, and $|\Sigma_+^0| \leq (M - r)\Gamma_c / (1 - \tau) \leq 2(M - r)\Gamma_c$. For $\tau \geq \frac{1}{2}$ write $\Sigma_+ - \Sigma_+^0 = \Sigma_-^0 - \Sigma_-$ and use part (2) for the roots in $\mathbb{C}_-$ and $|\Sigma_-^0| \leq r\Gamma_c / \tau \leq 2r\Gamma_c$. $\square$

Let $\mathcal{L}_+(\tau, c) = \sum \text{Log } y$, the sum over the roots y in $\mathbb{C}_+$ with multiplicity, and $\mathcal{L}_+^0(\tau, c) = \sum_k \text{Log } \frac{\mu_k + c}{1 - \tau}$, the same sum for H_d .

Lemma B.4 (the Burgers identity). On $(0, 1) \times \mathbb{C}_+$ the functions Σ_+ and $\mathcal{L}_+$ are continuously differentiable in τ and holomorphic in c , their partial derivatives are jointly continuous, and

$$\partial_{\tau} \mathcal{L}_+ = \partial_c \Sigma_+, \quad \partial_{\tau} \mathcal{L}_+^0 = \partial_c \Sigma_+^0 = \frac{M - r}{1 - \tau}.$$

Proof. Fix $(\tau_0, c_0) \in (0, 1) \times \mathbb{C}_+$, put $\eta = \frac{1}{2} \text{Im } c_0$ and $R_0 = 2(\|H\| + |c_0| + 1) / \min(\tau_0, 1 - \tau_0)$, and let U be a neighbourhood of (τ_0, c_0) on which $\text{Im } c \geq \eta$, $|c| \leq |c_0| + 1$ and $\min(\tau, 1 - \tau) \geq \frac{1}{2} \min(\tau_0, 1 - \tau_0)$. By Lemmas B.2(1) and B.3(1), for $(\tau, c) \in U$ the roots in $\mathbb{C}_+$ lie in $Q = \{y : \text{Im } y \geq \eta, |y| \leq R_0\}$. Let Ω be the open disc with centre iT and radius $T - \frac{\eta}{2}$, where $T = R_0^2 / \eta + 1$. Its closure lies in $\{\text{Im } y \geq \frac{\eta}{2}\} \subseteq \mathbb{C}_+$, and $Q \subseteq \Omega$: for $y = u + iw \in Q$, $|y - iT|^2 = |y|^2 - 2wT + T^2 \leq R_0^2 - 2\eta T + T^2 < (T - \frac{\eta}{2})^2$, because $\eta T > R_0^2$. So for $(\tau, c) \in U$ the roots in $\mathbb{C}_+$ lie in Ω , the roots in $\mathbb{C}_-$ lie outside $\bar{\Omega}$, and $p(\cdot; \tau, c)$ has no zero on $\partial\Omega$. Writing $p = \det(\tau - P) \prod_i (y - y_i)$, so that $p_y / p = \sum_i (y - y_i)^{-1}$, the Cauchy integral formula gives, for every φ holomorphic near $\bar{\Omega}$,

$$(B.1) \quad \sum_{y_i \in \mathbb{C}_+} \varphi(y_i) = \frac{1}{2\pi i} \oint_{\partial\Omega} \varphi(y) \frac{p_y(y; \tau, c)}{p(y; \tau, c)} dy.$$

We take $\varphi(y) = y$ and $\varphi = \text{Log}$. The integrand is a smooth function of $(y, \tau, \text{Re } c, \text{Im } c)$ on $\partial\Omega \times U$ and is holomorphic in c , so differentiation under the integral sign gives the regularity statements. Next, $p(y; \tau, c) = f(y, c + y\tau)$ with the polynomial $f(y, w') = \det(H + w' - yP)$, so $p_\tau = y p_c$. With $q = p_c/p$ we get $\partial_\tau(p_y/p) = \partial_y(p_\tau/p) = \partial_y(yq)$ and $\partial_c(p_y/p) = \partial_y q$, hence

$$\partial_\tau \mathcal{L}_+ - \partial_c \Sigma_+ = \frac{1}{2\pi i} \oint_{\partial\Omega} [\text{Log}(y) \partial_y(yq) - y \partial_y q] dy = \frac{1}{2\pi i} \oint_{\partial\Omega} \partial_y [(y \text{Log } y - y) q] dy = 0,$$

because $\text{Log}(y)(q + yq_y) - yq_y = \partial_y[(y \text{Log } y - y)q]$ is the derivative of a function that is holomorphic near $\partial\Omega$. For H_d the identity follows from the explicit formulas $\mathcal{L}_+^0 = \sum_k \text{Log}(\mu_k + c) - (M - r) \log(1 - \tau)$ and $\Sigma_+^0 = (\sum_k \mu_k + (M - r)c)/(1 - \tau)$. $\square$

For a simple root $y(\tau, c)$, formula (B.1) on a small circle around y (in place of $\partial\Omega$) reduces Lemma B.4 to $\partial_\tau y = y \partial_c y$, the complex inviscid Burgers equation; the lemma is the form of this equation, summed over the roots in $\mathbb{C}_+$, that remains valid at multiple roots.

Lemma B.5 (the two ends). *Fix $c \in \mathbb{C}_+$.*

- (1) *As $\tau \rightarrow 1^-$, $\mathcal{L}_+(\tau, c) - \mathcal{L}_+^0(\tau, c) \rightarrow 0$.*
- (2) *Let $T_0 = PHP + c - PHB(BHB + c)^{-1}BHP$ on $\text{ran } P$ (the matrix $BHB + c$ is invertible on $\text{ran } B$ since $\text{Im } c > 0$). Then $\text{spec } T_0 \subseteq \mathbb{C}_+$, and as $\tau \rightarrow 0^+$,*

$$\mathcal{L}_+(\tau, c) - \mathcal{L}_+^0(\tau, c) \rightarrow \theta_0(c) := \sum_{t \in \text{spec } T_0} \text{Log } t - \sum_k \text{Log}(\mu_k + c), \quad e^{\theta_0(c)} = \frac{\det(H + c)}{\det(H_d + c)}.$$

Proof. (1) $(1 - \tau)Y(\tau, c) = (P - \frac{1-\tau}{\tau}B)(H + c) \rightarrow P(H + c)$ as $\tau \rightarrow 1$. With respect to $\text{ran } P \oplus \text{ran } B$, $P(H + c)$ is block upper triangular with diagonal blocks $PHP + c$ and 0, so its eigenvalues are the numbers $\mu_k + c \in \mathbb{C}_+$ and 0 (r times). By continuity of eigenvalues, for τ near 1 exactly $M - r$ eigenvalues z_k of $(1 - \tau)Y$ lie near the $\mu_k + c$, hence in $\mathbb{C}_+$, labelled so that $z_k \rightarrow \mu_k + c$, and r lie near 0. Multiplication by $1/(1 - \tau) > 0$ preserves $\mathbb{C}_+$, and Y has exactly $M - r$ eigenvalues in $\mathbb{C}_+$ (Lemma B.2(2)); so the roots in $\mathbb{C}_+$ are the $z_k/(1 - \tau)$, and since $\text{Log}(z/(1 - \tau)) = \text{Log } z - \log(1 - \tau)$ for $z \in \mathbb{C}_+$, $\mathcal{L}_+ - \mathcal{L}_+^0 = \sum_k [\text{Log } z_k - \text{Log}(\mu_k + c)] \rightarrow 0$.

(2) By the Schur complement formula with respect to $\text{ran } P \oplus \text{ran } B$, $p_0(y) := \det(H + c - yP) = \det_B(BHB + c) \det_P(T_0 - y)$, a polynomial of exact degree $M - r$ whose roots form $\text{spec } T_0$. For real y the matrix $H - yP$ is Hermitian and $\text{Im } c > 0$, so $p_0(y) \neq 0$: $\text{spec } T_0$ contains no real number. As $\tau \rightarrow 0$, $p(y; \tau, c) = \det(H + c - yP + y\tau) \rightarrow p_0(y)$ coefficientwise. By Hurwitz's theorem, applied on a circle that encloses $\text{spec } T_0$ and on small circles around its points, for small τ exactly $M - r$ roots of $p(\cdot; \tau, c)$ lie in a fixed disc, and they converge to the points of $\text{spec } T_0$ with multiplicity. The other r roots leave every bounded set: $\tau Y(\tau, c) = (\frac{\tau}{1-\tau}P - B)(H + c) \rightarrow -B(H + c)$, whose eigenvalues are 0 ($M - r$ times) and the numbers $-(\beta_j + c)$, which lie in $\mathbb{C}_-$ at distance at least $\text{Im } c$ from $\mathbb{R}$; so r eigenvalues of Y are of the form $-(\beta_j + c + o(1))/\tau$ and lie in $\mathbb{C}_-$ for small τ . Since $\mathbb{C}_-$ contains exactly r roots, the $M - r$ bounded roots are those in $\mathbb{C}_+$. Their limits lie in the closure of $\mathbb{C}_+$ and are not real, so $\text{spec } T_0 \subseteq \mathbb{C}_+$ and, by continuity of Log on $\mathbb{C}_+$, $\mathcal{L}_+ \rightarrow \sum_{t \in \text{spec } T_0} \text{Log } t$. Also $\mathcal{L}_+^0 = \sum_k \text{Log}(\mu_k + c) - (M - r) \log(1 - \tau) \rightarrow \sum_k \text{Log}(\mu_k + c)$. Finally $e^{\theta_0} = \det_P T_0 / \det_P(PHP + c) = \det(H + c) / (\det_B(BHB + c) \det_P(PHP + c)) = \det(H + c) / \det(H_d + c)$. $\square$

Lemma B.6 (boundary values). *Let $c_0 \in \mathbb{R}$, $0 < \tau < 1$, and let $y_1, \dots, y_M$ be the roots of $p(\cdot; \tau, c_0)$. As $c \rightarrow c_0$ within $\mathbb{C}_+$, $\text{Im } \Sigma_+(\tau, c) \rightarrow \frac{1}{2} \sum_i |\text{Im } y_i|$ and $\text{Im } \Sigma_+^0(\tau, c) = (M - r) \text{Im } c / (1 - \tau) \rightarrow 0$. If $H + c_0 > 0$, all y_i are real.*

Proof. For real y , $p(y; \tau, c_0)$ is the determinant of a Hermitian matrix, hence real; so the non-real roots of $p(\cdot; \tau, c_0)$ come in conjugate pairs. The roots depend continuously on c , since the leading

coefficient does not. If z is a root of $p(\cdot; \tau, c_0)$ of multiplicity m_z , then for c near c_0 exactly m_z roots lie near z ; they lie in $\mathbb{C}_+$ if $z \in \mathbb{C}_+$, in $\mathbb{C}_-$ if $z \in \mathbb{C}_-$, and their imaginary parts tend to 0 if z is real. Summing over the roots in $\mathbb{C}_+$ gives $\text{Im } \Sigma_+(\tau, c) \rightarrow \sum_{\text{Im } z > 0} m_z \text{Im } z = \frac{1}{2} \sum_i |\text{Im } y_i|$. If $A = H + c_0 > 0$ and y is a non-real root with eigenvector v , then $\langle v, Av \rangle = y \langle v, (P - \tau)v \rangle$ with both inner products real, which forces $\langle v, (P - \tau)v \rangle = 0 = \langle v, Av \rangle$, a contradiction. $\square$

B.3. Proof of Theorem B.1. *Step 1.* For $c \in \mathbb{C}_+$ let $\Psi(c) = \int_0^1 [\Sigma_+(\tau, c) - \Sigma_+^0(\tau, c)] d\tau$; by Lemma B.3(3) the integral converges absolutely, uniformly for c in compact subsets of $\mathbb{C}_+$. For $0 < \delta < \frac{1}{2}$ the function $\Psi_\delta(c) = \int_\delta^{1-\delta} (\Sigma_+ - \Sigma_+^0) d\tau$ is holomorphic on $\mathbb{C}_+$ (Lemma B.4 and differentiation under the integral sign), and by Lemma B.4 and the fundamental theorem of calculus,

$$\Psi'_\delta(c) = \int_\delta^{1-\delta} \partial_c(\Sigma_+ - \Sigma_+^0) d\tau = \int_\delta^{1-\delta} \partial_\tau(\mathcal{L}_+ - \mathcal{L}_+^0) d\tau = (\mathcal{L}_+ - \mathcal{L}_+^0)(1 - \delta, c) - (\mathcal{L}_+ - \mathcal{L}_+^0)(\delta, c).$$

As $\delta \rightarrow 0$, $\Psi_\delta \rightarrow \Psi$ locally uniformly on $\mathbb{C}_+$, so Ψ is holomorphic and $\Psi'_\delta \rightarrow \Psi'$ (Weierstrass); by Lemma B.5, $\Psi'(c) = -\theta_0(c)$.

Step 2. The function $\theta_1(c) = \sum_j \text{Log}(\lambda_j + c) - \sum_j \text{Log}(\lambda_j^0 + c)$ is holomorphic on $\mathbb{C}_+$ and $e^{\theta_1} = \det(H + c)/\det(H_d + c) = e^{\theta_0}$. Hence $\theta_0 - \theta_1$ is continuous with values in $2\pi i\mathbb{Z}$, so it equals a constant $2\pi i k$ on the connected set $\mathbb{C}_+$. Let

$$\mathcal{E}(c) = \text{Tr}[(H_d + c) \text{Log}(H_d + c)] - \text{Tr}[(H + c) \text{Log}(H + c)],$$

that is, $\mathcal{E}(c) = \sum_j (\lambda_j^0 + c) \text{Log}(\lambda_j^0 + c) - \sum_j (\lambda_j + c) \text{Log}(\lambda_j + c)$. Since $\frac{d}{dc}[(\lambda + c) \text{Log}(\lambda + c)] = \text{Log}(\lambda + c) + 1$ and both spectra have M elements, $\mathcal{E}' = -\theta_1$. Hence $(\Psi - \mathcal{E})' = -\theta_0 + \theta_1 = -2\pi i k$, and $\Psi(c) = \mathcal{E}(c) - 2\pi i k c + \kappa_0$ on $\mathbb{C}_+$ for a constant $\kappa_0 \in \mathbb{C}$.

Step 3. Fix $c_0 \in \mathbb{R}$ and let $\epsilon \rightarrow 0^+$. By Lemma B.6, for every τ , $\text{Im}(\Sigma_+ - \Sigma_+^0)(\tau, c_0 + i\epsilon) \rightarrow \frac{1}{2} \sum_i |\text{Im } y_i(\tau; c_0)|$, where the $y_i(\tau; c_0)$ are the roots of $p(\cdot; \tau, c_0)$, that is, the roots of Theorem B.1 for $H + c_0$ in place of H ; and by Lemma B.3(3) the integrand is dominated by $M(\|H\| + |c_0| + 1)(2 + (\tau(1 - \tau))^{-1/2})$ for $\epsilon \leq 1$. By dominated convergence, $\text{Im } \Psi(c_0 + i\epsilon) \rightarrow \pi \mathbf{L}(H + c_0)$. On the other side, as $z \rightarrow t \in \mathbb{R}$ within $\mathbb{C}_+$, $z \text{Log } z \rightarrow t \log |t| + i\pi \min(t, 0)$ (with limit 0 at $t = 0$). Since $\min(t, 0) = t - t_+$ and $\sum_j \lambda_j = \sum_j \lambda_j^0$,

$$\begin{aligned} \text{Im } \mathcal{E}(c_0 + i\epsilon) &\rightarrow \pi \left[\sum_j \min(\lambda_j^0 + c_0, 0) - \sum_j \min(\lambda_j + c_0, 0) \right] \\ &= \pi \left[\sum_j (\lambda_j + c_0)_+ - \sum_j (\lambda_j^0 + c_0)_+ \right] = \pi \mathbf{R}(H + c_0). \end{aligned}$$

Taking imaginary parts in $\Psi = \mathcal{E} - 2\pi i k c + \kappa_0$ gives $\mathbf{L}(H + c_0) = \mathbf{R}(H + c_0) - 2k c_0 + \text{Im } \kappa_0 / \pi$ for every real c_0 .

Step 4. For $c_0 > \|H\|$ the matrices $H + c_0$ and $H_d + c_0$ are positive definite ($\|H_d\| \leq \|H\|$). By Lemma B.6 all roots are then real, so $\mathbf{L}(H + c_0) = 0$, and $\mathbf{R}(H + c_0) = \text{Tr}(H + c_0) - \text{Tr}(H_d + c_0) = 0$. Hence $-2k c_0 + \text{Im } \kappa_0 / \pi = 0$ for all $c_0 > \|H\|$, so $k = 0$ and $\text{Im } \kappa_0 = 0$, and $\mathbf{L}(H + c_0) = \mathbf{R}(H + c_0)$ for every real c_0 ; the case $c_0 = 0$ is the identity. Finally, $\tau \mapsto \sum_i |\text{Im } y_i(\tau)|$ is continuous on $(0, 1)$ by continuity of the roots, and every non-real root satisfies $|y| \leq \|H\| / \sqrt{\tau(1 - \tau)}$, by the argument of Lemma 3.3(3). $\square$

One can show in addition that $\kappa_0 = 0$, that is, $\Psi = \mathcal{E}$ on $\mathbb{C}_+$; we do not need this.

B.4. Theorem D from the Radon identity. Let g be Hermitian, let ρ and the roots $\xi_i(s, \tau)$ be as in Theorem D, and let $\lambda_j(\xi)$, $\lambda_j^0(\xi)$ and U_ξ be as in Lemma 3.5. The properties of ρ

stated in Theorem D are those of Lemma 3.3, whose proof is elementary. Fix $\xi, w \in \mathbb{R}$ and put $H = g - \xi P - w$. For $s = w + \xi\tau$,

$$g - s - x(P - \tau) = H - (x - \xi)(P - \tau),$$

so the roots of $p_{s,\tau}$ are the numbers $\xi + y_i(\tau)$, with $y_i(\tau)$ the roots of Theorem B.1 for this H , and $\rho(w + \xi\tau, \tau) = \frac{1}{2\pi} \sum_i |\operatorname{Im} y_i(\tau)|$. Moreover $H_d = g_d - \xi P - w$. Theorem B.1 therefore gives

$$(B.2) \quad \int_0^1 \rho(w + \xi\tau, \tau) d\tau = \sum_j (\lambda_j(\xi) - w)_+ - \sum_j (\lambda_j^0(\xi) - w)_+ = U_\xi(w),$$

the last equality because $(l - w)_+ - (w - l)_+ = l - w$ and $\sum_j \lambda_j(\xi) = \sum_j \lambda_j^0(\xi)$. Thus the integrals of ρ along all non-horizontal lines are the real slices U_ξ .

For real $a > 0$, $\int_{\mathbb{R}} e^{aw} (l - w)_+ dw = e^{al}/a^2$, and for real $a < 0$, $\int_{\mathbb{R}} e^{aw} (w - l)_+ dw = e^{al}/a^2$. Using the middle expression in (B.2) for $a > 0$ and the definition of U_ξ for $a < 0$, we get, for real $a \neq 0$,

$$\int_{\mathbb{R}} e^{aw} U_\xi(w) dw = a^{-2} \sum_j (e^{a\lambda_j(\xi)} - e^{a\lambda_j^0(\xi)}) = a^{-2} \mathcal{D}(a, a\xi),$$

as in Lemma 3.5(2). Since $\rho \geq 0$, Tonelli's theorem and the substitution $s = w + \xi\tau$ at fixed τ give

$$\int_{\mathbb{R}} e^{aw} U_\xi(w) dw = \int_0^1 \int_{\mathbb{R}} e^{aw} \rho(w + \xi\tau, \tau) dw d\tau = \int_0^1 \int_{\mathbb{R}} e^{as - a\xi\tau} \rho(s, \tau) ds d\tau.$$

Every real t is of the form $a\xi$, so $\mathcal{G}(a, t) = \mathcal{D}(a, t)/a^2$ equals $I(a, t) := \iint e^{as - t\tau} \rho(s, \tau) ds d\tau$ for all real $a \neq 0$ and all real t . Both functions are entire on $\mathbb{C}^2$: $\mathcal{G}$ by Step 0 of Section 3.5, and I because ρ is integrable with compact support. For fixed real t , $a \mapsto \mathcal{G}(a, t) - I(a, t)$ is entire and vanishes on $\mathbb{R} \setminus \{0\}$, hence identically; then for fixed $a \in \mathbb{C}$, $t \mapsto \mathcal{G}(a, t) - I(a, t)$ is entire and vanishes on $\mathbb{R}$, hence identically. This is (1.6). For the last statement of Theorem D, put $m(\tau) = \int_{\mathbb{R}} \rho(s, \tau) ds$, which is continuous on $(0, 1)$ and integrable. Since $\mathcal{G} = I$ on $\mathbb{C}^2$, Step 4 of Section 3.5 gives $\int_0^1 e^{-t\tau} m(\tau) d\tau = I(0, t) = \mathcal{G}(0, t) = \|BgP\|_F^2 \int_0^1 e^{-t\tau} d\tau$, that is, $\int_0^1 e^{-t\tau} (m(\tau) - \|BgP\|_F^2) d\tau = 0$ for every real t . Taking $t = 0, 1, 2, \dots$ and using that the polynomials in $e^{-\tau}$ are dense in $C[0, 1]$ (Weierstrass), $\int_0^1 \varphi(m - \|BgP\|_F^2) d\tau = 0$ for every continuous φ ; so $m = \|BgP\|_F^2$ almost everywhere, hence everywhere on $(0, 1)$ by continuity. $\square$

B.5. Theorem C without distributions. The proof of Theorem C in Section 3.6 uses Theorem D at $t = \mp ia$, Lemmas 3.1 and 3.2 (residues, geometric series, and the fact that two harmonic functions that agree on an open subset of the strip agree everywhere), and Tonelli's theorem. Only step (b) uses distributions and the uniqueness of Fourier transforms; we replace it by the following lemma.

Lemma B.7. *Let $Z: \mathbb{R} \rightarrow \mathbb{C}$ be continuous with $|Z(w)| \leq Ce^{-c|w|}$ for some $c > 0$. If $\int_{\mathbb{R}} e^{aw} Z(w) dw = 0$ for all a in a nonempty open interval $J \subseteq (-c, c)$, then $Z = 0$.*

Proof. The function $A(a) = \int_{\mathbb{R}} e^{aw} Z(w) dw$ is holomorphic on $\{| \operatorname{Re} a | < c\}$ (differentiation under the integral sign) and vanishes on J , hence on the whole strip. So $\hat{Z}(k) = \int e^{-ikw} Z(w) dw = A(-ik) = 0$ for all real k . For $\epsilon > 0$ and $w_0 \in \mathbb{R}$, Fubini's theorem and $\int_{\mathbb{R}} e^{iku - \epsilon^2 k^2/2} dk = (\sqrt{2\pi}/\epsilon) e^{-u^2/(2\epsilon^2)}$ give

$$0 = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{Z}(k) e^{ikw_0 - \epsilon^2 k^2/2} dk = \int_{\mathbb{R}} Z(w) \frac{e^{-(w-w_0)^2/(2\epsilon^2)}}{\epsilon\sqrt{2\pi}} dw,$$

and the right side tends to $Z(w_0)$ as $\epsilon \rightarrow 0$, since Z is bounded and continuous. $\square$

Step (b) without distributions. Let ω , σ , V and W be as in the proof of Theorem C. By (3.5), $\hat{\omega}(a) - \hat{\sigma}(a) = O(a^2)$ as $a \rightarrow 0$, so $\omega - \sigma$ has zero mass and zero first moment, and $W(w) = \int (w-u)_+ d(\omega-\sigma)(u) = \int (u-w)_+ d(\omega-\sigma)(u)$. For $0 < a < \pi$, the second expression and Fubini's theorem, which applies because $\iint e^{aw}(u-w)_+ dw d(\omega+\sigma)(u) = a^{-2} \int e^{au} d(\omega+\sigma)(u) < \infty$, give $\int e^{aw} W(w) dw = a^{-2}(\hat{\omega}(a) - \hat{\sigma}(a)) = \int e^{aw} V(w) dw$. The functions W and V are continuous and $O(e^{-\pi|w|})$ (step (b) of Section 3.6), so Lemma B.7 with $Z = W - V$, $c = \pi$ and $J = (0, \pi)$ gives $W = V$. Steps (c) and (d) are unchanged; step (d) uses only Theorem D and the formula $\mathcal{G}(0, t) = \|BgP\|_F^2(1 - e^{-t})/t$ of Step 4 of Section 3.5.

B.6. Remarks.

Remark B.8 (arbitrary Hermitian B ; Stahl's theorem). Let B be a Hermitian matrix with distinct eigenvalues $b_1 < \dots < b_n$ ($n \geq 2$) and spectral projections Π_k , let $H_d = \sum_k \Pi_k H \Pi_k$, and let $G_B = (b_1, b_n) \setminus \{b_2, \dots, b_{n-1}\}$. For $\tau \in G_B$ let $y_i(\tau)$ be the eigenvalues of $(B - \tau)^{-1}H$. The proof of Theorem B.1 extends to this setting: in a gap (b_l, b_{l+1}) exactly $\sum_{k>l} \text{rank } \Pi_k$ roots lie in $\mathbb{C}_+$, the bounds of Lemma B.3 hold with constants that depend on the gap and are integrable over it, and at an interior eigenvalue b_k the boundary terms coming from the two adjacent gaps cancel. The result is

$$\frac{1}{2\pi} \int_{G_B} \sum_i |\text{Im } y_i(\tau)| d\tau = \text{Tr } H_+ - \text{Tr}(H_d)_+.$$

The argument of Section B.4 then shows that, for Hermitian A and B ,

$$\text{Tr } e^{aA-tB} - \sum_k e^{-tb_k} \text{Tr}_{\Pi_k} e^{a\Pi_k A \Pi_k} = a^2 \iint e^{as-t\tau} \rho_{A,B}(s, \tau) ds d\tau,$$

where $\rho_{A,B}(s, \tau) = \frac{1}{2\pi} \sum_i |\text{Im } x_i(s, \tau)|$ for $\tau \in G_B$, with $x_i(s, \tau)$ the roots of $\det(A - s - x(B - \tau))$, and $\rho_{A,B} = 0$ otherwise. At $a = 1$ this says that $t \mapsto \text{Tr } e^{A-tB}$ is the Laplace transform of the positive measure $\sum_k \text{Tr}_{\Pi_k}(e^{\Pi_k A \Pi_k}) \delta_{b_k} + (\int_{\mathbb{R}} e^s \rho_{A,B}(s, \tau) ds) d\tau$ on $[b_1, b_n]$, which is Stahl's theorem [32], the former conjecture of Bessis, Moussa and Villani [3], here with a proof that uses neither distribution theory nor Fourier analysis. We make no priority claim about this by-product; for the history of the theorem and of the Fourier-slice mechanism see [32, 9, 23, 12]. The details are in the accompanying notes [26], and they were checked in the independent verification of Section 9.2; they are not used in this paper.

Remark B.9 (formalisation). The Lean theorem `QBP27.StripL3b.hyp_RI` is Theorem B.1, and `QBP27.StripL3b.theorem1_bmv2` is Theorem D, proved from it as in Section B.4. In the uniqueness step of the proof of Theorem C, the formal proof uses the Fourier inversion theorem of Mathlib in place of Lemma B.7.

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